TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering

Department of Electrical Engineering

Deep Learning Architectures for Two-Hop Relay Communication: A Comparative Study of Classical and Neural Network Relay Strategies

A thesis submitted toward the degree of

Master of Science in Electrical and Electronic Engineering

In Tel Aviv University

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This research work was carried out at Tel-Aviv University
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Abstract

This thesis compares classical and learned relay strategies for two-hop communication. The core study fixes one configuration: a single-relay SISO link with i.i.d. Rayleigh fast fading on both hops, complex baseband, Gray-coded QPSK and uncoded transmission, leaving the relay as the only variable. Eight strategies are evaluated, from AF and symbol-wise DF to generative and sequence models. Results are Monte Carlo estimates with 95% confidence intervals, calibrated against the closed-form composition.

Under this benchmark the best learned relays outperform AF at low SNR; at medium-to-high SNR, DF matches or exceeds every relay at no parameter cost, an empirical, uncoded per-symbol claim, not general optimality. A size sweep across every channel studied, replicated over three initializations, locates the smallest relay that costs nothing measurable and shows that channel memory rather than parameter count sets it: on a memoryless channel four parameters, a single hidden unit with no window, sit within 0.03 dB of the classical baseline, whereas three-tap channels need 73 to 145 and a window spanning the interference. Error does not rise again as capacity grows, so no overfitting is observed over the range swept. A hybrid relay switching to DF above the measured crossover is near-optimal at no extra cost if SNR is reliably estimated. A coded study raises that baseline: a rate-1/2 convolutional code with a *block*-DF relay, unlike symbol-wise DF, beats uncoded DF from 8 dB upward. Both learned relays overtake it by 20 dB (Mamba-S6 already at 16): a decoding relay spends the code's redundancy, a denoising one leaves it intact. Adapting modulation-and-coding makes rate dominate relay choice by an order of magnitude—unless latency reopens the gap.

The principal contribution is a ladder of four layers of progressively relaxed channel assumptions. At layer 1 classical processing wins: on a memoryless matched channel the MAP hard-decision DF rule is componentwise slicing after coherent equalization. Layer 2 adds an unknown three-tap ISI filter: the memoryless classical relays fail, DF's error rate rising with transmit power; a 169-parameter windowed MLP restores the link at constant per-symbol cost, 1–1.5 dB behind an exponential-cost Viterbi detector. Layer 3 degrades the channel estimate: the classical advantage holds to twenty pilots, lost by ten. Layer 4 redraws the channel every block, where the learned relay leads the best blind classical equaliser.

The results delineate operating regimes: classical processing where the model matches the channel, and the learned relay where the channel is unmodeled, poorly estimated, or missing per block.

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Contents

List of Symbols

Symbols are grouped as Latin capitals, Latin lowercase, Greek letters and other notation. Where a symbol carries more than one meaning, every meaning is listed and the context that disambiguates it is named.

Latin capitals

C	Ergodic Rayleigh (Shannon) capacity, bits per channel symbol
E_b	Energy per information bit
E_s	Average energy per transmitted symbol
G	Goodput, information bits per channel symbol; $G = \eta(1 - \text{FER})$
K	Constraint length of the convolutional code
L	Channel memory, i.e. the number of ISI taps (Chapter 6); *also* block length in symbols in the pilot-overhead study
L_{max}	Round-trip latency budget, in symbols
M	Constellation order (points per constellation)
N_0	One-sided noise power spectral density
P_e	Bit error probability
$Q(\cdot)$	Gaussian Q-function
R	Code rate, k/n
S	Number of trellis states, 2^{K-1} for a rate-$1/n$ code

Latin lowercase

$f_\theta(\cdot)$	Relay processing function; the only block varied in the canonical comparison
g	Unknown channel gain in the flat-channel control
$h_1[i]$, $h_2[i]$	Hop-1 and hop-2 fading coefficients at symbol index i
i	Symbol (time) index
k	Information bits per code input group; *also* bits per symbol, $\log_2 M$
n	Coded output bits per input group
$n_1[i]$, $n_2[i]$	Additive noise on hops 1 and 2
w	Window half-width; the relay observes $2w + 1$ samples
$x[i]$	Symbol transmitted by the source
$x_R[i]$	Symbol transmitted by the relay
$y_R[i]$, $y_D[i]$	Signal received at the relay and at the destination

Greek letters

$\alpha_k(s)$	BCJR forward recursion at trellis state s, step k
$\beta_k(s)$	BCJR backward recursion at trellis state s, step k
γ	Linear signal-to-noise ratio, $\gamma = 10^{\mathrm{SNR}_{\mathrm{dB}}/10} = E_s/N_0$
$\bar{\gamma}$	Average signal-to-noise ratio
$\gamma_k(s', s)$	BCJR branch transition-and-observation probability; distinct from the SNR γ above, disambiguated by its arguments
η	Spectral efficiency, $\eta = R \log_2 M$ bits per symbol
θ	Learned relay parameters; *also* unknown carrier phase in the flat-channel control (Section 6.1.1)
σ^2	Noise variance

Other notation

$\mathcal{CN}(0, 1)$	Circularly-symmetric complex Gaussian distribution, unit variance
$\mathbb{E}[\cdot]$	Expectation
$\mathbb{1}[\cdot]$	Indicator function
$\hat{\cdot}$	Estimate of the quantity beneath the hat

List of Figures

List of Tables

Chapter 1

Introduction

Relay communication is a key technique in contemporary wireless systems that uses intermediate nodes between a source and a destination to extend coverage, enhance reliability, and increase throughput. Classical relay strategies such as amplify-and-forward (`AF`) and decode-and-forward (`DF`) have been extensively analyzed and are widely used as baseline protocols; in several canonical relay-channel models, block-based `DF` is capacity-achieving or capacity-approaching under asymptotically long codewords [1, 2, 3, 4, 5]. As clarified in Remark 4.2, these results concern coded block operation and do not directly characterize the uncoded, symbol-wise setting studied in this thesis.

Recent advances in machine learning have motivated the application of neural networks to physical-layer communication problems, including detection, decoding, and end-to-end system optimization. Neural models can approximate nonlinear mappings that are analytically intractable and have been reported to offer advantages in low-SNR or non-ideal regimes.

This motivates the central question of this thesis: *under what conditions can neural network–based relays improve upon or complement classical relay strategies in two-hop relay communication?*

This thesis answers that question within a unified simulation framework in which, for the canonical comparison, the relay function is the only variable. Nine relay methods are implemented (Section 4.2), of which eight are evaluated in the canonical comparison (Section 5.2): two classical relay strategies (`AF` and `DF`) and six neural relays spanning supervised learning (`MLP`, SNR-adaptive `Hybrid` relay), generative modeling (variational autoencoder `VAE`), and sequence-based architectures (`Transformer`, `Mamba-S6`, and `Mamba-2 SSD` with structured state-space duality); a conditional GAN (`cGAN` with `WGAN-GP`) is implemented but excluded from the canonical comparison on cost grounds.

The objective is not to propose a new relay design but to answer the question above with a boundary rather than a verdict: to say where learning helps, where it does not, and why. Existing studies typically vary several things at once, a set of architectures and a set of channel models together, which makes it difficult to attribute a difference in error rate to model structure, to parameter count, or to the channel.

Method: one assumption at a time. This thesis answers the question with a monotone sequence of four layers, each relaxing a further assumption about the channel; the learned relay and the metric are held fixed throughout, while the classical baseline is strengthened at each layer to remain the toughest available opponent, and the channel, modulation, and evaluation budget are adapted as needed. Reading their outcomes in order *is* the answer to the central question.

At **layer 1** the channel is memoryless and perfectly known, which is the canonical setup of Chapter 5. Classical processing is not improved upon here, for structural rather than training-related reasons: on a memoryless matched channel the MAP symbol decision — and hence the optimal *hard symbol-regenerating* DF rule — is componentwise slicing after coherent equalization, which is precisely what decode-and-forward already does at no parameter cost. **Layer 2** (Chapter 6) adds channel memory and holds channel knowledge fixed. The memoryless classical relays now fail outright rather than degrade, and a windowed learned relay restores the link, but a sequence detector given the true channel is still ahead of it in BER for BPSK (QPSK reverses this, Section 6.1.2: sequence-optimal is not BER-optimal); what this layer isolates is therefore the *function class*, together with the cost of realizing it. **Layer 3** degrades the channel estimate to a finite pilot budget, and the classical receiver's advantage turns out to be real but purchased: it survives while enough pilots remain and is lost when they do not. **Layer 4** removes knowledge of the channel family altogether, redrawing the channel for every block, and here the learned relay leads.

The shape of that sequence is the contribution. What degrades from layer to layer is not the classical algorithms, which remain optimal for whatever channel they are handed, but their *preconditions*. The learned relay never becomes a better detector; it becomes the only one whose requirements are still satisfied. Hypotheses H1–H4 are settled at layer 1, and H5 at layers 2–4.

Scope of this thesis. This thesis studies one communication problem, held unchanged throughout, the classical source–relay–destination (two-hop) link, under *one canonical setup*: SISO on both hops, i.i.d. Rayleigh fading, complex baseband, QPSK, uncoded BER. The only axis varied is the *relay function*. BPSK is never transmitted on the canonical channel; it is the transmitted alphabet only in the unknown-channel experiments of Chapter 6, where the one-dimensional alphabet keeps the trellis small enough for an exact Viterbi benchmark, and it otherwise appears only in the closed-form expressions those chapters quote. Two dedicated departures from it are included: a supplementary coded study (Section 5.5) that measures a block-DF relay, soft-information forwarding, and an adaptive modulation-and-coding scheme against this baseline, and the unknown-channel contribution (Chapter 6). Other departures from the canonical setup are identified as future work in Chapter 7:

Table 1.1: Scope of the thesis: the canonical setup (QPSK on Rayleigh), its supplementary coded-study departure, and the unknown-channel contribution. No other topology, relay protocol, or performance metric is evaluated in the main line of this work; all such variations are deferred to future work

Axis	**Canonical value (future work in parentheses)**
Topology	SISO on both hops *(future work: MIMO second hop)*
Channel	i.i.d. Rayleigh fast fading (canonical), applied identically on both hops; AWGN solely as the noise term of the canonical model, the simulator being calibrated instead against the closed-form two-hop DF composition on the canonical channel itself (Table 5.2); unknown, mismatched, and time-varying channels are studied in the dedicated contribution (Chapter 6) *(future work: Rician and other channel families)*
Relay Function *(the studied axis)*	*Classical:* AF, DF. *Learned:* MLP, Hybrid, VAE, Transformer, Mamba-S6, Mamba-2 SSD; cGAN is implemented but excluded from the canonical comparison on cost grounds
Modulation	QPSK is paired with the canonical Rayleigh channel throughout (Chapter 5) and re-checked under the unknown-ISI channel of Chapter 6, whose relay comparisons otherwise use BPSK; 16-QAM additionally appears, not as a relay-architecture question, as a rung of the coded modulation-and-coding study (Section 5.5) *(future work: 16-QAM and denser as a relay-architecture question, 16-PSK, and beyond)*
Performance Metric	Uncoded BER vs. SNR, Monte-Carlo averaged

Of the classical relay protocols surveyed in the literature review, only those evaluated in this thesis are part of the studied system alongside the neural relays above: amplify-and-forward and symbol-wise decode-and-forward on the canonical comparison, the coded study's block-DF exception (Section 5.5), and the differential detection, pilot-aided Viterbi MLSE, and blind equalization baselines of the unknown-channel study (Chapter 6). Material outside that configuration, such as the general relay channel with a direct link, is background only.

1.1 Scope and Contributions

System model studied in this thesis. Throughout this work, a single canonical setup underlies the main line of argument: a half-duplex two-hop link, one relay, *no* direct source–destination path, SISO i.i.d. Rayleigh *fast* fading on both hops, complex baseband, Gray-coded QPSK, uncoded BER. Section 4.1 states it in full, once, and is the reference for every detail omitted here. The learned component of the system is confined to the *relay processing function* $f_\theta(\cdot)$, the

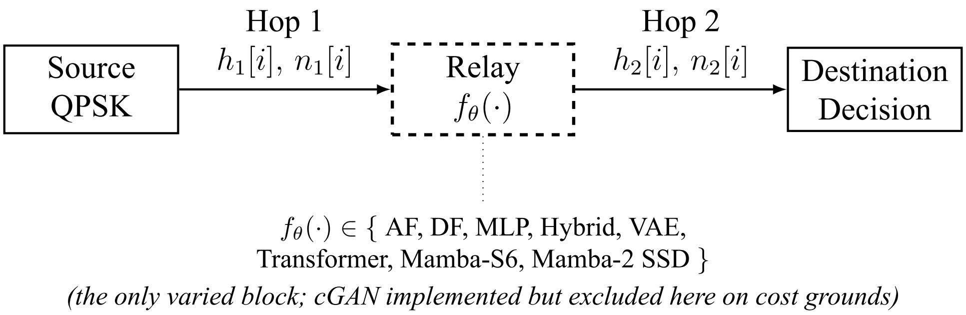


Figure 1.1: Canonical two-hop relay benchmark used throughout the thesis. The source, the channel model on both hops, and the destination are the same in every experiment of the canonical comparison; only the relay processing function $f_\theta(\cdot)$ (dashed) is varied among the eight candidate strategies compared there. The fading itself is not static: $h_1[i]$ and $h_2[i]$ are redrawn every symbol. Observed BER differences in the canonical comparison are therefore attributable to the relay strategy alone

block for which AF and DF are the classical alternatives; the source, the channel model, and the destination demodulator are held fixed in the canonical comparison, so observed BER differences there are attributable to the relay strategy alone; the coded and unknown-channel studies below deliberately vary the coding, channel, or destination processing as well. The simulator is calibrated on the canonical channel itself: symbol-wise DF's measured BER matches the closed-form two-hop composition closely, within 3.6% relative error at every tabulated SNR (Table 5.2; the largest gap is at 12 dB), once the QPSK constellation's E_s/N_0-to-E_b/N_0 conversion is applied. Beyond the canonical core, the thesis makes two further contributions of unequal weight. A smaller, supplementary study (Sections 5.5–5.5.4) revisits the uncoded/symbol-wise scope of the canonical setup itself: it adds a genuine information-theoretic block-DF baseline, coded with a rate-1/2 convolutional code and swept over constraint length on the QPSK/Rayleigh link, together with soft-decision counterparts of both the classical and the learned relay, to measure – rather than only assert, as Remark 4.2 does – where coding does and does not change the standing of symbol-wise DF against the learned relays. Its conclusion is architectural rather than algorithmic: a relay that decodes the code spends the code's redundancy before it transmits, so most of its residual errors cannot be repaired at the destination (only 16–25% are, Table 5.7), which is why a relay that merely denoises, leaving the redundancy intact, can end up ahead of one built on an optimal decoder. That advantage is then bounded, not erased, once the code rate is itself allowed to adapt to the channel: maximizing goodput over a modulation-and-coding grid shows the choice of rate dominates the choice of relay strategy by an order of magnitude or more, so the relay-decoding question this study opened matters far less once the link is doing the more basic thing correctly – and the envelope this adaptation reaches stays well short of the channel's own ergodic Shannon capacity throughout, under a third of it even at the top of the swept range, confirming the achieved goodput is limited by the finite constellation and coding grid rather than by anything left on the table for relay strategy to claim. Imposing a hard latency deadline on top of that adaptive-rate picture reopens the question the rate adaptation had closed: because a relay that decodes and re-encodes must buffer a full frame twice (once to decode, once again at the destination) against a denoising relay's once, a tight enough deadline locks the block-DF relay out of coding altogether while the denoising relay still has headroom, reversing the relay-strategy ordering by up to $1.5\times$ at operating points where the unconstrained comparison had favored the classical relay. The principal experimental contribution (Chapter 6) is a systematic study of learned relaying under *unknown and mismatched* channels that deliberately violate the assumptions of the canonical memoryless model, mapping, along every axis relevant to practical deployment, precisely when a minimal learned relay adds value over the strongest classical baselines and when it does not (H5). That chapter also revisits, under an unknown-ISI channel, whether the QPSK behaviour established on the canonical benchmark still holds: the memoryless-relay failure mode does, but the ordering between the learned relay and genie-CSI Viterbi MLSE does not (Section 6.1.2). All other departures (other channel families, a MIMO second hop, learned constellations) are identified as future work in Chapter 7. Capacity-theoretic results and the cooperative-diversity order are quoted in Chapter 2 as classical *background only*; they are not the operating point evaluated in the experiments.

Figure 1.1 summarizes this benchmark. The “identical” components named there are identical as *models*, not as particular channel realizations: the fading is *fast*, so $h_1[i]$ and $h_2[i]$ are redrawn independently for every symbol. Nothing in this thesis is a static or quasi-static link.

Chapter 2

Background and Related Work

Scope relative to the canonical model. Unless stated otherwise, the relay-processing methods reviewed in this chapter are discussed in the context of the canonical memoryless channel model adopted throughout Chapter 5. Channel-memory effects, sequence detection, and channel models with intersymbol interference (ISI) are intentionally deferred to the unknown-channel study of Chapter 6.

2.1 Classical Relay Strategies

The fundamental question that motivates this thesis is: can a learned relay function $f_\theta(\cdot)$ outperform the classical relay functions (AF scaling, DF regeneration) by exploiting patterns in the received signal that analytical methods do not capture? This question is particularly relevant at low SNR, where both AF (noise amplification) and DF (decoding errors) have well-known limitations, and where the non-linear decision boundary learned by a neural network may provide a potential advantage.

2.1.1 Amplify-and-Forward (AF)

The AF relay amplifies the received signal by a gain factor G that normalizes the output power: (Equation 2.1)

$$G = \sqrt{\frac{P_{\text{target}}}{\mathbb{E}[|y_R|^2]}} \tag{2.1}$$

The end-to-end SNR for AF relay is the standard harmonic-mean expression [3, 6]: (Equation 2.2)

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} \tag{2.2}$$

This form assumes *CSI-assisted variable-gain* AF, in which the relay knows its instantaneous first-hop gain and normalizes its output power per symbol (Equation 2.1); this is the variant implemented throughout this thesis. The additive $+1$ in the denominator originates from the relay's own noise being re-amplified along with the signal, and it is the term that distinguishes this convention from the fixed-gain (blind, long-term-power-normalized) variant, whose end-to-end SNR takes a slightly different form.

This expression reveals a fundamental limitation: even when one hop has high SNR, the effective SNR is bottlenecked by the weaker hop. Moreover, the relay amplifies both signal and noise from the first hop, leading to noise accumulation at the destination.

2.1.2 Decode-and-Forward (DF)

The DF relay demodulates the received signal, recovers the transmitted bits, and re-modulates clean symbols:

$$\hat{b}_R = \text{demod}(y_R), \quad x_R = \text{mod}(\hat{b}_R) \tag{2.3}$$

The end-to-end BER for DF is: (Equation 2.4)

$$P_e^{\text{DF}} = P_{e,1}(1 - P_{e,2}) + (1 - P_{e,1}) \cdot P_{e,2} \tag{2.4}$$

where $P_{e,1}$ and $P_{e,2}$ are the BER of the first and second hops, respectively: an end-to-end bit error occurs exactly when *one* of the two hops flips the bit (two flips cancel), so this expands to $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$, the same expression used in the AWGN analysis (Equation 2.27). For BPSK modulation over AWGN (real-valued noise with variance $1/\text{SNR}$), each hop's BER is $P_e = Q\left(\sqrt{\text{SNR}}\right)$. DF provides clean regeneration at high SNR but suffers from error propagation when the first-hop BER is non-negligible. As discussed in Remark 4.2, the "DF" baseline used here is the uncoded symbol-wise slicer, not information-theoretic block-DF; its optimality claims are therefore confined to the uncoded setting.

2.1.3 Other Classical Relay Techniques

AF and DF represent two canonical extremes of the relay processing spectrum: AF performs minimal processing (linear scaling) while DF performs maximal classical processing (full regeneration). Additional techniques (e.g. CF, EF, CoF) occupy intermediate points along this spectrum. In this thesis, the AI-based relay strategies are designed to *learn* the relay processing function from data, potentially discovering strategies that subsume or outperform these fixed classical schemes. The focus on AF and DF as classical baselines is therefore deliberate: on the canonical matched-channel benchmark they bracket the classical design space and serve as representative reference baselines against which the learned strategies are evaluated; they should not be read as strict lower and upper performance bounds, since under channel mismatch a learned relay can outperform DF (Chapter 6). These additional relay strategies are reviewed for completeness; they are not simulated in this thesis because the studied system assumes a two-hop link without a direct source-destination path.

2.2 Machine Learning in Wireless Communication

The application of machine learning to physical-layer wireless communication has gained significant momentum in recent years, driven by the ability of neural networks to learn complex non-linear mappings directly from data. Deep learning has been applied to channel estimation [7], signal detection [8], autoencoder-based end-to-end communication [9], and resource allocation [10]. These approaches learn complex mappings from data, potentially outperforming hand-crafted algorithms in scenarios where analytical solutions are intractable or suboptimal.

2.2.1 Theoretical Basis: Universal Approximation and Denoising

The theoretical justification for applying neural networks to relay signal processing rests on two pillars. First, the **universal approximation theorem** [11] guarantees that a feedforward network with a single hidden layer and a non-linear activation function can approximate any continuous function on a compact domain to arbitrary accuracy, given sufficient width. For relay denoising, the target function maps noisy observations to clean transmitted symbols:

$$f^* : \mathbb{R}^{2w+1} \to [-1, 1], \quad f^*(y_{i-w}, \ldots, y_{i+w}) = \mathbb{E}[x_i \mid y_{i-w}, \ldots, y_{i+w}] \tag{2.5}$$

This conditional expectation f^* is the Bayes-optimal denoiser: it minimizes the mean squared error (MSE) over all possible estimators. For BPSK symbols corrupted by AWGN, f^* reduces to the posterior mean:

$$f^*(y) = \tanh\left(\frac{y}{\sigma^2}\right) \tag{2.6}$$

which is a smooth sigmoid approaching the hard-decision signum function as $\sigma^2 \to 0$. A single tanh unit with the right input scaling represents this posterior *exactly* for a fixed σ^2. The deployed relay, however, is trained at SNR $\in \{5, 10, 15\}$ dB and evaluated over 0–20 dB, so one set of weights must approximate a whole *family* of posteriors indexed by σ^2, and on the fading channel a mixture over random effective SNRs. That a 169-parameter network approximates that family closely enough to track the classical baselines is an empirical finding of this thesis, not a consequence of the exact single-σ^2 representation.

Second, the **bias-variance decomposition** provides a framework for understanding model complexity:

$$\mathbb{E}[(\hat{x} - x)^2] = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}) + \sigma^2_{\text{irreducible}} \tag{2.7}$$

The irreducible term is the minimum achievable MSE, set by the channel noise. Adding parameters reduces bias but increases variance; since the target f^* here is essentially a soft threshold, the bias term is already small for modest networks, so on this reasoning additional capacity would buy little. This is the standard argument for preferring a small model on a simple target. Whether it is what actually happens in the measurements of this thesis is a separate question, and Section 7.2 is careful to report only that larger evaluated models did not improve BER, without attributing that to overfitting.

2.2.2 Prior Work in Deep Learning for Physical-Layer Processing

Ye et al. [7] demonstrated that deep neural networks can jointly perform channel estimation and signal detection in OFDM systems, achieving near-optimal performance with lower complexity than separate estimation and detection stages. Their key insight was that the DNN implicitly learns the channel's statistical structure, eliminating the need for explicit pilot-based estimation.

Samuel et al. [8] proposed DetNet, an unfolded projected gradient descent network for MIMO detection. By unrolling iterative optimization into a fixed number of neural network layers, DetNet achieves near-maximum-likelihood detection performance with $O(N_t^2)$ complexity instead of the exponential complexity of exhaustive ML search. This demonstrates the power of incorporating domain knowledge (the iterative structure of the detection problem) into the network architecture.

Dorner et al. [9] proposed treating the entire communication system: modulator, channel, and demodulator: as an autoencoder, trained end-to-end to minimize bit error rate. This approach jointly optimizes all components, potentially discovering novel modulation and coding schemes that outperform separately designed modules. The autoencoder paradigm is conceptually related to the relay processing task: both involve learning an encoder-decoder pair that maps through a noisy channel.

Sun et al. [10] applied deep learning to the NP-hard problem of resource allocation in interference networks, demonstrating that a DNN can learn near-optimal power control policies orders of magnitude faster than conventional iterative algorithms.

In the context of relay-assisted communication systems, most existing works employ neural networks primarily for *relay-selection* and *resource-allocation*, particularly in decode-and-forward (DF) relaying scenarios, where learning-based classifiers or predictors are used to identify the optimal relay based on channel state information, signal-to-noise ratios, or energy constraints, while the underlying relay operation remains conventional [12].

A broader perspective on the role of machine learning in wireless communications is provided in [13], which highlights the potential of learning-based methods across multiple protocol layers, including cooperative communications.

Another related line of work exploits the structural similarities between amplify-and-forward (AF) relay networks and neural networks, treating relay nodes as neurons with nonlinear transfer functions and using deep-learning optimization tools to tune relay gains and exploit hardware-induced nonlinearities [14, 15]. While these approaches demonstrate significant gains over classical linear AF schemes, they focus on analog gain optimization rather than symbol-level detection.

Classical cooperative communication frameworks based on AF and DF relaying remain well understood and widely adopted [5], but they rely on fixed linear forwarding or hard-decision rules. Consequently, despite substantial progress, relatively little attention has been devoted to learning the relay's symbol-level input-output mapping itself, motivating neural network-based detection at the relay as a generalized, data-driven alternative to conventional AF and DF processing.

2.2.3 Neural Network Relay Processing

For relay processing specifically, a supervised learning approach trains a neural network f_θ to minimize the empirical mean-squared error: (Equation 2.8)

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2, \quad \hat{x}_i = f_\theta(y_{i-w:i+w}) \tag{2.8}$$

where $\hat{x}_i = f_\theta(y_{i-w:i+w})$ is the network output based on a sliding window of $2w+1$ received symbols, and x_i is the clean transmitted symbol. This window-based approach provides temporal context that enables the network to exploit statistical dependencies in the noise-corrupted signal.

A well-known example of $f_\theta(\cdot)$ is the *multilayer perceptron (MLP)*, a feedforward artificial neural network composed of an input layer, one or more fully connected hidden layers with nonlinear activation functions, and an output layer, trained using backpropagation to model nonlinear input–output relationships. The MLP equation is shown in equation 2.9:

$$\mathbf{h} = \mathrm{ReLU}(\mathbf{W}_1\mathbf{w} + \mathbf{b}_1), \quad \hat{x} = \tanh(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2) \tag{2.9}$$

The sliding window formulation ($i \in [i-w : i+w]$) is motivated by the observation that adjacent received symbols share common noise statistics and channel conditions. Under the canonical model (i.i.d. fast fading, white noise), adjacent symbols are statistically independent, so a single-symbol estimator ($w = 0$) is already a sufficient statistic and a window confers no theoretical benefit. The window is therefore not motivated by temporal correlation in the canonical setting; it is retained because it is required by the unknown-channel study of Chapter 6, where a 3-tap ISI channel couples adjacent symbols and the relay must span that memory. Keeping a fixed windowed architecture across all experiments isolates the effect of the channel assumptions from the network structure.

Multi-SNR training is a key design choice: by training on data generated at multiple SNR levels (5, 10, 15 dB in this thesis), the network learns a denoising function that generalizes across operating conditions rather than specializing to a single noise level. This is analogous to training a robust estimator that operates well across a range of noise variances, a concept related to minimax estimation in statistical decision theory.

A critical question in applying neural networks to relay processing is the relationship between model complexity and performance. Prior work has generally assumed that larger models yield better performance, but this assumption has not been rigorously tested in the relay communication context. The bias–variance framework predicts that for a low-complexity target function like relay denoising, performance should plateau or degrade beyond a modest model size: a prediction investigated empirically in this thesis. Understanding this relationship is essential for practical deployment, where relay nodes may have limited computational resources.

2.3 Generative Models for Signal Processing

Generative models offer an alternative paradigm for relay signal processing. Rather than directly learning a denoising function (discriminative approach), generative models learn the distribution of clean signals $p(\mathbf{x})$ and use this knowledge to reconstruct the transmitted signal from noisy observations via Bayes' rule:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \tag{2.10}$$

The generative approach is theoretically appealing because it separates the modeling of the signal prior from the noise model, potentially enabling better generalization to unseen noise conditions.

2.3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [16] learn a latent representation by maximizing the evidence lower bound (ELBO). The generative model posits that data $\mathbf{x}$ is generated from a latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a decoder $p_\theta(\mathbf{x}|\mathbf{z})$. Since the true posterior $p(\mathbf{z}|\mathbf{x})$ is intractable, an encoder network $q_\phi(\mathbf{z}|\mathbf{x})$ approximates it. The ELBO objective is derived from the log-evidence decomposition: (Equation 2.11)

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]}_{\text{ELBO}(\theta,\phi;\mathbf{x})} + \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0} \tag{2.11}$$

Since the KL divergence is non-negative, the ELBO is a lower bound on the log-evidence. Maximizing the ELBO simultaneously trains the encoder to approximate the true posterior and the decoder to reconstruct the data. The ELBO decomposes into a reconstruction term and a regularization term:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction quality}} - \underbrace{\beta \cdot D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{Latent space regularity}} \tag{2.12}$$

The reconstruction term encourages the decoder to accurately reproduce the input from the latent code, while the KL term regularizes the latent space to be close to the standard Gaussian prior $p(\mathbf{z})$. The β parameter (β-VAE) [17] in Equation 2.12 controls the trade-off between reconstruction quality and latent space smoothness: $\beta < 1$ prioritizes reconstruction (beneficial for the relay task where accurate signal recovery is paramount), while $\beta > 1$ encourages disentangled latent representations.

The *reparameterization-trick* enables gradient-based optimization through the stochastic sampling layer:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \tag{2.13}$$

Instead of sampling from equation 2.13, the sample is expressed as:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \tag{2.14}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This moves the stochasticity outside the computational graph, allowing backpropagation through the encoder.

For relay signal processing, the VAE learns a compressed latent representation of the clean signal manifold. During inference, the noisy received signal is encoded into the latent space, and the decoder maps it back to a denoised estimate. The regularized latent space provides implicit denoising: noisy inputs that map to regions far from the learned signal manifold are pulled back toward high-probability regions during decoding. Sampling $\mathbf{z}$ stochastically at inference does, however, inject variance into an otherwise deterministic estimate, which is a plausible handicap for a deterministic BPSK denoising target; Section 2.3.3 reports what this is and is not found to cost in the measurements.

2.3.2 Conditional Generative Adversarial Networks

Conditional GANs (cGANs) [18] learn a denoising map by adversarial training: a generator conditioned on the noisy observation produces a clean estimate, and a discriminator is trained to separate generated from true clean symbols, the two optimizing a minimax objective. The original formulation is unstable to train, because the divergence the discriminator implicitly minimizes saturates when the two distributions do not overlap and stops supplying useful gradients. The Wasserstein GAN with gradient penalty (WGAN-GP) [19] replaces that objective with the Wasserstein distance, estimated through a critic constrained to be approximately 1-Lipschitz by penalizing the norm of its gradient at points interpolated between real and generated samples. This is the variant implemented in this thesis (Section 4.2).

The cGAN is described here and in Chapter 4 for completeness, but it is *excluded from every reported comparison* on cost grounds: training it took roughly two GPU-hours against the minimal MLP's few seconds on a CPU, which is not a defensible expenditure for a relay-denoising target this simple. No claim in this thesis rests on it, and the theory above is therefore kept deliberately brief.

2.3.3 Generative vs. Discriminative Paradigms for Relay Processing

The application of generative models to relay processing has, to the best of our knowledge, not been extensively studied. The key question is whether the generative inductive bias (learning the data distribution rather than just the input-output mapping) provides any advantage for the relay denoising task. For BPSK, the clean signal distribution is trivially simple (a discrete distribution on $\{-1, +1\}$), suggesting that the generative overhead may not be justified. This thesis compares VAE-based relay processing against classical and supervised learning methods on a common benchmark, and the results show that the generative paradigm provides no significant advantage for this particular task. A conditional GAN relay is implemented and described in Chapter 4 but is excluded from the reported comparison on cost grounds, so no claim is made about it here. On the canonical setup the VAE tracks the supervised MLP and symbol-wise DF rather than trailing them, reaching 0.00972 at 20 dB against DF's 0.00972 and the MLP's 0.00992 (Table 5.2), and it stays inside the feedforward group at every SNR in the equal-parameter comparison as well (Table 5.3). The generative bias is therefore neither an advantage nor a handicap here: it costs parameters and training time to arrive where a 169-parameter regressor already is. One qualification remains, unchanged by this: the relay samples $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference, and the deterministic variant that substitutes the posterior mean $\boldsymbol{\mu}$ at test time was not evaluated, so nothing here bears on whether that variant would do better.

2.4 Sequence Models: Transformers, State Space Models

Recent advances in sequence modeling have produced two competing paradigms with distinct computational properties and inductive biases. Both have achieved state-of-the-art results in natural language processing, but their relative merits for physical-layer signal processing (where sequences are real-valued temporal signals rather than discrete tokens) have, to the best of our knowledge, not been previously investigated.

2.4.1 Transformers and the Attention Mechanism

Transformers [20] use multi-head self-attention to capture global dependencies in sequences. The core attention mechanism computes a weighted combination of value vectors, where the weights are derived from the compatibility of query and key vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{2.15}$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{n \times d_v}$ are obtained from the input via learned linear projections W^Q, W^K, W^V. The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in magnitude, which would push the softmax into saturation regions with vanishing gradients.

Multi-head attention extends this by running h parallel attention heads, each with independent projections, and concatenating their outputs: (Equation 2.16)

$$\text{MultiHead}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \text{head}_i = \text{Attention}(\mathbf{X}W_i^Q, \mathbf{X}W_i^K, \mathbf{X}W_i^V) \tag{2.16}$$

Each head can attend to different aspects of the input: for signal processing, one head might focus on the immediate neighbors (local denoising) while another captures longer-range patterns. The total parameter count for multi-head attention is:

$$3d_{\text{model}}^2 + d_{\text{model}}^2 = 4d_{\text{model}}^2 \tag{2.17}$$

for Q, K, V projections and the output projection. The attention matrix:

$$\mathbf{A} = softmax(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}) \in \mathbb{R}^{n \times n} \tag{2.18}$$

computes pairwise interactions between all n positions, yielding a time and memory complexity of: $O(n^2)$. For the 11-symbol relay window used in this thesis, this is negligible ($11^2 = 121$ entries). However, the quadratic cost becomes prohibitive for longer sequences (e.g., $n = 1000$ symbols), motivating linear-time alternatives.

Positional encoding is necessary because the attention mechanism is permutation-equivariant (it treats input positions symmetrically). This thesis uses sinusoidal positional encoding: (Equation 2.19)

$$PE_{(pos,2k)} = \sin\left(\frac{pos}{10000^{2k/d}}\right), \quad PE_{(pos,2k+1)} = \cos\left(\frac{pos}{10000^{2k/d}}\right) \tag{2.19}$$

which injects position information into the embeddings, enabling the model to distinguish between symbols at different positions in the window.

2.4.2 Structured State Space Models

Structured state space models (SSMs) [21] treat a sequence model as a discretized linear dynamical system. A continuous system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$, $y(t) = \mathbf{C}\mathbf{x}(t) + Du(t)$ is discretized with a step size Δ (zero-order hold, $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$) to give the recurrence

$$\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k, \quad y_k = \mathbf{C}\mathbf{x}_k + Du_k \tag{2.20}$$

which is a linear recursive filter with a learned state transition. Two ingredients make this competitive with attention. The HiPPO framework supplies a principled initialization for $\mathbf{A}$ under which the state maintains a compressed polynomial approximation of the input history, and S4 [21] shows that a structured (diagonal or diagonal-plus-low-rank) $\mathbf{A}$ makes the recurrence cheap to evaluate. The resulting model is linear-time in sequence length, in contrast to attention's quadratic cost.

2.4.3 Mamba and Mamba-2

Mamba [22] makes the SSM parameters *input-dependent*, turning the time-invariant system into a time-varying one:

$$\Delta_k = \text{softplus}(W_\Delta u_k + b_\Delta), \quad \mathbf{B}_k = W_B u_k, \quad \mathbf{C}_k = W_C u_k \tag{2.21}$$

With $\mathbf{A}$ diagonal and negative, Δ_k acts as a learned gate: a large Δ_k drives $\bar{\mathbf{A}}_k = \exp(\Delta_k\mathbf{A})$ toward zero, resetting the state so the new input dominates, while a small Δ_k leaves $\bar{\mathbf{A}}_k \approx \mathbf{I}$, preserving the state and ignoring the input. This selectivity is the property of interest for relay processing: in principle the model can retain signal structure while suppressing noise, deciding per sample which of the two an observation carries. The selective layer is wrapped in a gated block with an expand-factor-2 projection and a short causal convolution.

Remark 2.1. These adaptive-filtering connections become directly relevant only in the unknown-channel study of Chapter 6; on the canonical memoryless channel the relay reduces to a per-symbol function and there is no temporal structure for a state to track.

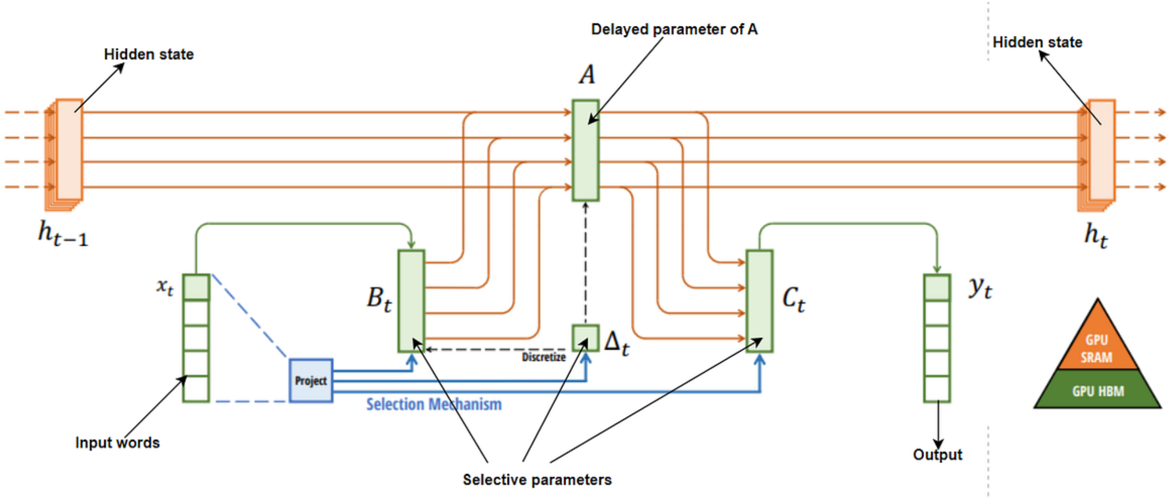


Figure 2.1: Structured SSMs map each channel independently through a higher-dimensional latent state; the selective variant makes the transition input-dependent so the state is reset or held per sample

Mamba-2 (SSD) [23] reformulates the same computation through an algebraic duality between the linear recurrence and a structured matrix multiplication: the output can be written $\mathbf{y} = M \cdot (\mathbf{B} \odot \mathbf{u})$ for a lower-triangular semi-separable matrix M. The practical consequence is that the sequence can be processed in chunks, with a batched matrix multiply inside each chunk and a single state hand-off between chunks, so the number of sequential steps falls from one per sample to one per chunk. Mamba-2 also forms the state-space parameters at the start of the block rather than sequentially along it. Both variants are evaluated here; at the short relay windows this thesis uses, the distinction is architectural rather than a source of measured advantage.

2.4.4 State Space Models for Signal Processing

The application of state space models to physical-layer signal processing is, to the best of our knowledge, largely unexplored. Classical signal processing relies extensively on LTI state space models (Kalman filters, Wiener filters), and the SSM framework provides a natural neural extension of these classical tools. The connection is direct: a trained SSM with fixed (input-independent) parameters implements a learnable linear filter, while the selective Mamba variant implements a learnable *adaptive* filter. For the relay denoising task, this means Mamba can potentially learn the optimal filter structure from data, without requiring manual specification of filter order, cutoff frequencies, or adaptation algorithms.

This thesis presents, to the best of our knowledge, the first comparison of Mamba-S6, Mamba-2

SSD, and Transformers for relay communication. The relay window used throughout is short ($n = 11$), which is precisely the regime where attention's quadratic cost is negligible and a linear-time recurrence has nothing to recover; the throughput advantages reported for the SSD formulation at longer context lengths [23] are therefore not exercised by this application and are not measured here.

2.5 Theoretical Foundations: Channel Models

This section presents the theoretical BER analysis for each channel model used in this study, derives the closed-form expressions, and validates them against Monte Carlo simulation. The theoretical-simulative comparison serves two purposes: (i) it verifies the correctness of the simulation framework, and (ii) it establishes the baseline performance that AI relays must beat.

2.5.1 AWGN Channel: Theoretical Analysis

The AWGN channel adds zero-mean Gaussian noise to the transmitted signal: (Equation 2.22)

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{P_s}{\mathrm{SNR}_{\mathrm{linear}}} \tag{2.22}$$

where $P_s = \mathbb{E}[|x|^2] = 1$ for unit-power BPSK symbols. The noise variance is inversely proportional to the linear SNR.

2.5.1.1 Single-hop BER

For BPSK ± 1 symbols and real-valued AWGN with variance $\sigma^2 = 1/\mathrm{SNR}$, an error occurs when the noise pushes the received sample past the decision boundary at the origin. The theoretical BER is: (Equation 2.23)

$$P_e^{\mathrm{AWGN}} = Q\left(\frac{1}{\sigma}\right) = Q\left(\sqrt{\mathrm{SNR}}\right) = \frac{1}{2}\,\mathrm{erfc}\left(\sqrt{\frac{\mathrm{SNR}}{2}}\right) \tag{2.23}$$

where $Q(x) = \frac{1}{2}\mathrm{erfc}(x/\sqrt{2})$ is the Gaussian Q-function and $\mathrm{SNR} = P_s/\sigma^2$ is the ratio of signal power to noise variance.

Remark 2.2 (Noise convention)**.** The standard textbook result $P_e = Q(\sqrt{2E_b/N_0})$ assumes that the one-sided noise PSD is N_0, giving a baseband noise variance of $N_0/2$ per real dimension. In our AWGN implementation the noise is purely real with variance $\sigma^2 = P_s/\mathrm{SNR}$, so $N_0 = 2\sigma^2 = 2P_s/\mathrm{SNR}$ and $E_b/N_0 = \mathrm{SNR}/2$. Substituting yields $Q(\sqrt{2 \cdot \mathrm{SNR}/2}) = Q(\sqrt{\mathrm{SNR}})$, which is the expression used throughout this work.

By contrast, the fading channels add **complex** Gaussian noise with total power $1/\mathrm{SNR}$ (i.e. $\sigma_I^2 =$

$\sigma_Q^2 = 1/(2\,\mathrm{SNR})$) and extract the real part after ZF equalization, halving the effective noise variance per decision dimension. This naturally introduces a factor of 2 inside the Q-function argument for all fading-channel BER expressions (the fading-channel analysis above), making them consistent with the standard textbook forms.

2.5.1.2 Two-hop AF BER

For amplify-and-forward with equal-SNR hops, the effective end-to-end SNR is:

$$\mathrm{SNR}_{\mathrm{eff}}^{\mathrm{AF}} = \frac{\mathrm{SNR}_1 \cdot \mathrm{SNR}_2}{\mathrm{SNR}_1 + \mathrm{SNR}_2 + 1} = \frac{\gamma^2}{2\gamma + 1} \tag{2.24}$$

where $\gamma = \mathrm{SNR}$ is the per-hop SNR. The resulting BER is

$$P_e^{\mathrm{AF}} = Q\left(\sqrt{\mathrm{SNR}_{\mathrm{eff}}}\right) \tag{2.25}$$

under the same real-noise convention as the single-hop case (Remark 2.2). At high SNR,

$$\mathrm{SNR}_{\mathrm{eff}} \approx \frac{\gamma}{2} \tag{2.26}$$

confirming the well-known 3 dB penalty of AF relaying.

2.5.1.3 Two-hop DF BER

For a decode-and-forward relay with equal-SNR hops, an error occurs at the destination if exactly one hop introduces an error:

$$P_e^{\mathrm{DF}} = P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2} \tag{2.27}$$

With equal hops (i.e. $P_{e,1} = P_{e,2} = P_e^{\mathrm{AWGN}}$), this becomes

$$P_e^{\mathrm{DF}} = 2P_e(1 - P_e) \tag{2.28}$$

At high SNR, $P_e \ll 1$ and

$$P_e^{\mathrm{DF}} \approx 2P_e \tag{2.29}$$

i.e., the two-hop penalty is approximately a factor of 2 in BER.

2.5.2 Rayleigh Fading Channel: Theoretical Analysis

The Rayleigh fading channel models non-line-of-sight (NLOS) propagation where the signal undergoes multiplicative fading:

$$y = hx + n, \quad h \sim \mathcal{CN}(0,1), \quad n \sim \mathcal{CN}(0,\sigma^2) \tag{2.30}$$

The fading coefficient h is a circularly-symmetric complex Gaussian random variable, so its magnitude $|h|$ follows a Rayleigh distribution:

$$f_{|h|}(r) = 2r \cdot e^{-r^2}, \quad r \geq 0 \tag{2.31}$$

with $\mathbb{E}[|h|^2] = 1$ (unit average power). The instantaneous SNR after equalization ($\hat{x} = y/h$) becomes $\gamma_h = |h|^2 \cdot \text{SNR}$, which is exponentially distributed.

2.5.2.1 Single-hop BER

Averaging the conditional BER $P_e(\gamma_h) = Q(\sqrt{2\gamma_h})$ over the exponential distribution of γ_h yields the closed-form [24, Eq. 14-4-15] (Equation 2.32):

$$P_e^{\text{Rayleigh}} = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right) \tag{2.32}$$

where $\bar{\gamma} = \text{SNR}$ is the average SNR. At high SNR ($\bar{\gamma} \gg 1$):

$$P_e^{\text{Rayleigh}} \approx \frac{1}{4\bar{\gamma}} \tag{2.33}$$

This $1/\text{SNR}$ decay is fundamentally slower than the exponential decay of AWGN ($Q(\sqrt{\gamma}) \sim e^{-\gamma/2}$), explaining why Rayleigh fading is significantly more challenging. The channel is **diversity-limited**: deep fades (where $|h| \approx 0$) cause errors regardless of the average SNR. Combating this diversity limitation is the primary motivation for the relay architectures studied in this thesis.

2.5.2.2 Two-hop DF BER

The two-hop DF relay over Rayleigh fading follows the same composition rule as AWGN:

$$P_e^{\text{DF,Rayleigh}} = 2P_e^{\text{Rayleigh}}(1 - P_e^{\text{Rayleigh}}) \tag{2.34}$$

2.6 Channel Coding: Convolutional Codes and Trellis Decoding

The channel models above assume an uncoded link. Chapter 5's supplementary coded study (Section 5.5) departs from that assumption, adding a convolutional code and a block-DF relay that decodes it. This section gives the minimal background needed to follow that study: what a convolutional code's rate and constraint length mean, how a receiver recovers the transmitted sequence from a noisy one, and how a single code is stretched across several rates.

2.6.1 Convolutional Encoding: Rate and Constraint Length

A convolutional encoder maps each group of k input bits to n coded output bits using a shift register of *constraint length* K – the number of k-bit input groups (including the current one) that influence each output. The code *rate* [24] is

$$R = \frac{k}{n} \tag{2.35}$$

so a rate-$1/2$ code, the mother code used throughout Chapter 5, emits two coded bits per information bit. This thesis uses $k = 1$ throughout, one information bit per trellis step; the encoder's output then depends on the current bit and the $K - 1$ preceding ones held in the shift register, so its evolution is a finite-state machine with [24]

$$S = 2^{K-1} \tag{2.36}$$

states, and the resulting state diagram unrolled over time is the code's *trellis*: every valid coded sequence corresponds to exactly one path through it. A larger K gives the code more memory to spread redundancy across, at the cost of a larger trellis to search; Chapter 5 measures this trade-off directly by sweeping $K \in \{3, 5, 7\}$, i.e. 4, 16 and 64 states (Section 5.5). A frame is conventionally *zero-tail terminated*: $K - 1$ zero bits are appended after the information bits to drive the encoder back to the all-zero state, so the trellis both starts and ends at a known point rather than an unknown one.

2.6.2 Maximum-Likelihood Sequence Decoding: The Viterbi Algorithm

Because every valid coded sequence is one path through the trellis, recovering the most likely transmitted sequence from a noisy received one is a shortest-path problem: find the trellis path whose hypothesized coded symbols are collectively closest to what was received. A brute-force search is infeasible – an L-bit information frame admits 2^L trellis paths. The Viterbi algorithm nonetheless solves the problem exactly, in time linear in L, by a dynamic-programming recursion over the trellis states: at each step it extends every surviving path by one branch and keeps only the lowest-cost path into each of the S states (add-compare-select), so the cost per step depends on S rather than on the number of paths [25, 24]. For a memoryless channel with an appropriate branch metric – squared Euclidean distance to the received sample under AWGN, as used by Chapter 5's coded block-DF relay [25] – the recovered sequence is

$$\hat{c} = \arg\min_{c \in \mathcal{C}} \sum_k |y_k - x(c_k)|^2 \tag{2.37}$$

where $\mathcal{C}$ is the set of trellis-valid coded sequences and $x(c_k)$ the modulated symbol for coded bits c_k – exactly the maximum-likelihood sequence estimate for the channel, with no approximation. Optimality is over the sequence, however, not over each bit independently, and this shapes how the decoder fails. When noise makes an incorrect path cheaper than the transmitted one, that path typically diverges from the correct path for several trellis steps before merging back, so a single decoding failure corrupts a *run* of information bits rather than an isolated one. Residual errors therefore arrive in bursts, and below the code's operating threshold these events become frequent enough that the burst penalty outweighs the coding gain. This is the mechanism behind the low-SNR coding penalty Chapter 5 measures (Section 5.5), where a fixed-rate code performs worse than no coding at all – a memoryless per-symbol slicer has no trellis to diverge along, and so cannot fail this way.

2.6.3 Soft-Decision Decoding: The Forward-Backward (BCJR) Algorithm

Viterbi decoding returns one hard path through the trellis and discards how close the alternatives were. The BCJR algorithm [26] instead computes, for every trellis transition, its posterior probability given the *entire* received sequence, using both a forward recursion $\alpha_k(s)$ (the likelihood of reaching state s at step k from everything received so far) and a backward recursion $\beta_k(s)$ (the likelihood of everything still to come, given state s at step k):

$$P(s_{k-1} = s',\, s_k = s \mid \mathbf{y}) \propto \alpha_{k-1}(s')\, \gamma_k(s', s)\, \beta_k(s) \tag{2.38}$$

where $\gamma_k(s', s)$ is the branch's transition-and-observation probability. Where Viterbi minimizes the probability of choosing the wrong *sequence*, BCJR minimizes the error probability of each *symbol* individually [26]; the two optimize different criteria, which is why neither dominates the other in Chapter 5's measurements. Marginalizing these joint probabilities over the transitions consistent with each coded bit or symbol gives a *soft* estimate – a posterior mean or log-likelihood ratio – in place of Viterbi's hard decision. This is the read-out Chapter 5 calls *soft block-DF* (Section 5.5): the relay forwards the BCJR posterior mean instead of committing to the single most likely codeword.

2.6.4 Puncturing, Spectral Efficiency, and Rate Adaptation

A code's rate and the modulation's constellation size M together set how much information it carries per channel use, its spectral efficiency [24]:

$$\eta = R \cdot \log_2 M \quad \text{[bits/symbol]} \tag{2.39}$$

so a rate-$1/2$ code halves the information rate of whatever modulation carries it, and comparing a coded link against an uncoded one at the same SNR without accounting for this is not a like-for-like comparison (Section 5.5.2 revisits Chapter 5's coded-versus-uncoded comparison on this basis). Rates other than a code's native k/n are obtained by *puncturing*: deleting a fixed, periodic subset of the encoder's output bits according to a puncturing pattern, and at the decoder re-inserting those positions as erasures (soft value zero) before running the identical trellis search – so one Viterbi or BCJR decoder implementation serves every punctured rate from the same mother code. This is how Chapter 5's rate-$1/2$ mother code is stretched to rates $2/3$ and $3/4$ for its link-adaptation study (Section 5.5.3).

Spectral efficiency alone is not the quantity a link adapts on, because it counts bits that were transmitted rather than bits that arrived. A higher-rate choice buys more bits per symbol but raises the frame error rate, and any real protocol discards a frame with residual errors. The operative objective is therefore *goodput*, the information rate actually delivered:

$$G = \eta \cdot (1 - \text{FER}) \quad \text{[information bits/symbol]} \tag{2.40}$$

Maximizing G over a grid of modulation-and-coding schemes at each SNR – the standard adaptive modulation and coding (AMC) problem – traces the link-adaptation envelope that Section 5.5.3 measures. The maximization is unconstrained: the FER penalty in Equation 2.40 already prices in unreliability, so no separate reliability constraint is imposed.

Chapter 3

Research Objectives

3.1 Main Objective

To determine *under which conditions* a learned relay surpasses classical relay processing in two-hop communication, and where it does not. A single canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER) serves as the controlled baseline on which classical and AI-based strategies are compared as a function of SNR regime and computational constraints. Two deliberate departures from that baseline then locate the boundary the canonical comparison cannot reach: a coded chain, which tests whether the conclusion survives when the classical relay is allowed to decode a channel code (Section 5.5), and channels whose structure lies outside the model class the classical relays assume, which is where the learned relay's value is actually established (Chapter 6). The canonical setup is therefore the control, not the scope.

3.2 Research Gap and Motivation

Despite growing interest in AI for wireless communication, a systematic comparison of relay processing paradigms (spanning classical signal processing, supervised learning, generative modeling, and modern sequence architectures) is, to the best of our knowledge, absent from the literature. The following specific gaps motivate this thesis:

1. **Gap 1: No cross-paradigm relay comparison.** Existing work on AI-based relay processing focuses on individual architectures in isolation. Ye et al. [7] demonstrated DNNs for OFDM detection, Dorner et al. [9] explored autoencoders for end-to-end communication, and Samuel et al. [8] applied unfolded networks to MIMO detection. However, no study systematically compares supervised feedforward networks, variational autoencoders, adversarial networks, attention-based Transformers, and state space models (Mamba [22], Mamba-2 SSD [23]) for the same relay denoising task under controlled conditions. State space models in particular have begun to be applied to receiver-side physical-layer tasks,

such as channel state prediction [27], but not, to the best of our knowledge, to the relay processing function itself, where the node must re-transmit rather than merely decide. Without such a comparison, practitioners cannot make informed architecture selections.

2. **Gap 2: The complexity–performance relationship is uncharacterized for relay processing.** Network sizing is a well-studied question in machine learning generally, and model-size trade-offs have been examined for other physical-layer learning tasks — automatic modulation classification, for instance, exhibits a parameter-count band beyond which accuracy degrades [28]; what is missing is the corresponding characterization for the relay, whose denoising task is unusually low-dimensional. Recovering one symbol from a noisy observation has a small minimum description length — the target is a soft threshold function — so it is not obvious that capacity helps here at all, and the question of how small a relay network can be before performance degrades has not, to the best of our knowledge, been answered. Nor has an equal-parameter comparison been reported for this task, which is what separates architectural effects from capacity effects.

This thesis addresses both gaps through a unified framework that implements nine relay strategies and compares eight of them on the canonical setup, with both original and normalized parameter counts, full statistical analysis, and a dedicated complexity study; Gap 1's adversarial paradigm (cGAN) is implemented but, being excluded from the canonical comparison on cost grounds, is addressed qualitatively rather than quantitatively. The following table summarizes the positioning of this work relative to the literature:

Table 3.1: Research positioning: comparison of this thesis against prior work in deep-learning-based relay communication

Aspect	Prior Work	This Thesis
Relay strategies compared	1–2 (typically AF vs. DF, or one AI method)	8 compared (2 classical + 6 AI); a ninth, cGAN, is implemented but excluded on cost grounds
AI paradigms	Single (e.g., supervised only)	4 implemented (supervised, generative, adversarial, sequential); adversarial (cGAN) excluded from the canonical comparison on cost grounds

Aspect	Prior Work	This Thesis
Evaluation discipline	Single ad-hoc configuration	One canonical setup (SISO, Rayleigh, QPSK) for the eight-relay comparison, chosen in advance and never varied there, calibrated against closed-form theory on that channel; a supplementary coded study deliberately varies the code, rate, and modulation against this baseline
Complexity analysis	None	Normalized ∼3K-parameter comparison + complexity study (does BER rise again with model size?)
State space models	Applied to receiver-side tasks; none as a relay processor	Mamba-S6, Mamba-2 SSD evaluated as relay processors
Statistical rigor	Often missing	Wilcoxon test, 95% CI, 10 trials per point

Relationship between the canonical study and the extension study. Hypotheses H1–H4 concern the canonical relay scenario introduced in Chapter 1: a memoryless two-hop channel with white noise and uncoded transmission. Hypothesis H5 studies a deliberately different regime in which the canonical model no longer represents the true channel. It therefore serves as a robustness extension rather than a direct continuation of the canonical experiments.

3.3 Research Hypotheses

Based on the theoretical background of Chapter 2, this thesis tests the following hypotheses. H1–H4 are posed and answered on the canonical setup. H5 is the central hypothesis of the thesis's principal contribution (Chapter 6), posed on channels *outside* the canonical family:

1. **H1 (AI advantage over AF at low SNR).** On the canonical setup, some AI-based relay strategies achieve lower BER than AF at low SNR (0–4 dB), where the non-linear denoising learned by the neural network provides an advantage over AF's noise amplification.

   Rationale: At low SNR, the MMSE-optimal (posterior-mean) denoiser of a BPSK symbol from a single AWGN observation, $f^*(y) = \mathbb{E}[x \mid y] = \tanh(y/\sigma^2)$, is a smooth nonlinear mapping that differs substantially from AF's linear scaling; a neural network can approximate f^* closely, whereas AF amplifies noise together with the signal. Note that f^* is optimal in the mean-square sense for the relay's local estimate; it is not claimed to be the end-to-end BER-optimal relay function, which is why H1 is posed as an empirical hypothesis rather than a theorem.

2. **H2 (DF dominance at high SNR).** The symbol-wise DF relay (Remark 4.2) achieves the lowest BER at medium-to-high SNR (≥ 6 dB), outperforming all AI methods.

   Rationale: At high SNR, $f^*(y) \approx \text{sign}(y)$, which is exactly the symbol-wise DF operation. AI relays introduce a small but non-zero approximation error, so DF should be optimal among the per-symbol relay functions considered in this uncoded setting. This dominance is asserted only within the uncoded, symbol-wise family evaluated here and is not a claim of information-theoretic optimality (cf. Remark 4.2).

3. **H3 (Overfitting at excess capacity).** There exists an optimal model size for relay denoising beyond which performance degrades due to overfitting, so that BER rises again once capacity exceeds what the mapping requires. Specifically, models with $\sim$100–200 parameters achieve performance comparable to models with 10–$100\times$ more parameters. Only the second, weaker part of this hypothesis was testable with the protocol originally adopted: establishing *overfitting* as the mechanism behind any upturn would require seed replication and train/test-gap measurement. The seed replication has since been performed for the MLP size axis by the sweep of Section 5.4, which finds no upturn above the across-initialization spread (Section 7.2.3); the train/test-gap measurement, and any replication of the *cross-architecture* comparison, remain outstanding and are recorded in Section 7.4.

   Rationale: The bias-variance analysis (Section 2.2.1) predicts that the low-complexity target function f^* requires minimal model capacity. Excess capacity increases variance without reducing bias.

4. **H4 (Architecture convergence at an equal parameter budget).** When all AI models are normalized to the same parameter count ($\sim$3,000), the performance differences between architectures narrow significantly, indicating that parameter count is a more important factor than architectural choice. This is an equal-*budget* comparison rather than an isolation of architecture from capacity: as Table 4.2 records, meeting the budget required adjusting width, depth, state dimension and input window size together, so those degrees of freedom co-vary with architecture and are not separately controlled.

   Rationale: All architectures are universal approximators and the target function is simple. At equal capacity, architectural inductive biases provide diminishing returns.

Unlike H1–H4, which assume the canonical memoryless channel throughout, H5 deliberately introduces impairments not present in the system model of Chapter 1, including channel memory, ISI, and nonlinear distortion.

5. **H5 (Principal contribution, unknown-channel mitigation).** On channels with structure unknown to the receiver chain, memory (intersymbol interference) or asymmetry (biased nonlinearity), the *memoryless* classical relays (AF, symbol-wise DF) fail to relay reliably, exhibiting error floors or severe degradation, whereas the minimal 169-parameter MLP relay restores reliable relaying by learning the impairment from data, approaching the performance of the optimal model-aware detector without requiring knowledge of the channel family.

   Rationale: AF forwards the channel impairment untouched, and symbol-wise DF applies a fixed memoryless zero-threshold decision; neither can compensate structure they do not model. Classical theory does provide a remedy *once the channel family is known*, estimate-then-MLSE (pilot-based channel estimation followed by Viterbi equalization) is optimal for a linear FIR channel and is included as a benchmark. The hypothesis is therefore not that learning beats classical theory, but that a single fixed learned architecture can approach model-aware performance with no channel-family knowledge, no explicit identification stage, and complexity independent of channel memory, and that it remains applicable on impairments (e.g., nonlinear bias) for which linear MLSE is mis-specified. How far this holds, and on which impairment structures it breaks down, is determined empirically in Chapter 6. Conversely, on channels that satisfy the classical assumptions, H2 predicts no learned relay can beat DF, so H5 also delineates the precise boundary of the learned relay's value.

3.4 Specific Objectives

1. **Implement nine relay strategies and compare eight of them** spanning four learning paradigms: no learning (AF, DF), supervised learning (MLP, Hybrid), generative modeling (VAE), and sequence modeling (Transformer, Mamba-S6, Mamba-2 SSD); the ninth, a conditional GAN (cGAN), is implemented but excluded from the comparison on cost grounds.
2. **Evaluate on the canonical setup:** SISO Rayleigh fast fading with QPSK, with symbol-wise DF checked against the closed-form two-hop composition on that same channel as the analytical calibration.
3. **Investigate the complexity–performance trade-off** by testing model architectures ranging from 0 parameters (classical) to 26,179 parameters (Mamba-2 SSD), and by conducting a normalized comparison at approximately 3,000 parameters.
4. **Test whether the canonical conclusion survives a coded chain** by replacing the uncoded symbol-wise DF baseline with a genuine block-DF relay — a convolutional code decoded and re-encoded at the relay — and comparing it against coded-aware learned relays at equal E_s/N_0 and again at equal throughput, since a code that halves the information rate

changes what "better" means (Section 5.5).

5. **Delineate the conditions under which a learned relay surpasses classical processing** on channels whose structure the classical relays do not model — unknown memory, non-linear asymmetry, and composite cascades of both — by locating the boundary rather than demonstrating a capability: a memoryless control where learning is expected to buy nothing, a matched-receiver bound (Viterbi MLSE with genie CSI), a pilot-budget sweep to find the crossover, and a blind regime where no channel posterior exists (Chapter 6).
6. **Identify practical deployment recommendations** for selecting the appropriate relay strategy given specific operational constraints (SNR range, computational budget, channel environment).

3.5 Scope and limitations

The following limitations define the scope of the canonical comparison, confined throughout to the setup below (SISO, Rayleigh fast fading, complex baseband, QPSK, uncoded); the supplementary coded and unknown-channel studies depart from it deliberately, as noted where relevant:

- **Modulation:** QPSK for the canonical relay comparison; the unknown-channel study (Chapter 6) mainly uses BPSK, with a QPSK re-check of the unknown-ISI result (Section 6.1.2), and 16-QAM enters only as a modulation-and-coding rung in the link-adaptation study. Higher-order constellations as a relay design question, and the joint 2D N-class formulation they require, are deferred to future work.
- **Channel knowledge:** Perfect CSI in the canonical core; estimation error is not modeled there. Chapter 6 lifts this with finite-pilot estimation, a swept pilot budget, and a zero-CSI blind regime.
- **Topology:** A single relay, two hops, half-duplex, SISO on both hops. Multi-relay, full-duplex and multi-antenna (MIMO) configurations are deferred to future work.

These delimitations isolate the relay processing function from protocol and system effects.

Chapter 4

Methods

4.1 System Model

The system under study is a two-hop relay network with a single relay node. All methods in this chapter are described for the canonical setup of Chapter 1: SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER. QPSK on the canonical Rayleigh channel is the pairing used here and throughout the canonical comparison; the coded and unknown-channel studies additionally use the pairings set out in Table 4.1.

Scope of the canonical system model. The system model defined in this chapter serves as the canonical benchmark used throughout the core experiments of the thesis (Chapter 5). It assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. Accordingly, all relay comparisons in the canonical study concern per-symbol processing and do not require sequence detection: the current observation $y_R[i]$ is already a sufficient statistic for detecting $x[i]$, and the classical AF and DF baselines operate per symbol. Some neural architectures nevertheless receive a finite window of neighboring observations (Section 4.2); this is an architectural input convention for the evaluated feedforward and sequence models, and under the canonical i.i.d. assumptions it neither introduces channel memory nor supplies additional information about $x[i]$. Channel memory, intersymbol interference (ISI), pilot-based channel estimation, blind equalization, and sequence-detection techniques such as Viterbi MLSE are introduced only later in Chapter 6 as deliberate departures from the canonical assumptions in order to study robustness under model mismatch. Channel coding is the other deliberate departure: the coded chain and its block-DF relay are specified where they are measured, in Section 5.5, rather than in this chapter.

Source bits are QPSK-modulated on the canonical setup, passed through Hop 1 (SISO Rayleigh), processed by the relay (classical or neural), transmitted over Hop 2 (SISO Rayleigh), and demodulated at the destination after one-tap coherent equalization at the corresponding receiver. Chapter 6 reuses this chain with BPSK (DBPSK where phase must be removed by differencing),

Figure 4.1: Two-hop relay system model: source transmits through two channels with a neural network relay in between. Each hop applies the channel operator followed by one-tap coherent equalization in the canonical setup

replacing Hop 1 with unknown ISI, a composite cascade, or a per-block redraw; Hop 2 is AWGN for the unknown-ISI comparison, Rayleigh otherwise. In the canonical setup, both hop channels are known perfectly at their respective receivers (Chapter 1); the unknown-ISI departure keeps perfect CSI, while the finite-pilot and blind departures relax it.

$$\text{Source} \xrightarrow{\text{Hop 1}} \text{Relay} \xrightarrow{\text{Hop 2}} \text{Destination} \tag{4.1}$$

4.1.1 Modulation

QPSK carries data in every canonical relay comparison; BPSK carries data mainly in the unknown-channel study (Chapter 6), which also re-checks the unknown-ISI result with QPSK (Section 6.1.2), 16-QAM additionally carries data as a rung of the coded link-adaptation study (Section 5.5), and denser constellations appear in Figure 4.2 for orientation only. The real-valued base case is BPSK, which maps a single bit to a real-valued symbol,

$$x = 1 - 2b, \quad b \in \{0, 1\} \implies x \in \{-1, +1\} \tag{4.2}$$

with hard-decision demodulation $\hat{b} = \mathbb{1}[\hat{x} < 0]$ at the destination. QPSK is defined in Section 4.5. Both are normalized to unit average symbol energy, $E_s = 1$, so they differ in bits per symbol rather than in transmit power: $k = 1$ and 2 respectively, giving $E_b = E_s/k$. Higher-order constellations as a relay design question are future work; 16-QAM appears only as a link-adaptation rung.

4.1.2 Hop Model

In the half-duplex two-hop model studied in this thesis, the relay cannot transmit and receive simultaneously, and there is no direct source–destination link. The communication proceeds in two time slots. On the canonical Rayleigh fast-fading channel each hop applies an i.i.d. complex fading coefficient followed by complex AWGN:

Slot 1 (Source → Relay), listening phase:

$$y_R[i] = h_1[i]\, x[i] + n_1[i], \quad h_1[i] \sim \mathcal{CN}(0, 1), \quad n_1[i] \sim \mathcal{CN}(0, \sigma^2) \tag{4.3}$$

Slot 2 (Relay $\rightarrow$ Destination), transmission phase:

$$y_D[i] = h_2[i]\, x_R[i] + n_2[i], \quad h_2[i] \sim \mathcal{CN}(0,1), \quad n_2[i] \sim \mathcal{CN}(0,\sigma^2) \tag{4.4}$$

where $x[i]$ is the complex symbol transmitted by the source at time i (QPSK on this canonical channel; the real BPSK alphabet $\{-1,+1\}$ never rides this complex channel, and is used in the real AWGN special case below and in the Chapter 6 experiments), $y_R[i]$ is the signal received at the relay, $x_R[i]$ is the signal transmitted by the relay, and $y_D[i]$ is the signal received at the destination. The relay output is written generally as

$$x_R[i] = f_\theta(\mathcal{Y}_{R,i})\,,$$

where $\mathcal{Y}_{R,i}$ denotes the observation supplied to the relay-processing function; for AF and symbol-wise DF, $\mathcal{Y}_{R,i} = y_R[i]$.

Each hop applies a channel function followed by optional equalization (Equation 4.5):

$$y = \mathcal{H}(x, \text{SNR}) + n \tag{4.5}$$

where $\mathcal{H}(\cdot)$ denotes the canonical Rayleigh fast-fading channel operator (the AWGN special case, $|h| \equiv 1$, is retained because the closed forms of Chapter 2 are stated in it). The calligraphic symbol distinguishes the channel *operator* from the scalar fading *coefficient* h used throughout the rest of this thesis.

Both hops use the same channel, canonically. $\mathcal{H}(\cdot)$ applies identically to both hops there, so a configuration is named by the pair (constellation, channel). The unknown-channel study is the exception, pairing an impaired Hop 1 with an unrelated Hop 2 – AWGN for the unknown-ISI comparison, Rayleigh otherwise (Chapter 6).

One-tap coherent equalization. Because the fading coefficient is known at each receiver, one-tap coherent equalization ($y \cdot h^*/|h|$ followed by $\Re\{\cdot\}$) reduces each hop to an equivalent real channel with a random per-symbol effective gain $|h[i]|$,

$$\tilde{y}_R[i] = |h_1[i]|\, x[i] + \tilde{n}_1[i], \qquad \tilde{y}_D[i] = |h_2[i]|\, x_R[i] + \tilde{n}_2[i], \tag{4.6}$$

with $\tilde{n}[i] \sim \mathcal{N}(0,\sigma^2)$ real per axis. Because the channel is complex, the canonical constellation is two-dimensional: QPSK places one bit on each axis, so both dimensions the channel acts in are used and the relay processes the pair; the real reduction above then applies per axis rather than discarding one. BPSK, being one-dimensional, is never transmitted over this complex canonical fading channel, which would require discarding the quadrature component; besides its use as the transmitted alphabet of the real-valued unknown-channel study (Chapter 6), it appears only in the real AWGN special case below and the closed-form expressions of Chapter 2. The *AWGN limit* referenced in those closed forms is the special case $|h[i]| \equiv 1$, i.e. $y_R[i] = x[i] + n_1[i]$ and $y_D[i] = x_R[i] + n_2[i]$.

The noise terms are independent and identically distributed with variance

$$\sigma^2 = \frac{P_s}{\gamma} \tag{4.7}$$

where $P_s = \mathbb{E}[|x|^2]$ is the transmit symbol power (unit for every constellation studied, per the normalization of Section 4.1.1) and γ denotes the per-hop linear signal-to-noise ratio (SNR), equal on both hops.

4.1.3 Evaluated Configurations

This thesis evaluates the primary (constellation, channel) configurations listed in Table 4.1, plus the QPSK/unknown-ISI generalization of Section 6.1.2 noted there. The governing principle is that **a real channel is paired with a real constellation and a complex channel with a complex constellation**: the signal space the constellation lives in and the signal space the channel acts on are the same space. Rayleigh with QPSK is the complex case, both living in the plane, where the fading coefficient $h \sim \mathcal{CN}(0, 1)$ rotates as well as scales and the constellation has a phase for it to act on.

The pairings excluded by this principle are precisely the ones that would force the simulation to patch over a mismatch. A real constellation on the complex Rayleigh channel must discard the imaginary part of the equalized sample after dividing by h, throwing away half the plane the channel acted in; a complex constellation on AWGN is not wrong, but it exercises none of what makes the complex channel interesting, since there is no phase rotation to undo. Neither is reported here.

Table 4.1: The primary evaluated (constellation, channel) pairings. The canonical and coded rows use the same channel on both hops; the unknown-channel row's Hop-1 filter is fixed throughout (not redrawn per block) and unknown to AF/DF/MLP, though the MLP is trained on it, matching the genie-CSI/pilot-aided Viterbi baselines; Hop-2 is AWGN (Chapter 6). A QPSK generalization of the unknown-ISI study, with the same Hop-1/Hop-2 split, is reported separately in Section 6.1.2

Constellation	Channel	Role	Why this pairing
QPSK (complex)	Rayleigh (complex)	Canonical	The complex case, and the operating point this thesis draws its conclusions on. QPSK places one bit on each axis, so both dimensions the fading coefficient acts in carry information and none is discarded.
BPSK (real)	Unknown 3-tap ISI	Extension	Used only in Chapter 6, where the impairment under study is channel *memory* rather than fading, and the one-dimensional alphabet keeps the trellis small enough for an exact Viterbi benchmark.
16-QAM (complex)	Rayleigh (complex)	Coded MCS rung	Used only as one rung of the modulation-and-coding grid of the coded study (Section 5.5), where the object of study is the rate-and-modulation choice rather than the relay's ability to process a denser constellation.

Beyond its role as the noise term of every channel model and its use in the closed-form expressions quoted as analytical background (Chapter 2), AWGN also appears as the Hop-2 channel of one unknown-channel relay comparison (Chapter 6); no standalone AWGN-only relay study is run.

A consequence worth stating plainly. The SNR argument sets the noise power from the average *symbol* energy, so it denotes E_s/N_0 in every configuration. Since $E_b = E_s/k$, E_b/N_0 is 3.01 dB below the stated value for QPSK. Where a BER figure is read against a BPSK closed form, this conversion is stated explicitly at the point of use.

Power Normalization. All relay strategies normalize their output power to ensure fair comparison:

$$x_R \leftarrow x_R \cdot \sqrt{\frac{P_{\text{target}}}{P_{\text{current}}}} \tag{4.8}$$

4.1.4 Relay Processing

On Hop 1 the relay's single receive antenna observes, as shown in Equation 4.9:

$$\tilde{y}_R[i] = |h_1[i]|\, x[i] + \tilde{n}_1[i], \quad \tilde{n}_1[i] \sim \mathcal{N}(0, \sigma^2) \tag{4.9}$$

where the effective observation follows one-tap coherent equalization of the Rayleigh fading coefficient (Section 4.1). Note that the per-symbol effective SNR is $|h_1[i]|^2\gamma$ (a random, exponentially distributed quantity) not a constant; the AWGN limit is recovered by setting $|h_1[i]| \equiv 1$. The relay applies its processing function to a sliding window of received samples and transmits a cleaner estimate $\hat{x}_R = f_\theta(\tilde{y}_{R,i-w:i+w})$; classical AF/DF use $w = 0$. This is purely a noise-removal task: in the canonical SISO setup there is no inter-stream interference at any stage, and the destination demodulates the Hop-2 output after that one-tap equalization with no further equalizer.

Remark 4.1 (Realizability of the symmetric window)**.** The canonical *channel* of this chapter is memoryless: $y_R[i]$ is already a sufficient statistic for $x[i]$, so $w = 0$ loses nothing, and the classical AF and symbol-wise DF baselines do use $w = 0$ and are strictly causal. The learned relays nevertheless take a symmetric window $y_R[i - w : i + w]$ with $w > 0$ everywhere they appear, the canonical experiments included: the MLP uses $w = 2$ (a 5-sample window, Section 4.2) and the sequence models $w = 5$. This is an *architectural input convention*, not a claim about the channel. Under the canonical i.i.d. assumptions the extra samples are redundant — they carry no additional information about $x[i]$ — and the convention is adopted so that the identical architecture carries over unchanged to the three-tap ISI channel of Chapter 6, where the window is what spans the memory and is no longer redundant. Holding the input format fixed across both settings is what allows the channel assumptions to be varied without also varying the network structure.

Two consequences follow, and both apply to the canonical experiments rather than only to Chapter 6. First, a symmetric window is not causal: it reads w samples ahead. It is admissible in the half-duplex protocol only because the relay buffers the whole listening-phase block before transmitting, and it costs a fixed w-symbol processing latency. Second, that non-causality is a limitation of the learned relays, recorded as such rather than justified: a strictly causal past-only window $y_R[i - 2w : i]$ is what a streaming deployment would require, and it is not evaluated in this thesis. Because the window is redundant on the canonical channel, none of the canonical results depend on the look-ahead; the cost is real only where the window is doing work, which is Chapter 6.

4.2 Relay Strategies

Nine relay strategies were implemented, spanning four learning paradigms. The selection of these nine strategies is designed to systematically explore the relay design space along three axes:

1. the degree of processing (from no learning to deep generative models).
2. the type of inductive bias (feedforward, recurrent, attention, generative).
3. the model capacity (from 0 to 26K parameters).

This section describes each strategy's architecture, training procedure, and the design rationale motivating each choice.

4.2.1 Selection and Classification of Relay Strategies

The selected nine relay strategies are drawn from four distinct architectural concepts:

(i) classical methods (AF and DF); (ii) supervised-learning models (MLP and hybrid MLP–DF); (iii) generative models (VAE and cGAN); and (iv) sequence models (Transformer, Mamba-S6, and Mamba-2 SSD).

1. **Classical AF:** Gain amplification as shown in Equation 2.1. AF serves as the minimal-processing classical baseline. It performs no learned processing and simply rescales the received signal, preserving both signal and noise. Its end-to-end effective SNR is given in Equation 2.2.

2. **Classical DF:** Demodulates, recovers bits, and re-modulates clean symbols (QPSK on the canonical channel, applied per axis). DF serves as the maximal-processing classical baseline. It performs full signal regeneration through symbol-wise demodulation and re-modulation. On the canonical matched channel and in the uncoded setting studied here, DF achieves the lowest BER among the classical baselines at medium-to-high SNR (see Remark 4.2). Its theoretical BER follows the error-composition formula of Equation 2.4.

3. **Supervised MLP (Minimal):** A two-layer feedforward neural network (multilayer perceptron, MLP) with 169 parameters following the relay architecture defined in Equation 2.9,

   where $\mathbf{w} \in \mathbb{R}^5$ is a sliding window of received symbols ($w = 2$ neighbors on each side), and the hidden layer has 24 neurons. Parameters: $(5 \times 24 + 24) + (24 \times 1 + 1) = 169$.

   Real network on a complex channel. The network above is real-valued, while the canonical channel and its QPSK constellation are complex. The two are reconciled the same way the DF baseline is, by operating *per axis*: after coherent compensation the equalized sample $\tilde{y}[i]$ decomposes into two independent real streams, its in-phase and quadrature components, each carrying one Gray-coded bit at ± 1 with equal noise variance (Section 4.1). The relay therefore applies the identical 169-parameter network twice per symbol, once to the I window and once to the Q window, and recombines the two outputs as $\hat{x}_R = \hat{x}_I + j\hat{x}_Q$ before power normalization. The parameter count is unchanged by this — one set of weights is shared across both axes, which is what makes the two sub-problems the *same* problem — but the arithmetic cost per transmitted symbol is twice the per-window figure. Every learned relay in the canonical comparison of Chapter 5 (MLP, hybrid, VAE, cGAN and the sequence models) is applied this way; AF and symbol-wise DF operate on the complex sample directly.

This is a design choice rather than the only option, and the coded study of Section 5.5 makes the opposite one for contrast: its relay interleaves the I and Q windows into a single input vector and classifies over the joint four-point QPSK alphabet, so it sees the constellation as a two-dimensional object rather than two independent axes. That choice is the main reason its parameter count is larger, and it is stated where it is used.

> **Architecture note:** The MLP relay is a standard discriminative two-layer perceptron trained with supervised learning (MSE loss on input–output pairs). In contrast, the generative models in this study are the VAE and cGAN (Section 4.2), which learn to *sample from* or *approximate* the data distribution. The MLP relay simply learns a deterministic mapping $f : \mathbb{R}^5 \to [-1, +1]$ from a noisy observation window to a denoised estimate of one axis: a classical regression task.

Design rationale: The tanh output activation naturally constrains the output to $[-1, +1]$, matching the binary ± 1 alphabet each axis carries under the per-axis application described above (and, in the real-valued unknown-channel study of Chapter 6, the BPSK symbol range directly). The ReLU hidden layer provides the non-linearity needed to approximate the Bayes-optimal soft threshold $\tanh(y/\sigma^2)$. With 24 hidden neurons and a 5-dimensional input, the network has approximately 34 parameters per input dimension: sufficient for the low-complexity denoising task while avoiding overfitting. He initialization is used for ReLU layers to maintain proper gradient flow.

Training uses MSE loss with multi-SNR training data (i.e., 5, 10, and 15 dB), 25,000 samples, and 100 epochs with a learning rate of 0.01.

4. **Hybrid:** SNR-adaptive relay that switches between MLP (low SNR) and DF (high SNR) based on a learned threshold. Combines the AI advantage at low SNR with DF's low-BER, zero-parameter operation at high SNR. Same 169 parameters as MLP.

   Design rationale: The Hybrid relay is motivated by the observation (confirmed empirically) that AI relays outperform DF only at low SNR, while DF achieves the lowest BER among the evaluated symbol-wise relay strategies at high SNR. By introducing a switching threshold, the Hybrid relay automatically selects the better strategy for each operating condition. The threshold is determined empirically from the training data by finding the

SNR at which MLP and DF BER curves cross. This approach requires no additional parameters beyond those of the MLP sub-network.

Training points at the relay node: SNR = **5, 10, 15 dB** (same trained MLP sub-network as the Minimal MLP relay).

5. **Generative VAE:** Probabilistic relay with encoder $q_\phi(\mathbf{z}|\mathbf{x})$ mapping to a latent space and decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructing the signal. Architecture: encoder $(7 \to 32 \to 16 \to \mu, \sigma^2(8))$, decoder $(8 \to 16 \to 32 \to 1)$. Total: 1,777 parameters. Trained with β-VAE loss ($\beta = 0.1$) for 100 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The low $\beta = 0.1$ prioritizes reconstruction quality over latent space regularization, appropriate for a task where accurate signal recovery is paramount. The 8-dimensional latent space provides sufficient representational capacity for the BPSK signal manifold while maintaining a tractable KL divergence. The encoder window of 7 symbols provides local context.

6. **Generative cGAN (WGAN-GP):** Adversarial relay with a generator conditioned on the noisy signal and a critic providing the training signal. Generator: $(7 + 8 \to 32 \to 32 \to 16 \to 1)$, Critic: $(1 + 7 \to 32 \to 16 \to 1)$. Total: 2,946 parameters. Trained with Wasserstein loss, gradient penalty ($\lambda = 10$), and L1 reconstruction loss ($\lambda_{\text{L1}} = 100$) for 200 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The high $\lambda_{\text{L1}} = 100$ ensures that the generator is primarily supervised by the reconstruction loss, with the adversarial term acting as a regularizer that encourages outputs to lie on the clean signal manifold. The noise vector $\mathbf{z} \in \mathbb{R}^8$ provides the stochastic input needed by the GAN framework. The 5:1 critic-to-generator update ratio follows the WGAN-GP recommendation for stable critic training. The doubled epoch count (200 vs. 100) compensates for the slower per-step convergence of adversarial training.

7. **Sequential models: Transformer:** Multi-head self-attention over a window of 11 symbols. Architecture: $d_{\text{model}} = 32$, 4 attention heads, 2 encoder layers, feedforward dimension 128. Total: 17,697 parameters. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The Transformer is included as the dominant sequence architecture in modern deep learning. The 11-symbol window provides the same temporal context as the Mamba-S6/SSD models. With $d_{\text{model}} = 32$ and 4 heads, each head operates in an 8-dimensional subspace, which is sufficient for capturing the local noise structure. The feedforward dimension of 128 ($4 \times d_{\text{model}}$) follows the standard Transformer expansion ratio.

8. **Sequential models: Mamba-S6** Selective state space model with input-dependent state transitions. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 Mamba blocks with residual connections. Total: 24,001 parameters. Each block applies: LayerNorm $\rightarrow$ expand $(32 \rightarrow 64) \rightarrow$ split to Conv1D/SiLU/S6 main branch and SiLU gate $\rightarrow$ contract $(64 \rightarrow 32) \rightarrow$ residual. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The expand factor of 2 doubles the internal dimension during the S6 scan, and the state dimension $N = 16$ makes each layer a 16th-order adaptive filter, more expressive than a classical Wiener or matched filter. LayerNorm, residual connections and SiLU gating are standard stabilizers carried over from the reference implementation.

9. **Sequential models: Mamba-2 SSD** Structured State Space Duality model that replaces the sequential S6 recurrence with a chunk-parallel structured matrix multiply. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, chunk size 8, 2 Mamba-2 blocks with SiLU gating and residual connections. Total: 26,179 parameters. Each block applies: LayerNorm $\rightarrow$ parallel gate/SSD branches $\rightarrow$ SiLU gate $\rightarrow$ contract $(64 \rightarrow 32) \rightarrow$ residual. The SSD layer builds a lower-triangular causal kernel M per chunk and applies it via batched matmul, with inter-chunk state passing for continuity. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$) and gradient clipping ($\|\nabla\| \leq 1$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The chunk size of 8 balances parallelism against the quadratic memory cost of the $L \times L$ chunk matrix, giving two chunks at the 11-symbol window. Gradient clipping at norm 1.0 and the S4D-style initialization $A_{\text{log}} = \log(1, 2, \ldots, N)$ are carried over from the reference implementation.

Remark 4.2 (Terminology: symbol-wise DF vs. information-theoretic DF)**.** The term "decode-and-forward" is used in two distinct senses in the literature, and the distinction matters for interpreting the results of this thesis. In its *information-theoretic* sense [1, 2], DF is a *block* strategy: the source protects each message block with a strong (ideally capacity-achieving) channel code, the relay decodes the entire block, and only then re-encodes and forwards the recovered message. This block-DF is inherently *not* memoryless and, over a single link, can operate reliably at any rate below the first-hop capacity.

The canonical system studied in the core of this thesis is *uncoded*: one QPSK symbol carries two information bits, one per axis, and no outer error-correction code is applied there. Consequently, the "DF" baseline throughout the canonical comparison refers to *symbol-wise slicing*: per-symbol Gray demodulation on the coherently equalized I and Q components followed by QPSK re-modulation, with $w = 0$. This is the appropriate classical counterpart to the learned relays under the uncoded BER metric used there, but it is strictly weaker than block-DF with coding; the reported DF results should not be read as bounds on coded block-DF performance. This caveat is measured directly, not left as a standing assumption: Section 5.5 adds a real information-theoretic block-DF baseline (a rate-1/2 convolutional code with soft-decision Viterbi decoding, swept over constraint length on the QPSK/Rayleigh link) and reports where it does and does not beat the symbol-wise DF used elsewhere in this thesis.

4.3 Simulation Framework

4.3.1 Monte Carlo BER Estimation

BER is estimated through Monte Carlo simulation, which provides an unbiased estimate of the true BER at each SNR point. The simulation parameters are:

- **Bits per trial:** 10,000
- **Trials per SNR:** 10 (independent random seeds)
- **Total bits per SNR point:** 100,000
- **SNR range:** 0 to 20 dB (step: 2 dB), yielding 11 evaluation points
- **Random seed control:** Each trial uses a unique seed for bit generation and noise realization

BER Computation:

$$\hat{P}_e = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(b_i \neq \hat{b}_i) \tag{4.10}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $N = 10{,}000$ is the number of bits per trial. By the central limit theorem, for large N and moderate BER (say $P_e \geq 10^{-3}$), the per-trial BER estimate is approximately normally distributed with variance $P_e(1 - P_e)/N$.

Confidence Intervals. The 95% confidence interval is computed from the $M = 10$ independent trial estimates as:

$$\mathrm{CI}_{95\%} = \bar{P}_e \pm t_{0.025,M-1} \cdot \frac{s}{\sqrt{M}} \tag{4.11}$$

where $\bar{P}_e$ is the sample mean BER across trials, s is the sample standard deviation, and $t_{0.025,9} = 2.262$ is the critical value of the Student's t-distribution with 9 degrees of freedom. At low BER ($\hat{P}_e \lesssim 10^{-4}$), the normal approximation may become unreliable due to the small number of observed errors; however, in this regime the relay methods converge and the relative ranking is stable.

BER resolution. With $N \cdot M = 100{,}000$ total bits per SNR point, the minimum detectable

BER is approximately $1/N = 10^{-4}$ per trial, or 10^{-5} for the aggregate. BER values below this threshold are reported as 0.

Per-experiment simulation budgets. The budget above ($M = 10$ trials $\times N = 10{,}000$ bits $= 100{,}000$ bits per SNR point) applies to the canonical relay comparison and the parameter-normalization/complexity study (E2–E3). Other experiments use different budgets, stated in their respective sections: the main unknown-ISI study, the flat-channel control, and the QPSK generalization check (Chapter 6, Sections 6.1, 6.1.1, 6.1.2) use $10 \times 100{,}000 = 1{,}000{,}000$ bits per SNR point, matching the project-wide $M = 10$ standard; the composite cascade (Section 6.1.3) likewise uses $10 \times 100{,}000$; and the blind and partial-posterior sub-studies (Sections 6.1.5–6.1.4) use $50 \times 20{,}000 = 1{,}000{,}000$ bits per SNR or operating point. The larger trial count in those last two is deliberate rather than incidental: their hop-1 channel redraws its ISI, amplifier and phase realization for every block, so a trial is a draw from the impairment family and the ensemble average is limited by the trial count, not by bits per trial. Increasing bits per trial sharpens the estimate for one particular channel; only increasing trials averages over the family the study is about. Every experiment therefore uses $M \geq 10$ trials, so the Wilcoxon signed-rank test is attainable throughout (at $M = 5$ its smallest two-sided p is $2^{-4} = 0.0625$, which could not reach $p < 0.05$).

4.3.2 Statistical Significance Testing

Differences between relay methods are assessed using the **Wilcoxon signed-rank test**, a non-parametric paired test. For each pair of relay strategies (A, B) at each SNR point, the test compares the $M = 10$ paired BER observations (Equation 4.12):

$$H_0 : \text{median}(P_{e,A} - P_{e,B}) = 0 \quad \text{vs.} \quad H_1 : \text{median}(P_{e,A} - P_{e,B}) \neq 0 \tag{4.12}$$

The Wilcoxon test is preferred over the parametric paired t-test for two reasons: (i) BER distributions are bounded ($[0, 0.5]$) and potentially skewed, violating the normality assumption, and (ii) the test is robust to outlier trials. The significance level is set at $\alpha = 0.05$ *per comparison*; no multiple-comparison correction (e.g. Holm or Bonferroni, or false-discovery-rate control) is applied across the many relay-pair and SNR-point comparisons made throughout this thesis, so the reported significance claims should be read at the per-comparison level rather than as holding jointly across the full set of tests performed.

4.3.3 Training Protocol

All AI relays are trained **once** at the beginning of each experiment using: - **Multi-SNR training data:** Signals generated at SNR = 5, 10, 15 dB in equal proportions - **Training samples:** 25,000 (supervised), 20,000 (generative), 10,000 (sequence models) - **Single training per channel:**

The same trained model is evaluated across all SNR points

The multi-SNR training protocol is critical: training at a single SNR would produce a model that performs well at that SNR but poorly at others (overfitting to a specific noise level). By training at three representative SNR values spanning the low-to-moderate range, the network learns a robust denoising function that generalizes across operating conditions. The SNR values 5, 10, 15 dB were chosen to cover the regime where AI relays provide the most benefit (low-to-medium SNR).

Impact of sparse SNR training grid. Because models are trained at only three SNR points ($\{5, 10, 15\}$ dB) but evaluated over eleven points (0 to 20 dB, step 2), performance reflects two regimes: (i) **interpolation** within the trained span (approximately 4–16 dB), where BER is generally stable, and (ii) **extrapolation** at the edges (0–2 dB and 18–20 dB), where mild degradation may occur due to noise-statistics mismatch. In the present results, this sparse-grid effect is secondary: the relay ranking remains consistent across the full evaluated SNR range.

Training method (what is trained, where it is trained). Training is performed **offline** before BER sweeps. Only the seven AI relays are optimized (MLP, Hybrid's MLP sub-network, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD); AF and DF are analytical baselines and have no trainable parameters. Training data are synthetically generated source/channel pairs under the multi-SNR protocol, with model-specific losses (MSE for supervised relays, ELBO for VAE, WGAN-GP + L1 for cGAN, Adam-based supervised loss for sequence models). Compute placement follows implementation: NumPy models (MLP/Hybrid) train on CPU, while PyTorch models train on CPU or CUDA depending on device availability/configuration. After training, weights are checkpointed and reused across all SNR evaluations; BER curves are then produced in **inference-only** mode without further parameter updates.

Weight initialization. He initialization [29] is used for all layers with ReLU activation ($W \sim \mathcal{N}(0, 2/n_{\text{in}})$), while Xavier initialization is used for layers with tanh activation ($W \sim \mathcal{N}(0, 1/n_{\text{in}})$). These initialization schemes maintain proper gradient magnitude through the network layers, preventing both vanishing and exploding gradients during training.

Optimizer settings. All PyTorch-based models use the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ (defaults). The MLP and Hybrid relays use a simple SGD implementation in NumPy with constant learning rate $\eta = 0.01$.

4.3.4 Reproducibility and Weight Management

All experiments use controlled random seeds at both the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results. Specifically, for seed s and trial t, the random state is initialized as seed$(s \cdot 1000 + t)$, ensuring that each trial is independent while the overall experiment is deterministic.

A **weight checkpoint system** saves trained model parameters to disk after training completes, enabling experiment resumption without retraining. This is particularly important for the cGAN (2-hour training time) and sequence models (24–37 minutes). The checkpoint system supports: - Automatic save after training with architecture metadata - Resume from saved weights with architecture validation - Seed-specific weight directories (e.g., `trained_weights/seed_42/`)

The framework includes 159 automated tests (pytest) covering all modules: channels, modulation, coding, relay strategies, simulation, statistics, and weight management, with 100% pass rate.

Code and reproducibility archive. The complete simulation framework (`relaynet`), every experiment script, and the committed result files each figure and table is generated from are maintained under version control (git) in the `Gilzuk/relaynet2` repository; each numerical claim in this thesis traces to a specific committed script and result file rather than to an unarchived intermediate run, and `verify_thesis_tables.py` in that repository re-derives every numerical table from its source file and flags any mismatch. The exact commit this thesis was typeset from is recorded in the repository's history rather than reproduced here, since stating a fixed hash in the text it describes is self-referential; the repository's tags and commit log identify the revision corresponding to any given PDF build.

4.4 Normalized Parameter Comparison

A fundamental confound in comparing AI architectures is that different models have vastly different parameter counts: from 169 (MLP) to 26,179 (Mamba-2 SSD): a ratio of 155:1. Performance differences between these models could reflect either (a) the superiority of one architecture's inductive bias, or (b) the simple advantage of having more learnable parameters. To reduce this confound, all seven AI models were scaled to approximately 3,000 parameters, giving a comparison at an equal parameter budget. The control is over capacity alone: as Table 4.2 shows, meeting a common budget required changing width, depth, state dimension and, for the feedforward models, the input window size, so those factors co-vary with architecture and the comparison does not isolate architectural inductive bias from them. The MLP-3K in particular sees an 11-sample window against its canonical 5, so it is not fed the same context as the models whose windows were left unchanged.

Choice of target parameter count. The target of ~3,000 parameters was chosen as a compromise: it is large enough that all architectures can express a reasonable denoising function (avoiding underfitting), yet small enough that the normalization represents a meaningful reduction for the larger models (Transformer: 5.9× reduction, Mamba-S6: 7.9×, Mamba-2 SSD: 8.7×). The MLP model is scaled **up** from 169 to 3,004 parameters, testing whether additional capacity benefits the simple feedforward architecture.

Scaling methodology. Each architecture's hyperparameters were adjusted to reach the target while preserving its essential structure:

Table 4.2: Normalized 3K-parameter model configurations: parameter count and scaling method for each architecture

Model	Parameters	Scaling Method
MLP-3K	3,004	Increased window (5→11) and hidden (24→231)
Hybrid-3K	3,004	Same as MLP-3K + DF switching
VAE-3K	3,037	Increased window (7→11), adjusted latent/hidden
cGAN-3K	3,004	Increased window (7→11), adjusted hidden layers
Transformer-3K	3,007	Reduced d_model (32→18), heads (4→2), layers (2→1)
Mamba-S6-3K	3,027	Reduced d_model (32→16), d_state (16→6), layers (2→1)
Mamba-2-SSD-3K	3,004	Reduced d_model (32→15), d_state (16→6), layers (2→1)

Parameter counting convention. All learnable parameters are counted, including biases, normalization parameters (LayerNorm gain/bias), and projection matrices. Non-learnable buffers (e.g., positional encoding tables, fixed **A** matrices) are excluded. The counts are verified programmatically via `sum(p.numel() for p in model.parameters() if p.requires_grad)`

Unified input window. All 3K models use a window size of 11 (vs. 5 for original MLP/Hybrid, and 7 for original VAE/cGAN), providing a common input context. This ensures that differences in performance reflect architectural inductive biases rather than differences in the amount of input information available.

This normalization removes parameter count as an explanation for any residual gap, which is what makes the comparison informative: if performance converges at equal parameter budgets, the conclusion is that capacity (not architecture) was the dominant factor in the unnormalized comparison. If significant gaps persist, the conclusion is weaker than “architectural inductive bias provides an advantage”, because the scaling changed more than the architecture label: it is that something other than raw parameter count accounts for the gap, with architecture, depth, width, state dimension and window size still confounded among themselves.

4.5 Modulation: QPSK

The experiments use QPSK, the canonical constellation. This section defines it and the I/Q-splitting technique that lets real-valued relays process a complex constellation.

4.5.1 QPSK: Gray-Coded Quadrature Phase-Shift Keying

Quadrature Phase-Shift Keying maps pairs of bits (b_0, b_1) to complex symbols [24]:

$$x = \frac{(1 - 2b_0) + j(1 - 2b_1)}{\sqrt{2}} \tag{4.13}$$

yielding four constellation points at $\{(\pm 1 \pm j)/\sqrt{2}\}$ with unit average power ($E_s = 1$). The Gray coding ensures that adjacent constellation points differ by exactly one bit:

Table 4.3: QPSK Gray-coded symbol mapping: bit pairs to complex symbols

Bit pair (b_0, b_1)	Symbol	Quadrant
00	$(+1 + j)/\sqrt{2}$	I
01	$(+1 - j)/\sqrt{2}$	IV
11	$(-1 - j)/\sqrt{2}$	III
10	$(-1 + j)/\sqrt{2}$	II

Demodulation applies independent sign decisions on each component: $\hat{b}_0 = \mathbb{1}(\mathrm{Re}(\hat{x}) < 0)$ and $\hat{b}_1 = \mathbb{1}(\mathrm{Im}(\hat{x}) < 0)$. The spectral efficiency is 2 bits/symbol, double that of BPSK.

Theoretical QPSK BER. For uncoded QPSK over an AWGN channel, the BER per bit equals the BPSK BER at the same E_b/N_0 because each I/Q component carries an independent BPSK stream [24]:

$$P_b^{\mathrm{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b^{\mathrm{BPSK}} \tag{4.14}$$

The advantage of QPSK is doubled throughput for the same BER and per-bit energy.

4.5.2 Constellation Diagram Summary

Figure 4.2 presents the constellation diagrams for the modulation schemes appearing in this thesis. The progression from BPSK (2 points on the real axis) through QPSK (4 points on the unit circle) to denser grids illustrates the fundamental trade-off between spectral efficiency and noise robustness: higher-order constellations pack more bits per symbol but reduce the minimum Euclidean distance between points, increasing the BER at a given SNR. QPSK is the constellation used in every canonical relay-architecture comparison; 16-QAM appears only as a rung of the coded link-adaptation study (Section 5.5), and BPSK only in the unknown-channel study's relay comparisons (Chapter 6). The others are shown for orientation.

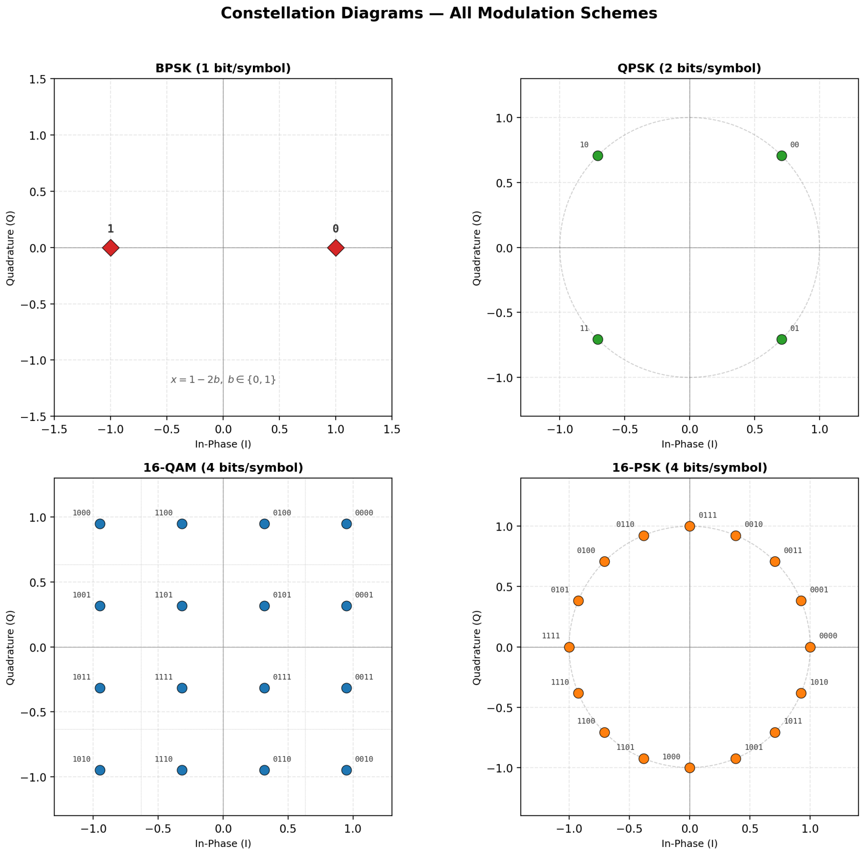


Figure 4.2: Constellation diagrams. (a) BPSK: 2 real-valued symbols at ± 1. (b) QPSK: 4 Gray-coded symbols on the unit circle. (c) 16-QAM: 16 Gray-coded symbols on a 4×4 rectangular grid with decision boundaries (dotted lines). (d) 16-PSK, shown for completeness only; it is not studied in this work. All constellations are normalised to unit average power

4.5.3 I/Q Splitting for AI Relay Processing of Complex Constellations

$$\hat{x}_R = f_\theta(\mathrm{Re}(y_R)) + j \cdot f_\theta(\mathrm{Im}(y_R)) \tag{4.15}$$

Justification and caveat. For rectangular constellations (QPSK, QAM), the I and Q components carry independent information and are corrupted by independent noise, so per-axis denoising with a shared real-valued function is a natural and parameter-efficient decomposition. It is, however, a *restriction* of the hypothesis class to axis-separable functions, not an equivalence to joint 2D processing: per-axis errors accumulate independently. For QPSK, where each axis carries a single binary decision, the restriction costs nothing; for denser constellations it becomes a structural limitation that a joint two-dimensional relay formulation would be needed to remove, which is left to future work.

Relay-specific handling:

Table 4.4: Relay-specific complex signal handling strategies and rationale

Relay type	Complex signal processing	Rationale
AF	Amplifies complex signal directly	Power normalization ($\|y\|^2$) is valid for complex vectors
AI relays	I/Q splitting (process Re and Im separately)	Real-valued networks; independence of I/Q in rectangular constellations

Limitation beyond binary-per-axis constellations. I/Q splitting works here because QPSK places a single binary decision on each axis, which is exactly what a tanh-output relay trained on $\{\pm 1\}$ represents. Constellations with more than two levels per axis (16-QAM and denser) break that correspondence and would require either modulation-specific retraining or a joint two-dimensional relay formulation; neither is evaluated here, and both are identified as future work.

Chapter 5

Core Experiments: The Canonical Setup

5.1 Experiments summary

Scope relative to the system model. The core experiments in this chapter (E2, E3) are conducted under the canonical system model defined in Chapter 4. In particular, the channel remains memoryless, the noise is white, transmission is uncoded, and relay processing is evaluated on a per-symbol basis. Consequently, the objective of that core is to compare relay functions under matched-model conditions rather than to address channel estimation, sequence detection, or channels with memory. Channel memory and channel estimation are introduced only later in Chapter 6 as deliberate departures from the canonical assumptions in order to study model mismatch and robustness. Coding is introduced earlier, within this chapter, by the supplementary coded study of Section 5.5, which deliberately departs from the uncoded, symbol-wise core to measure the Remark 4.2 caveat directly.

Following the single-model convention fixed in Chapter 1, the core of this chapter evaluates the relay strategies on the *canonical setup*: a two-hop SISO link, i.i.d. Rayleigh fast fading on both hops, complex baseband, QPSK, uncoded BER. A supplementary coded study (Section 5.5) deliberately departs from that setup within this same chapter, adding a rate-1/2 convolutional code, a block-DF relay, and an adaptive modulation-and-coding scheme. Two experiments constitute the core: the canonical relay comparison (E2) and the parameter-normalization and complexity study (E3), which together answer every canonical hypothesis (H1–H4). The principal contribution (Chapter 6) is the unknown-channel study (E6), which tests hypothesis H5. All other departures from the canonical setup are identified as future work in Chapter 7. The core (E2, E3) and unknown-channel (E6) experimental configurations are summarised in the table below; the supplementary coded study's configuration is described separately in Section 5.5.

Each experiment subsection reports the key results inline (BER tables and figures); supporting material and model hyperparameters are collected in the appendices (Appendices 9.1–9.3).

Table 5.1: Summary of core and unknown-channel experimental configurations

Exp.	Topology	Modulation	Channel / Equalization	Focus
E2	SISO	QPSK	Rayleigh (canonical)	Canonical relay comparison (H1, H2)
E3	SISO	QPSK	Rayleigh (canonical)	Parameter normalization & complexity (H3, H4)
E6	SISO	BPSK, QPSK	Unknown (ISI; composite; blind; partial-posterior); control: Rayleigh	Extension: unknown-channel failure of AF/DF and learned mitigation (H5)
E7	SISO	QPSK	Rayleigh (canonical), rate-1/2 convolutional code	Supplementary: coded block-DF against symbol-wise DF and coded-aware learned relays (Section 5.5)
E8	SISO	BPSK, QPSK	All nine families above, plus the coded scenario	Minimum relay size and the overfitting hypothesis H3, with three initializations per configuration (Section 5.4)

5.2 Canonical Relay Comparison (SISO, Rayleigh, QPSK)

Goal: Evaluate the classical and AI-based relay strategies on the canonical setup of this thesis.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **Reported learned relay architectures (six):** Supervised (MLP, Hybrid), Generative (VAE), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** ReLU hidden, tanh output

Conclusion: On the canonical Rayleigh channel, AI relays match but do not surpass symbol-wise DF: DF achieves the lowest or tied-lowest BER at essentially every SNR point, including the low-SNR regime, while the best neural relays (MLP, Hybrid, VAE) track it within statistical uncertainty and clearly outperform AF. This confirms H1 (AI relays beat AF at low SNR) and H2 (DF dominance at ≥ 6 dB with zero parameters) on the canonical channel.

5.2.1 Results Tables

Table 5.2: BER on the canonical setup: QPSK over i.i.d. Rayleigh fast fading, SISO on both hops, uncoded, $10 \times 10{,}000$ bits per SNR point. "DF th." is the closed-form two-hop composition of Eq. 2.34 (Section 2.5.2.2), included for direct comparison against the measured DF column. Column abbreviations: Transf. is the Transformer relay, Mamba2 the Mamba-2 SSD relay

SNR (dB)	AF	DF	DF th.	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2
0	0.4076	0.3337	0.3333	0.3349	**0.3329**	0.3359	0.4014	0.4071	0.4032
4	0.3129	**0.2219**	0.2216	0.2233	0.2220	0.2254	0.2758	0.2812	0.2772
8	0.1852	**0.1218**	0.1203	0.1229	0.1220	0.1235	0.1424	0.1448	0.1416
12	0.0822	0.0580	0.0560	0.0586	**0.0579**	0.0589	0.0628	0.0637	0.0621
16	0.0309	**0.0245**	0.0239	0.0247	**0.0245**	0.0247	0.0260	0.0259	0.0251
20	0.0110	**0.0097**	0.0098	0.0099	**0.0097**	**0.0097**	0.0101	0.0100	0.0098

Two features of Table 5.2 deserve comment. First, symbol-wise DF tracks the closed form (the "DF th." column above): reading the abscissa as E_s/N_0 and converting to E_b/N_0 for QPSK ($k = 2$, so 3.01 dB lower), the two-hop expression $2P(1 - P)$ with $P = \frac{1}{2}\big(1 - \sqrt{\bar{\gamma}/(1 + \bar{\gamma})}\big)$ gives 0.0098 at 20 dB against a measured 0.00972 (0.8% relative), and 0.0239 at 16 dB against 0.02451 (2.6% relative): agreement to within a few percent relative error at every tabulated point (worst case 3.6%, at 12 dB), which validates the simulator end to end against the analysis of Section 2.5.2.2. Figure 5.1 plots this closed form alongside the measured curves, together with the single-hop floor P itself – the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – which no relay may cross and none does at any SNR.

Second, the relays divide by family rather than by paradigm. The three feedforward relays, the MLP, the Hybrid and the VAE, are statistically indistinguishable from DF from 4 dB upward and from each other throughout; the VAE reaches 0.00972 at 20 dB, matching DF exactly at the resolution of this budget. The two sequence models are the ones that separate, trailing by roughly 0.07 in BER at 0 dB before converging by 16 dB. Capacity is not the explanation, since the sequence models are the larger of the two groups; the memoryless channel simply offers no temporal structure for their inductive bias to exploit, a reading the equal-parameter comparison of Section 5.3 tests directly and confirms.

Two comments on Table 5.2. First, symbol-wise DF is the best or tied-best relay at every tabulated point, and the learned relays sit within a few thousandths of it from 4 dB upward: on a matched, known channel the learned relay matches DF rather than beating it, at 169 parameters against DF's none. This is H2, and it is the result the rest of the thesis is measured against. Second, the two sequence models trail the feedforward relays at low SNR (0.4014 and 0.4032 at 0 dB against the MLP's 0.3349) and converge with them by 16 dB, so their extra capacity buys nothing on a memoryless channel; Section 5.3 separates that from the parameter-count confound.

Why QPSK and not BPSK. The canonical channel is complex, so the canonical constellation is complex: QPSK places one bit on each axis the Rayleigh coefficient acts in, and the I/Q-split relays process it without modification. BPSK on this channel would require discarding the imaginary part of the equalized sample, throwing away half the plane the channel acted in, which is why BPSK is not used on this channel at all. One consequence must be kept in view when reading these BER figures against BPSK closed forms: the SNR axis here is E_s/N_0, and QPSK carries two bits per symbol, so a given abscissa corresponds to an E_b/N_0 that is 3.01 dB lower than the same number on a BPSK axis. The constellations are not on a common axis and their error rates are not directly comparable.

5.2.2 Results Figures

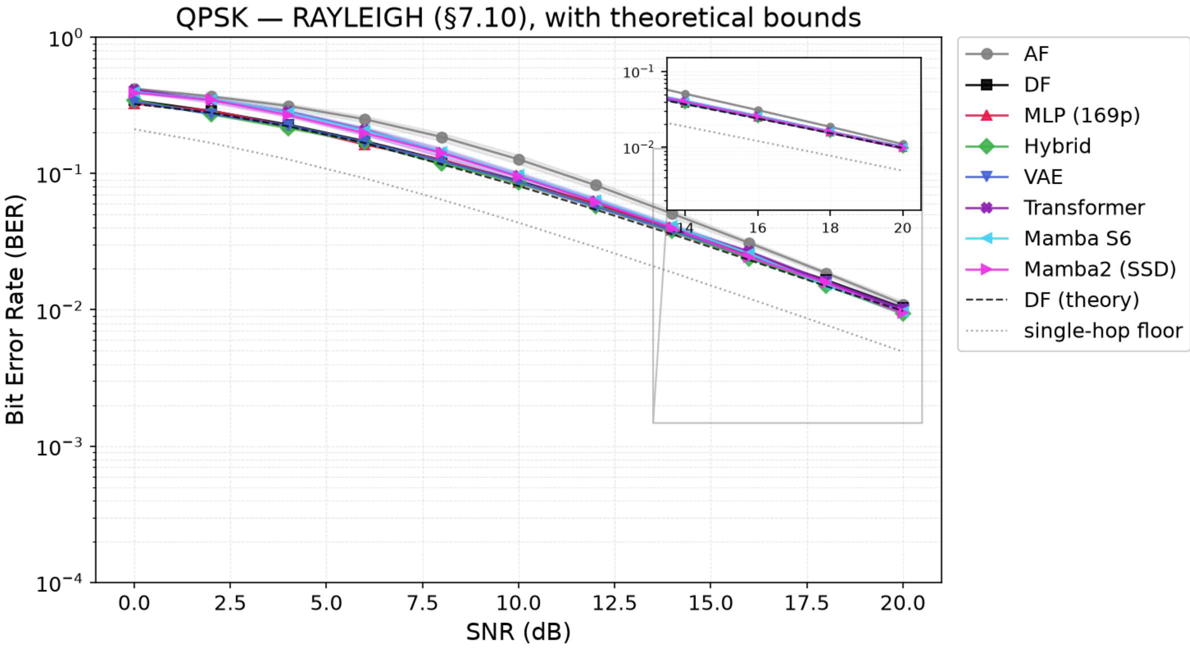


Figure 5.1: The canonical setup: QPSK over i.i.d. Rayleigh fast fading, BER vs. E_s/N_0 for the eight relays of Table 5.2, with 95% confidence intervals, plus two theoretical reference curves: "DF (theory)" (dashed), the closed-form two-hop composition, which the measured DF curve tracks closely, within a few percent relative error everywhere (worst case 3.6%, at 12 dB); and the "single-hop floor" (dotted), the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – a physical lower bound no relay crosses. AF is uniformly worst; symbol-wise DF and the feedforward learned relays are indistinguishable from 4 dB upward; the two sequence models trail at low SNR and converge by 16 dB

5.3 Parameter Normalization & Complexity Trade-off

This study tests H3 (complexity versus BER) and H4 (architecture convergence at an equal parameter budget) on the canonical setup.

Goal: Isolate architectural inductive biases from parameter count effects and characterize the complexity-performance trade-off for relay denoising.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** All normalized to ~3,000 parameters, plus original sizes (169 to 26K) - **NN activation:** ReLU hidden, tanh output

Conclusion: A minimal 169-parameter MLP matches models roughly 140× larger, and no larger model evaluated here improves on it; whether the curve eventually turns upward, and whether overfitting would be the reason, is not tested by *these* experiments (Section 7.2). It is tested by the size sweep of Section 5.4, which replicates every configuration across three initializations and finds no upward arm above the initialization spread. At an equal budget of 3,000 parameters the gap between feedforward and sequence architectures is strongly SNR-dependent: it falls monotonically from 17.4% at 0 dB to 4.1% at 20 dB (Table 5.3). Parameter count therefore accounts for the comparison at high SNR, where the choice of model is nearly irrelevant, but not at low SNR, where the six equal-sized models separate cleanly into a feedforward group tracking DF and a sequence group tracking AF. Since all six carry the same budget, that separation is not explained by capacity; it is attributable to the remaining differences between the models, of which architecture is the most salient but not the only one, since depth, width, state dimension and window size were adjusted to meet the budget (Table 4.2). The generative VAE falls inside the feedforward group at every SNR rather than apart from it, ending at 0.0101 against MLP-3K's 0.0099, and it is ahead of all three sequence models from 0 to 16 dB; the generative paradigm is therefore not a handicap at this budget (Section 2.3.3).

5.3.1 Results Tables

Table 5.3: Equal-parameter-budget comparison on the canonical setup (QPSK over Rayleigh). All learned columns are the ∼3,000-parameter variants; every learned model is scaled to approximately 3,000 parameters, so parameter count is removed as an explanation for the residual differences; width, depth, state dimension and window size still co-vary with architecture (Table 4.2). AF and DF are the unparameterized references. Column abbreviations: Transf. is the Transformer relay, Mamba2 the Mamba-2 SSD relay

SNR (dB)	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2	AF	DF
0	0.3361	0.3391	0.3424	0.4068	0.4007	0.4001	0.4076	**0.3337**
4	0.2244	0.2271	0.2291	0.2818	0.2751	0.2742	0.3129	**0.2219**
8	0.1229	0.1229	0.1267	0.1458	0.1406	0.1405	0.1852	**0.1218**
12	0.0586	0.0580	0.0604	0.0640	0.0620	0.0622	0.0822	**0.0580**
16	0.0247	**0.0245**	0.0251	0.0259	0.0250	0.0251	0.0309	**0.0245**
20	0.0099	**0.0097**	0.0101	0.0100	0.0097	0.0098	0.0110	**0.0097**

5.4 Minimum Relay Size Across Channel Families

The complexity study of Section 5.3 establishes that no evaluated model improves on the 169-parameter MLP, but leaves two questions open: how far *below* 169 parameters the relay can be

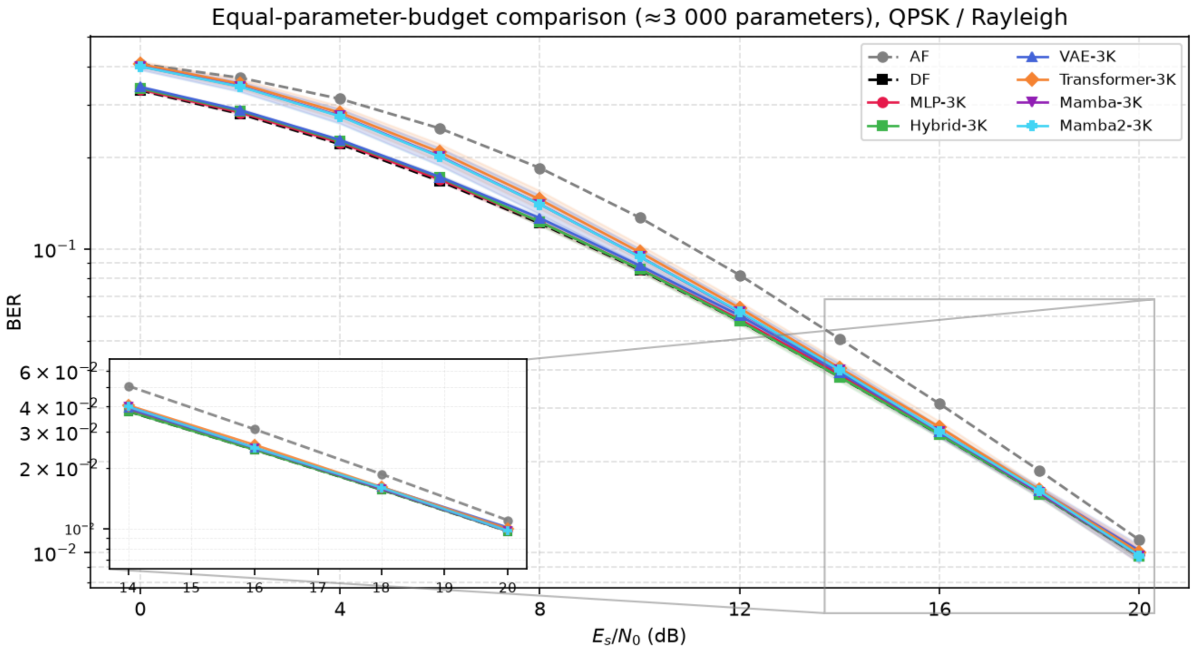


Figure 5.2: BER vs E_s/N_0 for all relay variants normalised to ≈3,000 parameters, QPSK over Rayleigh fading. AF and DF (dashed) are the unparameterised references. Shaded bands show 95% confidence intervals for the learned relays. The feedforward group (MLP-3K, Hybrid-3K, VAE-3K) tracks DF across the full range; the sequence models (Transformer-3K, Mamba-3K, Mamba2-3K) converge with them only at high SNR

taken, and whether the answer is a property of the architecture or of the channel. This section answers both by sweeping relay size downward on every channel family used in this thesis.

Goal: Locate the smallest relay that costs nothing measurable against the thesis's own published relay, per channel, and determine which architectural axis — window, depth or width — sets that floor.

Trials: - **Topology:** SISO, two hops, hop 2 per each channel family's own convention (AWGN or coherently-compensated Rayleigh) - **Modulation scheme:** BPSK or QPSK, per channel, as used elsewhere in this thesis - **Channels:** the nine families of Chapters 5 and 6, plus the coded scenario of Section 5.5 - **Comparators:** symbol-wise DF where the channel is memoryless, Viterbi/MLSE where it has memory, block DF for the coded scenario; each channel additionally compared against its own published relay (MLP-169, or MLP-756 coded) - **Relay architecture:** window $\in \{1, 3, 5, 7\}$, depth $\in \{1, 2, 3\}$, width from 1 to 48 - **Replication:** three independent initializations per configuration, each configuration held to its worst - **Metric:** additional SNR required to reach a given BER

Conclusion: The floor is set by channel memory, not by parameter count, and it is far lower than 169 parameters only where the channel is memoryless. Table 5.4 gives the smallest configuration meeting a range of degradation budgets against each scenario's own published relay. At zero degradation, three of the five memoryless channels are matched by four parameters — a single hidden unit with no window, $42\times$ smaller than MLP-169 — whereas the three-tap channels require 73 to 145 parameters, only 1.2× to 2.3× smaller. Reading the window column rather than the parameter column explains why: every memoryless row settles at window 1 and every three-tap row needs window 5 or 7 until a full decibel of degradation is permitted. Figure 5.3 shows the same effect directly — widening the window is worth 8 to 11 dB on the three-tap channels and nothing at all on the memoryless ones — and Figure 5.4 plots the size required against the degradation the designer is willing to concede. Depth was varied from one to three hidden layers and improved no configuration; where a two-layer network matched, a single layer matched at fewer parameters. The complexity question is therefore not ”how many parameters” but ”how much window”, and the answer is fixed by the channel’s impulse response rather than by the architecture. Section 7.2.3 takes up what this implies for the overfitting hypothesis H3, and reports the across-initialization spread that bounds the resolution of these figures.

Table 5.4: Smallest relay meeting a given degradation budget against each scenario’s own published relay (MLP-169, or MLP-756 for the coded row), written as parameters and window. The reader should observe the window column rather than the parameter column: every memoryless row settles at window 1 while every three-tap row requires window 5 or 7 until a full decibel is conceded, which is the sense in which channel memory rather than model size sets the floor. A dash indicates that no smaller configuration meets that budget

Channel	Taps	≤0 dB	≤0.1 dB	≤0.25 dB	≤0.5 dB	≤1 dB	≤2 dB
awgn	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
rayleigh	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
flat gain	1	21p w3	21p w3	21p w3	4p w1	4p w1	4p w1
branch asym.	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
nonlinear bias	1	29p w5	4p w1	4p w1	4p w1	4p w1	4p w1
ISI	3	73p w7	73p w7	57p w5	57p w5	37p w7	21p w3
ISI (complex)	3	73p w7	73p w7	73p w7	73p w7	57p w5	21p w3
ISI + Rayleigh	3	145p w7	73p w7	73p w7	73p w7	37p w7	21p w3
composite	3	145p w7	73p w7	57p w5	21p w3	21p w3	6p w3
coded	code	372p w9	244p w5	244p w5	18p w1	18p w1	18p w1

5.4.1 Results Figures

5.5 Coded Block-DF: Measuring the Remark 4.2 Caveat

A note on constellation scope. The canonical and coded-study fixed-MCS relay-decoding comparisons of this thesis (block-DF vs. the learned relay at a fixed code rate and constellation) are conducted on QPSK throughout; the unknown-channel study’s relay comparisons mainly use BPSK (Chapter 6), with one exception (Section 6.1.2) that repeats the unknown-ISI relay comparison with QPSK in place of BPSK. 16-QAM is not studied as a relay-architecture question (Section 4.5): it does appear as one rung of the modulation-and-coding ladder that the

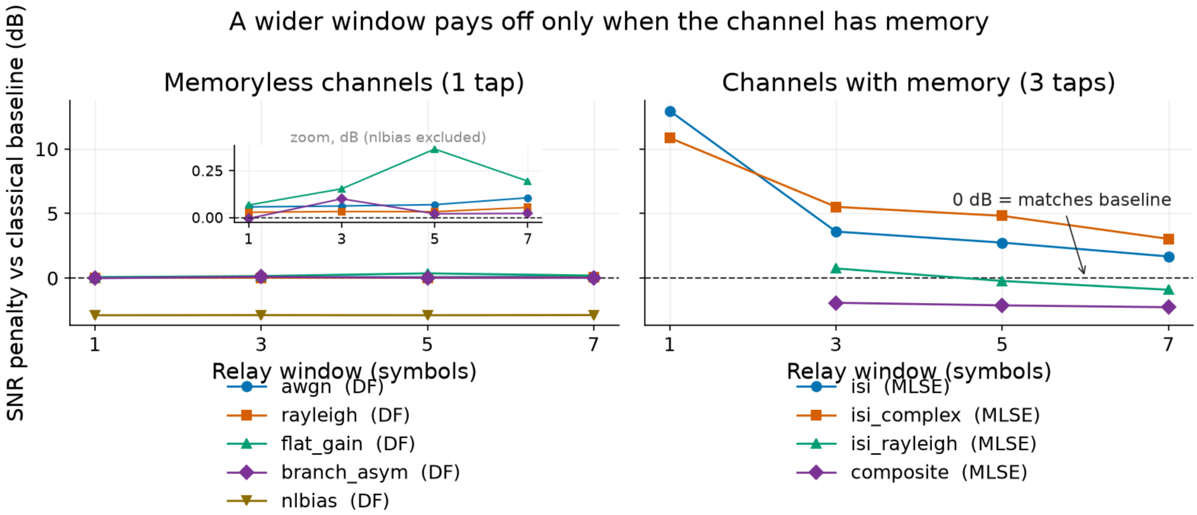


Figure 5.3: SNR penalty against each channel's classical comparator as the relay window widens, for the best configuration at each width. The reader should observe that the two panels behave oppositely: on the memoryless channels the penalty is flat to within a fifth of a decibel across every window (see inset), so the window buys nothing, while on the three-tap channels it falls by 8 to 11 dB between window 1 and window 7. This is the sense in which the relay's window must be matched to the channel's impulse response

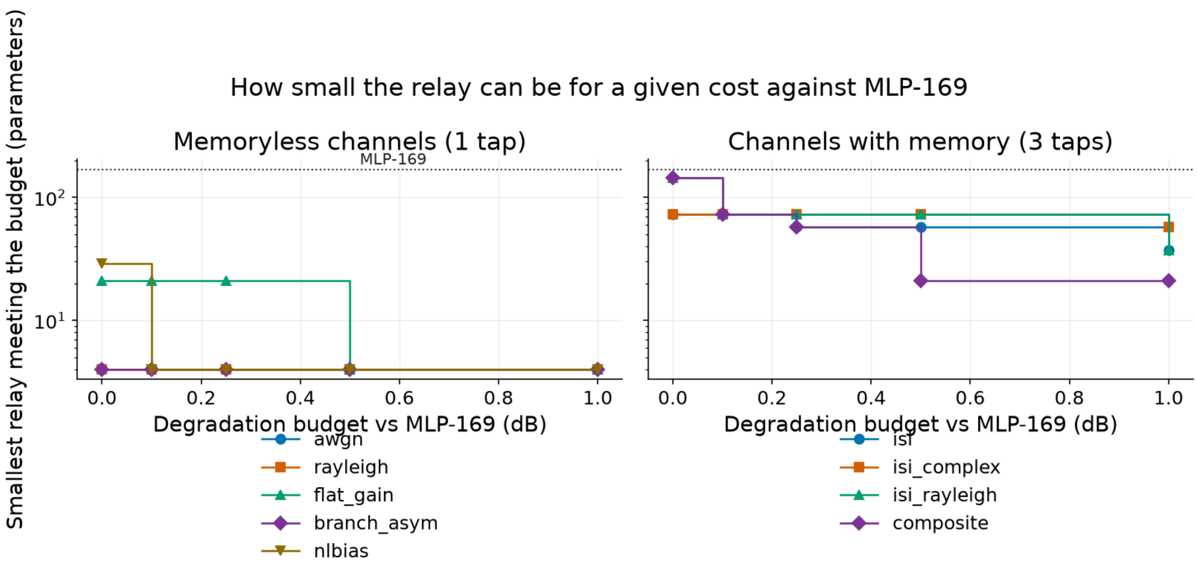


Figure 5.4: Smallest relay meeting a given degradation budget against MLP-169, drawn as a step function because the horizontal axis is a budget the designer chooses rather than a measured quantity. The reader should observe the two-order-of-magnitude separation between the panels: the memoryless channels are served by four parameters at every budget shown, while the three-tap channels need between 21 and 145 depending on how much degradation is conceded

link-adaptation study of Section 5.5.3 selects from, and that study does compare the block-DF and denoise-only relay strategies at 16-QAM rungs (e.g. Table 5.12), but the object of study there is the *rate-and-modulation choice* rather than the relay's ability to process a denser constellation. Nothing in that study rests on how a learned relay handles four levels per axis, which is the question left to future work.

Remark 4.2 draws a distinction the rest of this thesis relies on: the DF baseline used everywhere here is *symbol-wise* (uncoded, per-symbol hard slicing), not the *block* DF of the information-theoretic relay-channel literature, and the remark explicitly cautions that "the reported DF results should not be read as bounds on coded block-DF performance." That caveat was asserted, not measured. This section measures it: a real information-theoretic block-DF relay, built from a rate-1/2 convolutional code and a soft-decision Viterbi decoder, added as a supplementary study on the canonical channel rather than as a revision of the core comparison above.

Setup. A standard rate-1/2 convolutional code (constraint length K, 2^{K-1} trellis states, zero-tail terminated per 200-information-bit frame) is applied before QPSK modulation on the canonical Rayleigh channel; `CodedDecodeAndForwardRelay` decodes the full frame at the relay via soft-decision Viterbi (squared-distance branch metric on the coherently-compensated I/Q samples), then re-encodes and re-modulates a fresh, clean codeword before forwarding – genuine block DF, as opposed to the per-symbol slicing used elsewhere in this thesis. Two coded-aware learned relays are trained on the same task for comparison: a windowed 756-parameter MLP classifier (window 21, $\approx 5 \times K$ for $K = 3$) and the Mamba-S6 architecture already used in Table 5.2 (24,084 parameters at this configuration), both predicting the transmitted QPSK symbol from a window of noisy coded observations with no explicit knowledge of the code. All results use the thesis-standard budget of 100 trials $\times$ 100,000 information bits per SNR point.

Table 5.5: Coded block-DF vs. uncoded symbol-wise DF and coded-aware learned relays, canonical QPSK/Rayleigh, rate-1/2 K=3 code. "coded-AF" forwards the noisy codeword unmodified (the relay does not decode), isolating the code's own gain from any relay intelligence

SNR (dB)	uncoded DF	coded AF	coded DF	MLP-coded	Mamba-coded
0	**0.3337**	0.4902	0.4204	0.4439	0.4505
4	**0.2216**	0.4468	0.2229	0.2974	0.3214
8	0.1205	0.2722	**0.0638**	0.1010	0.1174
12	0.0560	0.0539	**0.0154**	0.0214	0.0226
16	0.0239	0.0099	0.0042	0.0045	**0.0041**
20	0.0098	0.0020	0.0014	0.0011	**0.0010**

Two findings, reported as measured rather than smoothed toward either a "coding always helps" or a "the learned relay wins" narrative:

The caveat holds precisely where it should, and no further. At equal E_s/N_0, coded block-

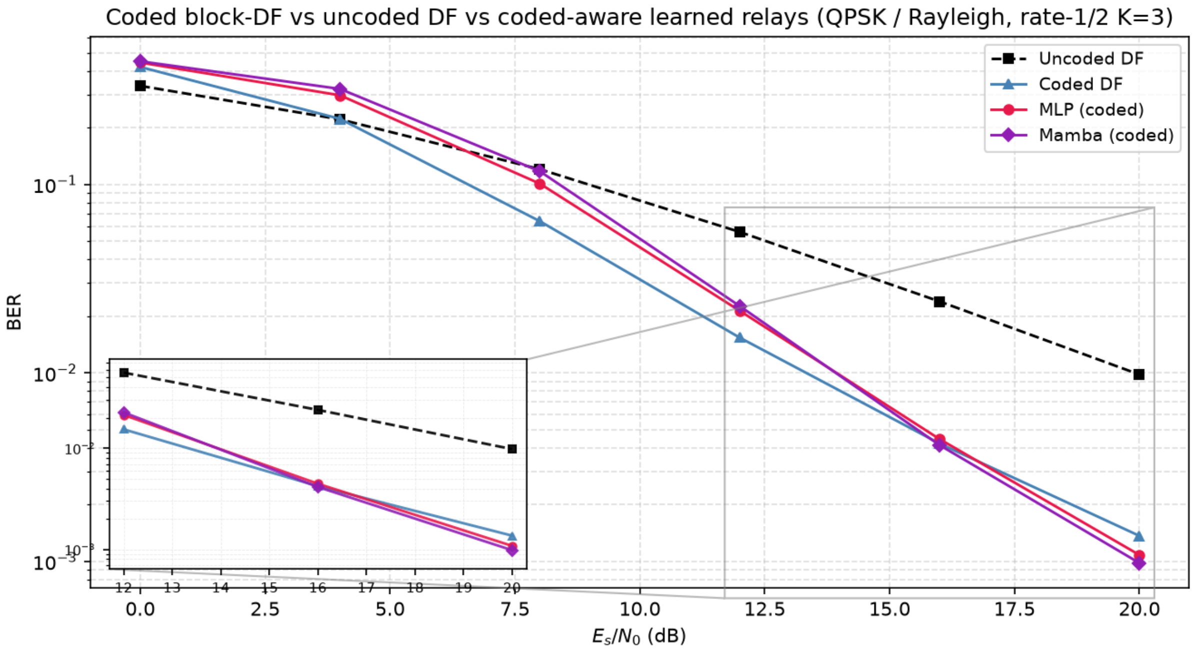


Figure 5.5: BER vs E_s/N_0 for coded block-DF, uncoded symbol-wise DF, and the two coded-aware learned relays (MLP and Mamba), canonical QPSK/Rayleigh, rate-1/2 K=3. Coding hurts below the error-propagation threshold ($\lesssim$6 dB) and helps decisively above it; the learned relays close on coded-DF only at the top of the sweep

DF beats uncoded symbol-wise DF decisively from 8 dB upward – a $5.7\times$ reduction in BER at 16 dB, $7.2\times$ at 20 dB – closing the gap the remark left open. Those factors are the right comparison for a detection question but the wrong one for a link budget, because the rate-1/2 code halves the information carried per channel use; Section 5.5.2 redoes the comparison at equal throughput and the advantage largely disappears. But below roughly 6 dB, coding *hurts* even on the favourable axis: both coded AF and coded DF are worse than plain uncoded DF at 0 and 4 dB. This is the well-known convolutional-code error-propagation threshold, not an artifact: a fixed-rate code below its operating threshold produces decoding errors that cluster and propagate across the trellis, which a memoryless per-symbol slicer cannot do. The caveat is real, but it cuts both ways – coded block-DF is a strictly stronger baseline only above its own threshold, which itself has to be measured, not assumed.

At low-to-mid SNR the classical decoder is far ahead, and the learned relay's extra capacity buys nothing. Unlike the canonical memoryless channel of Table 5.2, where Section 5.3 found "the memoryless channel simply offers no temporal structure for [the sequence models'] inductive bias to exploit," a convolutional code *is* genuine temporal structure. It does not rescue the learned relays where the code is doing the most work: both are far behind coded-DF at 4–8 dB (Mamba-coded 0.1174 vs. 0.0638 at 8 dB). Nor does the $32\times$-larger Mamba-S6 relay show a clear advantage over the 756-parameter MLP there, being worse at 4–8 dB (0.3214 vs. 0.2974 at 4 dB). Giving the sequence model a task with real memory to exploit therefore does not overturn the thesis's minimal-capacity finding (H3): extra capacity still buys no reliable advantage over the minimal architecture.

At high SNR the ordering reverses, and it is not noise. The differences at 16–20 dB in Table 5.5 are small in absolute terms, and an earlier version of this section dismissed them as trial-to-trial variation. That was wrong, and the correction is worth stating explicitly because it changes the conclusion. Re-measuring those two points with *paired* trials – every relay driven by identical information bits and identical hop-1/hop-2 fading realizations within each trial, so the channel-draw variance that dominates an unpaired comparison cancels in the per-trial difference – and testing with the Wilcoxon signed-rank test used elsewhere in this thesis gives Table 5.6. At 20 dB both learned relays beat coded-DF in essentially every one of 100 paired trials ($p < 10^{-4}$). At 16 dB the result splits by architecture: Mamba-coded wins (68 of 100 trials) while the MLP significantly *loses* (23 of 100), both at $p < 10^{-4}$. The effects are real in both directions, and no single sentence of the form "the learned relay beats/does not beat Viterbi" describes them.

Table 5.6: Paired re-measurement of the two high-SNR points of Table 5.5, 100 paired trials × 100,000 information bits. "Diff" is the mean per-trial BER difference against coded-DF (negative favours the learned relay); "W/L" counts trials won/lost; p is Wilcoxon signed-rank

SNR	Relay	BER	Diff vs. coded-DF	W/L	p
16 dB	coded-DF	0.004245	—	—	—
	MLP-coded	0.004437	+0.000192	23/77	$< 10^{-4}$
	Mamba-coded	**0.004124**	−0.000121	68/32	$< 10^{-4}$
20 dB	coded-DF	0.001362	—	—	—
	MLP-coded	0.001066	−0.000295	99/1	$< 10^{-4}$
	Mamba-coded	**0.000967**	−0.000395	100/0	$< 10^{-4}$

This is not a learned relay out-decoding Viterbi, and the mechanism says why. Viterbi is the maximum-likelihood *sequence* decoder for a known code on a known channel, so a learned decoder beating it at its own task would be a surprising claim requiring more than a BER table. It is not what happens here. Instrumenting the relay output directly (Table 5.7) shows coded-DF is the *better* decoder by a wide margin: it puts roughly $2.1\times$ *fewer* symbol errors on the air than the learned relay at both SNRs, exactly as the optimality result predicts. What separates them is what happens to those errors afterwards. Coded-DF decodes and then *re-encodes*, so a frame it gets wrong leaves the relay as a different but perfectly *valid* codeword: the code's redundancy has been spent and then regenerated around the error, and the destination's decoder has no way to detect or repair it. The learned relay never re-encodes, so its (more numerous) mistakes remain isolated bad symbols inside an otherwise-valid codeword, and the destination still holds the redundancy needed to fix them. Only 16–25% of coded-DF's relay errors are repaired downstream against 58–73% of the learned relay's. That also explains the SNR-dependent split above: at 16 dB the $2.1\times$ raw-error penalty still outweighs the repair advantage for the MLP, and by 20 dB it no longer does.

Table 5.7: Where the errors go, 100 trials × 500 frames. "Relay symbol ER" counts symbols the relay put on the air that differ from the transmitted ones; "repaired" is the fraction of those that do not survive the destination's decoder. The oracle relay forwards the clean codeword and so isolates the hop-2-only floor

SNR	Relay	Relay symbol ER	Repaired	Final BER
16 dB	coded-DF	**0.00505**	15.7%	0.004257
	MLP-coded	0.01065	**57.9%**	0.004487
	oracle	—	—	0.002140
20 dB	coded-DF	**0.00182**	24.7%	0.001369
	MLP-coded	0.00385	**73.0%**	0.001039
	oracle	—	—	0.000668

Does a stronger code help? The $K{=}3$ code used above is deliberately small. Sweeping the constraint length to $K \in \{5, 7\}$ on the same QPSK/Rayleigh link does *not* improve matters: the larger-K codes are worse than $K{=}3$ across 0–8 dB, a sharper error-propagation cliff for the bigger trellis, and by 20 dB all three are within trial-to-trial noise of one another (values 0.0014, 0.0015, 0.0012). Within this frame length and trial budget, the relay's re-encoding liability is therefore not something a stronger code repairs; the full sweep is available in `results/coded_df_experiment.json`.

Does reliable decoding remove it? Remark 4.2 defines block-DF with "a strong (ideally capacity-achieving) channel code", which the measurements above do not approach: at 20 dB the destination fails on 20.23% of frames under coded-DF and on 10.58% even under an *oracle* relay forwarding the clean codeword, so one frame in ten is lost for reasons unrelated to the relay. Since the liability tracks the relay's frame-error rate rather than the code family behind it, the link is extended up the SNR axis, code fixed, until decoding is reliable (Table 5.8); two learned relays are run, one on the recipe above and one retrained on 5–30 dB, which brackets three of the four evaluated points and leaves only the 32 dB reading as a mild, 2 dB extrapolation, so no high-SNR reading rests on a large train/test mismatch.

Table 5.8: Block-DF into the reliable-decoding regime: the rate-1/2 $K{=}3$ code and canonical QPSK/Rayleigh link of Table 5.5, extended up the SNR axis, 100 trials × 1,000 frames (20,000,000 information bits) per point. "MLP (5/10/15)" uses that table's training recipe; "MLP (5–30)" is the same architecture retrained on 5–30 dB, bracketing all but the 32 dB point, hence a different instance whose 20 dB entry need not reproduce it. Bold marks the best *relay* in each row (both marked where they tie); the oracle forwards the clean codeword and is the single-decode floor they are measured against, not a competitor

SNR (dB)	coded-DF	MLP (5/10/15)	MLP (5–30)	oracle	coded-DF FER
20	0.001378	**0.001064**	0.001093	0.000704	0.1998
24	0.000485	**0.000308**	0.000318	0.000253	0.0804
28	0.000187	**0.000105**	0.000105	0.000092	0.0324

SNR (dB)	coded-DF	MLP (5/10/15)	MLP (5–30)	oracle	coded-DF FER
32	0.000069	**0.000037**	**0.000037**	0.000034	0.0122

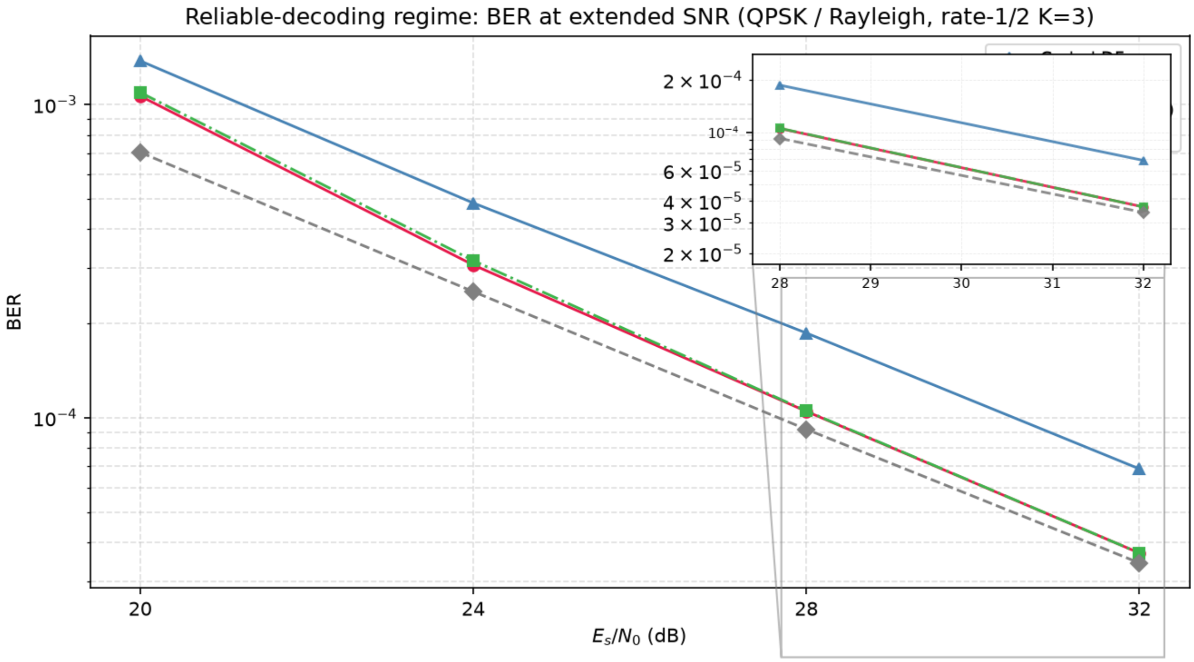


Figure 5.6: BER vs E_s/N_0 at extended SNR (20–32 dB), rate-1/2 K=3 QPSK/Rayleigh. Coded DF's BER remains near twice the oracle floor at every point; the retrained learned relay (MLP 5–30 dB) closes on that floor monotonically and reaches statistical parity with it at 32 dB, demonstrating that the high-SNR reversal survives into the reliable-decoding regime

The liability does not dissolve. Across 20–32 dB coded-DF's frame-error rate falls $16.3\times$ (0.1998 to 0.0122), yet its BER stays near twice the oracle floor with no trend at all (1.96, 1.92, 2.03, 2.00; the scatter is trial noise, the absence of a downward slope is the point), while the retrained learned relay (MLP 5–30) closes on that floor (1.55, 1.26, 1.15, 1.08) and reaches statistical parity with it by 32 dB (636 against 607 frame errors in $100{,}000$, $z = +0.8$) while still beating coded-DF there ($z = +13.7$). The cause is structural, not code strength: block-DF decodes once per hop, so two independent frame-error events accumulate and neither decoder repairs the other's, whereas denoising leaves a single decoding operation at the destination. The reversal is thus no artefact of a weak code or short frame; it survives, and relatively widens, as decoding becomes reliable. One limit stays out of reach: driving per-hop failure to zero makes the sum of two vanishing terms vanish, so a capacity-achieving code at long blocklength would recover the floor. No operating point here does: even at 32 dB the oracle's own decode – one hop's worth, by construction, and equal to the other hop's by the link's symmetry – still fails on 0.6% of frames.

5.5.1 Soft-Decision Relaying, and What the Liability Actually Is

The mechanism of Table 5.7 suggests an obvious repair. If the trouble with block-DF is that it commits to a codeword, replace the commitment with soft information: run BCJR (forward-backward MAP) at the relay instead of Viterbi and forward the posterior-mean symbol per trellis step, so a symbol the relay is unsure of is shrunk toward the origin rather than asserted. The same idea applies to the learned relay, whose softmax already carries a posterior it was throwing away: forwarding $\sum_k p_k s_k$ instead of $\arg\max_k p_k$ costs nothing, needs no retraining, and reuses the identical weights, so any difference isolates the read-out rule alone. Table 5.9 evaluates both against their hard counterparts and against an oracle relay that forwards the clean codeword, on paired trials.

Table 5.9: Soft- versus hard-decision relaying, canonical QPSK/Rayleigh, rate-1/2 K=3, 100 paired trials $\times$ 100,000 information bits. The MLP hard/soft pair shares one set of trained weights and differs only in the read-out. The oracle relay forwards the clean codeword, giving the hop-2-only floor. Both block-DF variants are marked at 8 dB, where they are a statistical tie

SNR (dB)	hard block-DF	soft block-DF	MLP hard	MLP soft	oracle
0	0.4208	**0.4150**	0.4443	0.4343	0.3022
4	0.2231	**0.2218**	0.2969	0.2761	0.1278
8	**0.0636**	**0.0636**	0.1008	0.0878	0.0327
12	**0.01537**	0.01537	0.02142	0.01768	0.00768
16	0.004222	0.004222	0.004403	**0.003633**	0.002105
20	0.001354	0.002348	0.001056	**0.000912**	0.000682

Soft read-out helps the learned relay everywhere (Figure 5.7). At every SNR the posterior mean beats the argmax, on identical weights: 0.4343 against 0.4443 at 0 dB, 0.0878 against 0.1008 at 8 dB, 0.000912 against 0.001056 at 20 dB. The gain is largest exactly where the mechanism predicts, since a relay that reports its uncertainty gives the destination's decoder more to work with than one that reports a confident guess. It is enough to change a verdict: at 16 dB the hard MLP significantly *loses* to block-DF (19 wins, 80 losses and one tie) while the soft MLP significantly *beats* it (100 wins, no losses), and at 20 dB the soft relay comes within 0.00023 of the oracle floor that no relay of any kind can pass.

Soft block-DF does not rescue block-DF, and that is the more interesting result. The prediction going in was that BCJR would combine the code-aware cleanup of Viterbi with the repairability of a non-committing relay, and therefore beat both. It does not. It edges hard block-DF at 0–4 dB, ties it from 8–16 dB, and at 20 dB is clearly worse (0.002348 against 0.001354, losing all 100 paired trials). The 20 dB degradation turns out to be a calibration artefact rather than a property of soft forwarding: the relay is given the *nominal* noise variance, but post-equalization Rayleigh noise has variance $1/(2\gamma|h|^2)$, so at high SNR, where the nominal value is small, BCJR becomes badly overconfident on deep-fade symbols. Viterbi is structurally

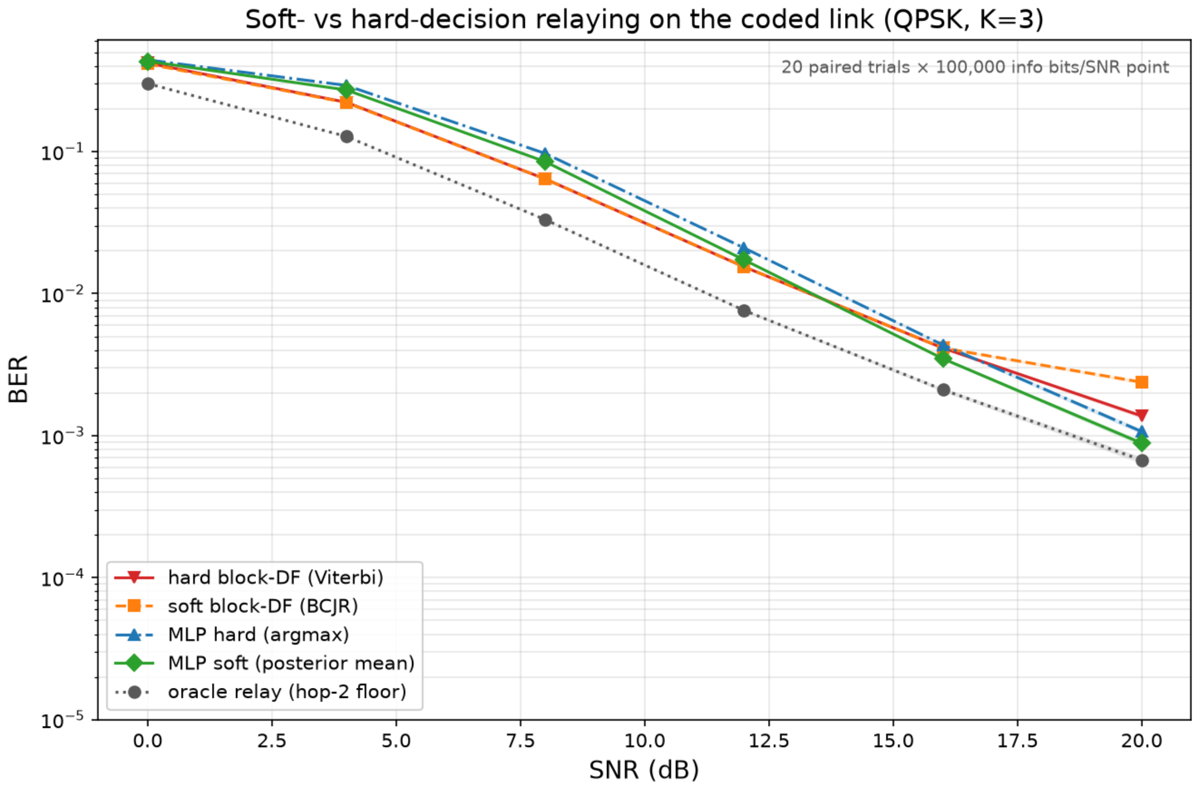


Figure 5.7: Soft- versus hard-decision relaying on the coded link (canonical QPSK/Rayleigh, rate-1/2 K=3, 100 paired trials). The soft MLP (posterior mean) tracks below the hard MLP at every SNR, with the largest gain visible at mid-to-high SNR where the relay's residual uncertainty is most useful to the destination decoder. The two block-DF variants (hard Viterbi, soft BCJR) converge at high SNR and are never separated by a reliable margin, confirming that the liability is consuming the code's redundancy at the relay rather than hard quantization per se. The oracle relay (dashed) is the hop-2-only floor

immune, its squared-distance metric being invariant to any uniform scaling of the assumed variance. Deliberately inflating the assumed variance confirms this directly: at 20 dB a factor of two removes the entire gap ($0.002404 \to 0.001376$), and every larger factor up to $100\times$ holds there.

What that correction exposes is the substantive point. Even properly calibrated, soft block-DF only *ties* hard block-DF (0.001376 against 0.001375 at 20 dB; 0.004255 against 0.004255 at 16 dB) – it never beats it. The reason is that at high SNR the BCJR posteriors saturate, so the posterior mean degenerates to the constellation point and soft block-DF reproduces Viterbi's decisions; when a code-exploiting relay is wrong, it is confidently wrong whether or not it quantizes. The liability identified in Table 5.7 is therefore *not* hard quantization at the relay, as the soft-forwarding hypothesis assumed. It is consuming the code's redundancy at the relay at all. Any relay that decodes the code – Viterbi or BCJR, hard or soft – has spent that redundancy by the time it transmits, and whatever it got wrong arrives at the destination unrepairable. The learned relay's high-SNR advantage comes from the opposite discipline: it never decodes the code, only denoises symbol by symbol, so the full redundancy survives intact for the destination to spend. Soft read-out then adds a second, smaller gain on top by preserving the confidence that the destination can weigh.

This sharpens the thesis's recurring claim rather than overturning it. A learned relay still does not out-decode a matched classical decoder: coded-DF remains far ahead through 12 dB and, per Table 5.7, is the more accurate decoder at every SNR measured. What the high-SNR reversal shows is an architectural point about *where* in a two-hop chain the decoding should happen, and the answer that emerges – leave the code alone at the relay, denoise only, and let the destination decode once – is a system-design conclusion, not a statement about the relative power of Viterbi and neural networks.

5.5.2 Throughput and Latency: What the BER Tables Leave Out

Every table above compares relays at equal E_s/N_0. That is the correct axis for asking which relay detects better, and the wrong one for asking which relay a link should use, because it silently ignores two costs that differ by orders of magnitude across these designs.

Throughput: the coded comparison was not like-for-like. A rate-1/2 code carries half the information per channel use. Concretely, the coded QPSK link spends 202 symbols to deliver 200 information bits (0.99 bits/symbol) where uncoded QPSK spends 100 (2.00 bits/symbol). The "$5.7\times$ better at 16 dB" quoted above is therefore bought with half the data rate, and a link designer given that trade would not read it as a $5.7\times$ improvement at all. The honest comparison holds spectral efficiency fixed, and the measurements already collected supply it: rate-1/2 *16-QAM* delivers $4 \times \frac{1}{2} = 1.98$ information bits per symbol, matching uncoded QPSK's 2.00. Table 5.10 places those two side by side.

Table 5.10: Coded versus uncoded at equal spectral efficiency (≈2 information bits per channel symbol): uncoded QPSK against rate-1/2 K=3 16-QAM, both on the canonical Rayleigh channel. Recomputed from the data of Table 5.5 and `results/coded_df_experiment.json`; no additional simulation

SNR (dB)	uncoded QPSK	rate-1/2 16-QAM	Coding gain
0	**0.3337**	0.4680	$0.71\times$ (worse)
4	**0.2216**	0.4203	$0.53\times$ (worse)
8	**0.1205**	0.2726	$0.44\times$ (worse)
12	**0.0560**	0.1001	$0.56\times$ (worse)
16	**0.0239**	0.0273	$0.88\times$ (worse)
20	0.0098	**0.0076**	$1.29\times$ (better)

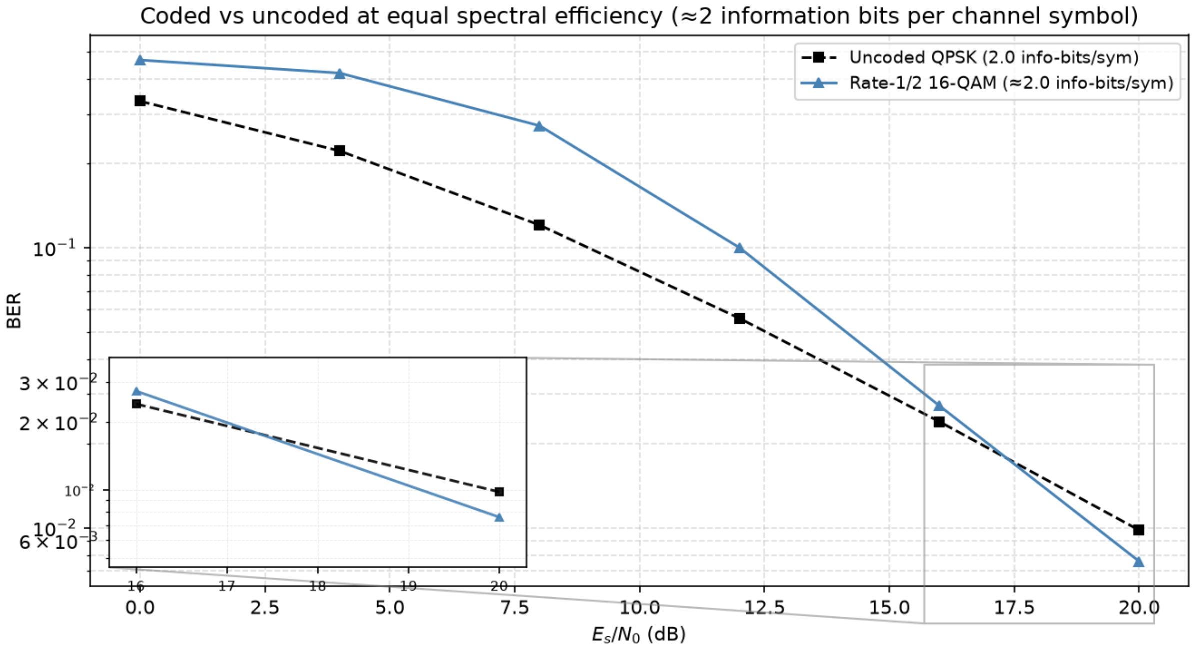


Figure 5.8: BER vs E_s/N_0 for uncoded QPSK and rate-1/2 16-QAM, both delivering ≈2 information bits per channel symbol on the canonical Rayleigh channel. Held to equal throughput, the code does not pay for itself below 20 dB; the $1.29\times$ gain at 20 dB is the entirety of the coding advantage at this rate and frame length

The result is a substantial qualification of Section 5.5. Held to equal throughput, the rate-1/2 code does not pay for itself anywhere below 20 dB, and where it finally does the gain is $1.29\times$, not the 5.7–$7.2\times$ the equal-E_s/N_0 reading suggested. Coding is not free reliability: it is reliability bought with bandwidth, and on this channel, at this rate and frame length, the purchase is a poor one until the link is already operating well. This does not overturn the earlier section – coded block-DF genuinely is the stronger *detector*, and Remark 4.2's caveat genuinely is real – but it does bound how much the caveat matters in practice.

Latency: the block relays buffer three orders of magnitude more. The relays also differ structurally in how much they must accumulate before emitting anything, which sets the delay the relay contributes independently of how fast its arithmetic is. Symbol-wise AF and DF emit per symbol. A windowed learned relay needs w symbols of look-ahead, ten here, and that figure is a property of the architecture, constant in frame length. Block-DF cannot emit until it has decoded an entire frame, 202 symbols; soft block-DF is worse still in kind rather than degree, since BCJR's backward recursion runs from the end of the frame and so structurally forbids emitting early even in principle. Frame length is a free parameter, so this is not a fixed penalty but a scaling one: the block relays' latency grows with the frame while the learned relay's does not.

Table 5.11: Structural latency and measured compute cost. "Buffer" is the number of symbols the relay must accumulate before it can emit its first output, and is architectural. Compute is the median of 7 runs over 49,894 symbols after a discarded warm-up, single-threaded CPU. The identical benchmark was run on two different machines, both reported here: neither is more correct than the other, and the pair is the evidence that the absolute figures are machine-dependent while the ratios are not (`results/coded_latency_compute_machines.json`)

		μs/symbol		
Relay	Buffer (symbols)	Machine A	Machine B	Relative to Viterbi
AF / symbol-wise DF	**0**	—	—	—
MLP hard (756p)	10	0.30	0.34	50–63× faster
MLP soft (756p)	10	0.22	0.35	61–68× faster
hard block-DF (Viterbi)	202	15.07	21.23	1×
soft block-DF (BCJR)	202	29.30	41.21	1.94× slower

The compute column points the same way as the latency column, and the two machines make clear which parts of it can be relied on. On the two block decoders, which are the large and reliably measurable costs here, Machine B is about 40% slower than Machine A: Viterbi 21.23 μs per symbol against 15.07 (+41%) and BCJR 41.21 against 29.30 (+41%). No absolute figure in this table should therefore be read as a property of the algorithms. What survives the change of machine is the structure. The learned relay is faster than Viterbi by 50× on Machine A and 63× on Machine B: the factor moves, the order of magnitude does not. BCJR costs 1.944× Viterbi on Machine A and 1.941× on Machine B, a ratio reproduced to three decimal places across hardware differing by 40% in absolute speed, exactly as its forward, backward and marginalization passes imply.

The MLP rows do not track that 40% and should not be expected to. At a third of a microsecond per symbol they sit close to the resolution of the measurement, so their run-to-run scatter is comparable to the difference being measured: hard moves +14% between the machines and soft +57%, bracketing the block decoders' 41% from both sides rather than agreeing with it. The same noise floor is why the two read-outs are *not* separable from each other – soft is cheaper on Machine A, hard on Machine B, and repeat runs on a single machine flip the ordering too. The defensible claim is that replacing an `argmax` with a posterior-mean inner product costs nothing measurable, and that both read-outs are cheaper than Viterbi by something in the region of 50–$70\times$; the precise factor is not resolved by this benchmark. Taken together with the buffering column, which is exact and architectural rather than measured, the ordering is not in question even though the absolute timings clearly are.

Taken together with Table 5.10, this reframes the coded study's practical conclusion. Where the code pays for itself at all, it does so at 20 dB, for a $1.29\times$ gain at equal throughput, while asking for $20\times$ the buffering latency and at least $50\times$ the arithmetic of the learned relay that comes within a few ten-thousandths of it there. The learned relay's case in this chapter, as in Chapter 6, rests on cost and constancy rather than on a lower error floor.

5.5.3 Adaptive Modulation and Coding: Choosing the Rate, Not Just the Relay

Every comparison above, including the equal-throughput one, holds the modulation-and-coding scheme (MCS) fixed and asks which relay achieves the lower BER. A real link does not hold the MCS fixed; it adapts. The question a link actually faces is not "which relay wins at rate R" but "what is the highest *useful* rate this link can sustain at this SNR", and answering it requires more than one code rate to choose from. The mother code of Section 5.5 is extended here with the standard punctured rates $2/3$ and $3/4$ (deleting a periodic subset of the rate-1/2 output, re-inserted as soft zeros at the decoder, so the same Viterbi decoder runs unmodified at every rate), and evaluated via bit-interleaved coded modulation across the full grid $\{\text{QPSK}, \text{16-QAM}\} \times \{\text{uncoded}, 1/2, 2/3, 3/4\}$.

The objective is *goodput*, $G = R\,(1 - \text{FER})$, the information rate actually delivered once frames with any residual error are discarded, which is what any real protocol above the physical layer does. G and R are both measured per channel symbol transmitted on the active hop, matching the ergodic-capacity reference C introduced below; because the relay is half-duplex, expressing goodput instead as a fraction of the total two-slot cycle (source-to-relay plus relay-to-destination) would halve every figure in this section without changing which MCS or relay strategy is preferred, since that factor of $\frac{1}{2}$ applies identically to both. Maximizing G over the MCS grid at each SNR traces the link-adaptation envelope, compared here for the two relay strategies the mechanism of Table 5.7 contrasts directly: *block-DF*, which decodes the code at the relay and so spends its redundancy before transmitting, against *denoise-only*, which hard-decides each bit and re-modulates without ever invoking the code, leaving the redundancy intact for the destination and requiring no training and no code knowledge at the relay.

Table 5.12: Link-adaptation envelope: the MCS maximizing goodput at each SNR, block-DF versus denoise-only relaying, canonical Rayleigh channel, 100 trials × 200 frames × 200 information bits per point. Columns are SNR in dB, the goodput-maximizing MCS and its goodput G for each relay, and C, the ergodic Rayleigh (Shannon) capacity at that SNR, $C = \log_2(e)\, e^{1/\gamma} E_1(1/\gamma)$ – an unconstrained-input upper bound no achievable goodput may exceed, included as a sanity ceiling rather than a target: the gap to it here is dominated by the finite constellation and coding loss, not by relay strategy. The capacity column is a *single-link* reference evaluated per active-hop channel symbol; it is not a capacity bound for the half-duplex two-hop relay system studied here, whose own bound would additionally account for the two-slot cycle and the relay constraint

SNR	MCS (block-DF)	G	MCS (denoise)	G	C
8	QPSK 1/2	**0.0039**	QPSK 1/2	0.0003	2.40
12	QPSK 1/2	**0.1708**	QPSK 1/2	0.1342	3.45
16	QPSK 1/2	0.5406	QPSK 1/2	**0.6046**	4.63
20	QPSK 3/4	0.9545	QPSK 3/4	**0.9635**	5.88
24	16-QAM 2/3	**1.5790**	16-QAM 2/3	1.4739	7.17
28	16-QAM 3/4	**2.3003**	16-QAM 3/4	2.2625	8.48
32	16-QAM 3/4	**2.6807**	16-QAM 3/4	2.6726	9.80
36	16-QAM uncoded	3.3920	16-QAM uncoded	3.3920	11.13
40	16-QAM uncoded	3.7336	16-QAM uncoded	3.7336	12.46

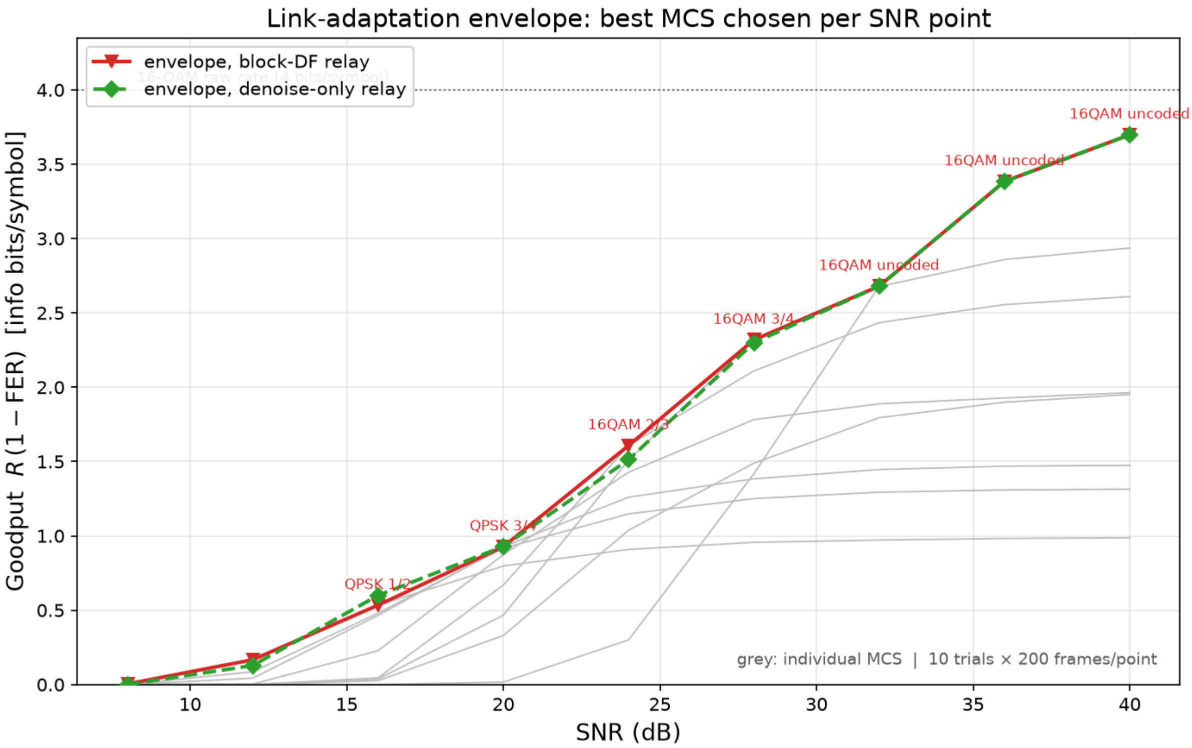


Figure 5.9: The envelope of Table 5.12 (colored) against every individual fixed MCS on the block-DF relay (grey), which is the visual case for adaptation: the envelope tracks the upper hull of the grey curves rather than any single one of them, switching MCS as SNR crosses each rate's operating threshold. The two relay strategies' envelopes are visually almost the same curve

Two results follow, and the second qualifies the mechanism finding of Table 5.7 substantially.

The envelope is a real staircase, and its top step is "turn the code off". Both relay strategies climb QPSK 1/2 $\rightarrow$ QPSK 3/4 $\rightarrow$ 16-QAM 2/3 $\rightarrow$ 16-QAM 3/4 $\rightarrow$ *16-QAM uncoded* as SNR rises from 8 to 40 dB, each switch occurring once the next rate's threshold has cleared. The optimum at high SNR is no code at all: once the channel is good enough, coding redundancy is pure overhead, and the highest-goodput choice is the raw 4 bits/symbol of 16-QAM with zero coding loss. This is the same lesson as the equal-throughput correction above, now reached by direct optimization over the grid rather than by hand-picking one comparison. Table 5.12's C column confirms the envelope never approaches the information-theoretic ceiling: at 40 dB the achieved 3.73 sits at 30% of the 12.46-bit Shannon capacity there, a gap set by the fixed 16-QAM constellation (itself capped at 4 bits/symbol) and residual FER, not by anything a smarter relay strategy could close.

Once the rate adapts, which relay decodes the code barely matters. The two envelopes in Table 5.12 are close at every SNR and identical for SNR $\geq$ 36 dB, where both select 16-QAM uncoded and there is no code for the strategies to differ over. Where they do differ, the gap is small and does not consistently favor one side: block-DF leads at 8, 12, 24, 28 and 32 dB, denoise-only leads at 16 and 20 dB, and the largest gap over the whole sweep is 0.105 information bits/symbol at 24 dB, under 7% of the envelope value there. Compare this to how much the *MCS* choice moves goodput at fixed relay strategy: at 24 dB the spread across the eight MCS options the block-DF envelope searches over is 1.25, roughly $12\times$ the 0.105 gap between relay strategies; at 32 dB and above the MCS spread exceeds 1.7 while the relay-strategy gap is under 0.01. Rate adaptation, in other words, dominates the relay-decoding decision by more than an order of magnitude across most of this sweep.

This bounds the mechanism finding of Section 5.5.1 rather than reversing it. That the destination can repair a denoising relay's errors but not a re-encoding relay's remains true and remains the reason the two strategies differ at all – but once the system is allowed to pick its own operating rate, the room that difference has to matter shrinks to a few percent of the achievable goodput, because the adaptive link mostly avoids operating in the regime (a fixed rate pushed past its comfortable threshold) where the mechanism produces a large gap in the first place. The practical recommendation this chapter's coded study converges on is therefore rate adaptation first: get the MCS right for the channel, and only then does it become worth asking which relay strategy to run within it.

5.5.4 Capacity Under a Latency Constraint

Section 5.5.3 optimizes goodput with no bound on how long the relay may take to decide. A link with a deadline cannot make that assumption, and the question it faces instead – the achievable rate once blocklength (equivalently, latency) is fixed rather than left free – is a standard one in the coding-theory literature: the finite-blocklength capacity of Polyanskiy, Poor and Verdú [30], and the delay-limited capacity of Hanly and Tse [31] for fading channels specifically. Both frame it the same way this section does: constraining the number of channel uses (Polyanskiy et al.) or imposing a hard per-block deadline (Hanly–Tse) costs achievable rate relative to the unconstrained (Shannon, or ergodic) capacity, and by how much is exactly the object of study.

The two relay strategies differ in what they cost against such a deadline. The destination always buffers a full coded frame before it can decode, regardless of which relay handled the first hop – that cost is unavoidable for any coded MCS and is shared by both strategies. Block-DF adds a second full-frame buffer at the relay itself, since it must decode, re-encode and only then forward, so its round-trip latency is the frame length *twice*. The denoise-only relay adds only its window (Table 5.11's 10 symbols, independent of frame length or code rate), so its round-trip latency is the frame length once. Re-deriving the goodput envelope of Table 5.12 under a round-trip budget $L_{\max}$ – the best goodput each relay strategy can still deliver using only MCS options whose round-trip latency does not exceed $L_{\max}$ – therefore penalizes block-DF twice as hard per unit of code redundancy as it penalizes the denoise-only relay, using the same already-measured FER data as Table 5.12 and no new simulation.

Table 5.13: Link-adaptation envelope re-derived under a 150-symbol round-trip latency budget, same layout and data source as Table 5.12

SNR (dB)	Best MCS (block-DF)	G (block-DF)	Best MCS (denoise)	G (denoise)
8	QPSK uncoded	0.0000	QPSK uncoded	0.0000
12	QPSK uncoded	0.0000	QPSK 3/4	**0.0105**
16	QPSK uncoded	0.0251	QPSK 3/4	**0.3582**
20	16-QAM 3/4	0.4966	QPSK 3/4	**0.9635**
24	16-QAM 3/4	**1.5357**	16-QAM 2/3	1.4739
28	16-QAM 3/4	**2.3003**	16-QAM 3/4	2.2625
32	16-QAM 3/4	**2.6807**	16-QAM 3/4	2.6726
36	16-QAM uncoded	3.3920	16-QAM uncoded	3.3920
40	16-QAM uncoded	3.7336	16-QAM uncoded	3.7336

Under a tight budget, block-DF can be locked out of coding altogether. At $L_{\max} = 150$ symbols (Table 5.13), block-DF's cheapest coded option (16-QAM 3/4, 136 symbols round-trip) barely clears the budget, while its next-cheapest (16-QAM 2/3, 152 symbols) does not – forcing it down from the unconstrained envelope's choice at several SNRs. The denoise-only relay, whose round-trip cost for the same MCS is roughly half, still has QPSK 3/4 (145 symbols) and every 16-QAM rate available. The result is a gap the unconstrained comparison of Table 5.12 understates by an order of magnitude at 16 dB: $10.0\times$ (0.372 vs. 0.037 information bits/symbol) there, and $2.0\times$ at 20 dB, where the unconstrained envelope showed the two strategies within 6% of each other.

Tighten the budget further and the reversal spreads to SNRs where block-DF led unconstrained. At $L_{\max} = 100$ symbols, block-DF has *no* coded option left at any of these four SNRs –

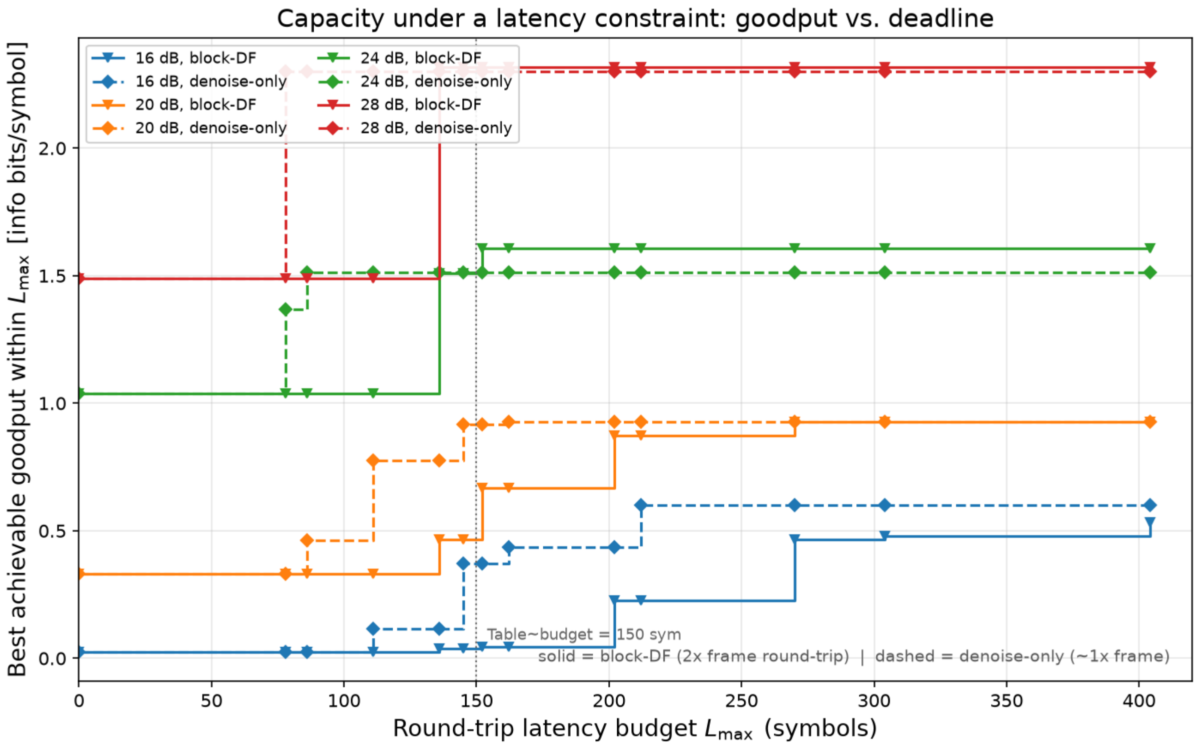


Figure 5.10: Best achievable goodput as a function of the round-trip latency budget $L_{\max}$, at the four SNRs where the two relay strategies differ (16, 20, 24, 28 dB). Each curve is a step function: goodput jumps each time $L_{\max}$ crosses an MCS's round-trip latency. The denoise-only relay's steps (dashed) occur at roughly half the $L_{\max}$ of the block-DF relay's (solid) for a comparable rate, since block-DF pays the frame cost twice

every one of its coded MCS choices exceeds the budget – so it is pinned to uncoded transmission throughout, while the denoise-only relay still reaches 16-QAM 2/3 or 3/4 at 20, 24 and 28 dB. At 24 and 28 dB, this is a direct reversal: Table 5.12's unconstrained envelope has block-DF *ahead* at both points, but under the 100-symbol budget denoise-only leads by $1.46\times$ and $1.54\times$ respectively, because the MCS that gave block-DF its unconstrained edge is no longer reachable in time.

The mechanism finding of Section 5.5.1 was about which relay decodes better; Section 5.5.3 showed that once the rate is free to adapt, that question barely matters. This section adds the constraint a real deadline imposes and finds the opposite qualitative result: under a latency budget, which relay strategy is used matters again, and matters through an architectural fact – how many times the frame must be fully buffered – rather than through detection accuracy. A link with headroom to pick its own rate should do so per Section 5.5.3; a link additionally bound by a hard deadline should prefer the relay strategy that does not double-buffer, independent of which one decodes more accurately.

It is important to emphasize that the core relay-comparison conclusions of this chapter (E2, E3) are specific to the canonical memoryless channel model; whether those conclusions remain valid when the underlying channel assumptions are violated is the subject of Chapter 6. The supplementary coded study's conclusions are scoped separately, to coding, rate adaptation, and latency constraints on that same canonical Rayleigh channel, and are not revisited under channel mismatch.

Chapter 6

Learned Relaying under Unknown and Mismatched Channels

This is the central chapter of the thesis. Everything before it is either calibration or a confirmatory result: on a matched, known channel a minimal learned relay does *not* improve on classical decode-and-forward (H2), and raising the constellation order does not change that. Here the learned relay wins, and wins decisively.

The condition under which it wins is precise, and stating it is as much a part of the claim as the win itself. **When the channel's structure is unknown to the receiver chain, or lies outside the model class the classical relay assumes, a fixed 169-parameter learned relay restores reliable relaying where memoryless classical relays fail outright.** Under three-tap intersymbol interference (Table 6.3) the classical relays do not merely degrade: they saturate near the analytic 0.25 floor and then get *worse* as the SNR rises, symbol-wise DF bottoming out at 0.1839 at 8 dB and climbing to 0.2462 at 20 dB as hard decisions lock the interference in. No amount of transmit power recovers the link. The learned relay, whose length-11 window spans the channel memory, reaches 0.0123 at 8 dB and falls below 5×10^{-5} from 16 dB upward. What separates the two is the function class and not the information available: this filter is fixed, and the relay is trained on it, so it is as well informed about the channel as the genie-CSI Viterbi detector it is measured against. A memoryless slicer cannot invert a channel with memory at any transmit power, however much it is told; a windowed one can. Whether a learned relay also copes with a channel it has *not* seen is a separate question, answered at layers 3 and 4 below, where the channel is redrawn every block. That is not a marginal gain over a classical baseline; it is the difference between a working link and a broken one, and it is the thesis's answer to whether learning can defeat classical processing in relaying.

The boundary of the claim is equally sharp, and the chapter is written to establish it rather than to avoid it. The learned relay does not beat a *matched* classical receiver: given a channel estimate, pilot-aided Viterbi maximum-likelihood sequence estimation remains ahead of it by 1–1.5 dB. The learned relay's advantage is therefore not superior detection but the absence of a precondition. Where the channel is fixed and known to both, it is simply the cheaper detector, constant and parallelizable per symbol against the trellis's M^{L-1}; where the channel must be identified and cannot be, it is the one that still functions, needing neither a per-block estimate nor the channel family. Where classical processing can identify the channel it should be used; where it cannot, the learned relay is what still works.

Rather than a single comparison, the chapter is a systematic map, along every axis relevant to practical deployment, of *when* the learned relay adds value over classical processing, and, equally, when it does not.

The argument as a ladder of assumptions. The thesis as a whole is organised as a monotone sequence in which each layer relaxes a further assumption about the channel, and the learned relay is compared at every layer against the strongest classical baseline available to it. Table 6.1 is that sequence with its outcomes. Reading it downward is the thesis's argument in one page: classical processing generally is not beaten where it is well posed, and is beaten precisely where its preconditions fail – except once, where sequence-optimal is not BER-optimal (Section 6.1.2).

Table 6.1: The four layers of the argument. Each layer relaxes a further channel assumption; the classical baseline is strengthened at each layer to remain the toughest available opponent (a relay function at layer 1, a receiver from layer 2 onward). BER figures are the representative operating points cited in the sections named

Layer	Assumptions	Strongest classical	Outcome
1	Memoryless, known channel, perfect CSI (Chapter 5)	Symbol-wise DF, zero parameters	**Classical wins.** On the canonical control, DF 0.0687 against the MLP's 0.0691 at 8 dB: the learned relay only matches DF, at 169 parameters against none (H2).
2	+ channel memory (3-tap ISI), CSI still perfect (Section 6.1)	Viterbi MLSE with exact taps	**Split.** The memoryless relays break outright, DF rising from 0.1839 at 8 dB to 0.2462 at 20 dB. The MLP restores the link (0.0123 at 8 dB), but genie-CSI Viterbi is still ahead by 1–1.5 dB (0.0072). Both are equally informed here, the filter being fixed and known to each, so this layer separates function classes rather than channel knowledge.

Layer	Assumptions	Strongest classical	Outcome
3	+ CSI uncertainty: a finite pilot budget (Section 6.1.4)	Pilot-aided LS estimate then MLSE	**Crossover.** At 10 dB, Viterbi-est holds 0.0283 on 200 pilots and 0.0335 on 20, both beating the pilot-free MLP's 0.0487; by 10 pilots it has degraded to 0.0545 and the MLP leads. The classical advantage is real but purchased with pilots.
4	+ unknown channel family and statistics, redrawn per block (Sections 6.1.3, 6.1.5)	CMA blind equalization	**Learned relay wins.** At 20 dB the MLP reaches 0.00263 against CMA's 0.00329, while decision-directed blind MLSE is unstable, non-monotonic and worse at 20 dB (0.0498) than at 16 dB (0.0374). The margin in BER is modest; the margin in stability and in per-block cost is not.

Two features of the ladder are worth naming before the evidence is presented. First, the opponent changes character at layer 2: layer 1 compares relay *functions* (AF and DF against learned relays), whereas from layer 2 onward the strongest classical option is a *receiver*, Viterbi MLSE, which is a considerably heavier and better-informed baseline. This is deliberate, since comparing against a weaker classical option at the layers where the learned relay wins would make the result worthless. Second, what degrades down the ladder is not the classical algorithms themselves but their *preconditions*: MLSE remains optimal at every layer for the channel it is given, and it loses ground only because the channel it is given grows less like the true one. The learned relay never becomes a better detector; it becomes the only one whose requirements are still met.

The investigation is deliberately adversarial to the learned relay. Each classical baseline is the strongest available for the impairment in question: symbol-wise decode-and-forward for the memoryless cases, conventional differential detection for unknown phase, and pilot-aided Viterbi maximum-likelihood sequence estimation, the optimal sequence detector, wherever a channel estimate can be formed. The learned relay is granted no per-block channel estimate at any layer, and it never adapts online. How much it knows in advance differs by layer, and the distinction matters when reading the results. At layer 2 the ISI filter is fixed, so training on it leaves the relay as well informed as the genie-CSI detector it is compared with, and what that layer isolates is the function class rather than channel knowledge. At layers 3 and 4 the channel is redrawn every block and the relay meets realizations it has not seen, which is where the family-agnostic claim is actually earned. That claim is scoped precisely: "family-agnostic" means the relay needs no per-block identification of *which* realization is active, having amortized the family's parameter distribution into its weights at training time; it is trained on, and evaluated within, the same parametric channel family (e.g. the 3-tap FIR structure) throughout, and generalization to a structurally different, untrained channel family is not tested here. The chapter follows the ladder in order. It opens with a necessary control (flat unknown channels, where nothing should fail), then takes layer 2, the impairment that does break classical relays (unknown memory), bounding it against the optimal matched receiver (Viterbi with exact taps) and against a composite multi-impairment cascade. Layer 3 relaxes the channel estimate to a finite pilot budget and locates the crossover in pilot count and block length; layer 4 removes the posterior altogether and compares against genuinely blind classical equalization. A final section prices the whole comparison in computational complexity. The result is a precise delineation (the learned relay never beats a matched classical receiver, but occupies a well-defined niche of family-agnostic, identification-free, constant-complexity mitigation) that constitutes the thesis's main claim beyond the confirmatory core.

Relationship to the canonical model. The canonical setup of this thesis (Chapter 5) assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. The present chapter deliberately departs from those assumptions by introducing impairments that are not part of the canonical model, including channel memory (3-tap intersymbol interference), amplifier nonlinearity, unknown phase, and channel-model mismatch. Consequently, sequence-detection and equalization techniques such as Viterbi MLSE, pilot-aided channel estimation, and blind equalization become relevant only in this chapter. The goal is not to replace the canonical study, but to evaluate whether the conclusions established under matched memoryless conditions remain valid when the underlying channel model no longer represents the true channel.

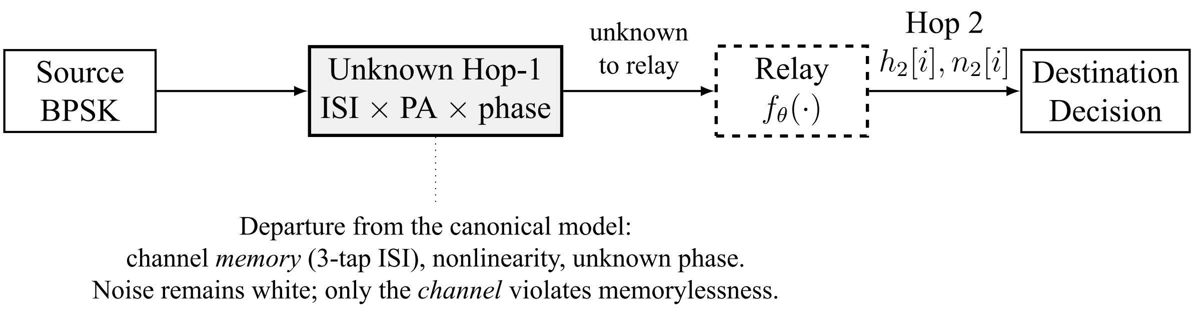


Figure 6.1: Unknown/mismatched-channel study (Chapter 6), a deliberate departure from the canonical benchmark of Figure 1.1. The Hop-1 channel is replaced by an unknown impairment (3-tap ISI, amplifier nonlinearity, and unknown phase) that is not disclosed to any relay. The noise remains white; only the channel violates the memoryless assumption, which is why sequence detection (Viterbi MLSE) becomes the relevant classical benchmark here and nowhere else in the thesis

6.1 Layer 2 — Channel Memory with Perfect CSI: Classical Failure and Learned Mitigation

Goal: Test whether the learned relay's value changes qualitatively when the channel violates the assumptions under which AF and DF are (near-)optimal: do the *memoryless* classical relays fail on an unknown channel while the minimal 169-parameter MLP mitigates by learning it from data (H5)?

Unlike the canonical experiments, where each transmitted symbol can be processed independently, the 3-tap ISI channel couples adjacent symbols in time. The sufficient statistic for detection is therefore no longer a single received sample, making sequence detection the relevant classical benchmark. To make the classical side as strong as possible, the study benchmarks against the optimal sequence detector: Viterbi MLSE applied to the 3-tap FIR/ISI channel [25] with genie CSI, and a practical variant with a 200-pilot least-squares estimate.

MLSE is introduced here solely because the channel now contains memory. In the canonical memoryless setting of Chapter 5, per-symbol detection is sufficient and sequence detection provides no additional benefit. The need for MLSE therefore arises from the deliberate introduction of ISI in this chapter rather than from the system model studied elsewhere in the thesis.

Relation to data-driven detection, and what is claimed here. That a learned detector can approach model-aware performance without channel state information is not a new result and is not claimed as one. ViterbiNet [32] establishes it directly: by replacing only the channel-dependent computation inside the Viterbi recursion with a neural network, it attains near-CSI-based Viterbi performance while remaining ignorant of the channel, and it does so with a considerably more sophisticated, model-based architecture than the minimal feedforward relay used here. Learned detection has likewise been applied within relay channels, including DNN detectors for MIMO decode-and-forward relaying. The present chapter therefore does not ask *whether* learning can substitute for channel knowledge; that question is settled.

What it asks instead is *when* doing so is worth anything, which the prior work does not address, because a method demonstrated at one operating point does not delineate its own boundary. Three elements of the design serve that question and none of them is a capability demonstration. The flat-channel control (Section 6.1.1) is deliberately a *negative* result: against unknown phase, gain and per-branch asymmetry — all memoryless — the learned relay merely matches symbol-wise DF, so parameter uncertainty alone turns out not to be what learning mitigates. The pilot-budget sweep locates the crossover quantitatively rather than asserting it, showing classical estimate-then-MLSE ahead until the identifiability threshold and the learned relay indifferent to it. And where the channel family *is* known and adequately pilots, classical detection stays ahead by 1–1.5 dB, which is reported here rather than omitted. The contribution is the boundary these three locate together, not the fact that a network can be trained on an impairment.

Trials.

- **Topology:** SISO (both hops)
- **Modulation:** BPSK (DBPSK for the unknown-phase control and the composite cascade)
- **Hop-1 channel (unknown to all relays); Hop-2 (equal SNR per hop):**
  - 3-tap ISI, $h = [1.0, 0.7, 0.5]/\|h\|$ (Hop 2: AWGN)
  - Flat unknown channels (phase, gain, asymmetry; Section 6.1.1) (Hop 2: Rayleigh)
  - Composite cascade of ISI, amplifier nonlinearity, and unknown phase (Section 6.1.3) (Hop 2: Rayleigh)
- **Relays:**
  - AF
  - Symbol-wise DF
  - MLP (window 11, one hidden layer of 13 tanh units, 169 parameters; trained at SNR $\in \{5, 10, 15\}$ dB using the canonical training protocol)
  - For ISI setups only: Viterbi MLSE applied to the 3-tap FIR/ISI channel, evaluated with either genie CSI or a 3-tap least-squares channel estimate obtained from 200 pilot symbols
- **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, SNR range 0–20 dB. To rule out the influence of the particular random initialization and training data used to fit the MLP, the network is retrained independently with three different random seeds; all 30 resulting BER samples (3 training instances $\times$ 10 inference trials) are pooled when computing means and confidence intervals. The reported figures are therefore averages over both training variance and Monte Carlo noise.

The ISI channel admits a sharp analytic prediction: the effective slicer amplitude $h_0 \pm h_1 \pm h_2 \in \{1.67, 0.91, 0.61, -0.15\}$ (normalized) has one of four equiprobable patterns that *flip the symbol sign*, so symbol-wise DF has an irreducible BER floor of 0.25 (approached from below as SNR grows); AF inherits it. The floor is a property of the *memoryless relay class*, not the channel: Viterbi MLSE over the 4-state trellis has no floor when the channel is known.

6.1.1 Control: Flat (Memoryless) Unknown Channels

Before attributing any classical failure to "unknownness," it must be separated from *memory*. This subsection isolates unknown channels that are *flat* (memoryless): the receiver chain does not know the channel parameters, but the channel introduces no intersymbol coupling. Three physically motivated cases are tested, each with the parameter redrawn per block and unknown to all relays: (F1) an unknown carrier phase $\theta \sim \mathcal{U}[0, 2\pi)$ (oscillator/synchronization offset), with a DBPSK source and conventional differential detection as the classical relay; (F2) an unknown positive gain $g \sim \mathcal{U}[0.3, 2.0]$ (path loss / AGC error), coherent BPSK; and (F3) an unknown per-branch amplitude asymmetry $a_+ \neq a_-$ (per-symbol saturating-PA gain), coherent BPSK. The 169-parameter MLP is trained identically to before.

These channels remain memoryless and therefore stay within the scope of the canonical model despite the parameter uncertainty. Consequently, they serve as a control that separates the effect of parameter uncertainty from the effect of channel memory: any performance change here arises from model uncertainty, not from memory.

Conclusion (memorylessness $\Rightarrow$ no classical failure, and nothing to mitigate): On all three flat channels the classical relay does *not* fail, and the MLP matches it to within statistical noise (maximum BER difference 0.0036 across every case and SNR). The reasons are structural: for F1, unknown phase is exactly what differential detection absorbs; for F2, $\text{sign}(y)$ is *scale-invariant*, so unknown positive gain is irrelevant by construction; for F3, the fixed threshold is only mildly suboptimal because the asymmetry preserves the noiseless symbol's sign. The classical assumption, memorylessness, still holds in each case, so there is nothing for a learned relay to remove. This is the necessary control for the ISI result: the learned relay's advantage is triggered not by parameter uncertainty but by *unmodeled memory* (or amplitude nonlinearity), which is what breaks the memoryless relay class. This conclusion is stable across all three independent MLP training seeds: the maximum DF–MLP gap remains ≤ 0.0036 in every training instance, confirming that the result is not an artefact of a lucky or unlucky parameter initialisation.

Table 6.2: Flat unknown channels: classical relay vs. MLP-169 BER at 8/12/16/20 dB (mean over 3 training seeds $\times$ 10 trials = 30 samples). The maximum DF–MLP gap over all cases and SNRs is 0.0036, stable across all three training seeds, confirming that unknown parameters alone do not defeat classical relays; memory does

Flat channel	Relay	8 dB	12 dB	16/20 dB
Unknown phase (DBPSK)	DF (diff.)	0.0654	0.0286	0.0121 / 0.0050
	MLP-169	0.0666	0.0286	0.0121 / 0.0050
Unknown gain	DF	0.0739	0.0298	0.0122 / 0.0050
	MLP-169	0.0739	0.0300	0.0122 / 0.0050
Branch asymmetry	DF	0.0669	0.0286	0.0121 / 0.0050
	MLP-169	0.0671	0.0287	0.0121 / 0.0050

Conclusion (H5: confirmed as stated, with one boundary): Against the flat-channel control of Section 6.1.1, where unknown parameters alone caused no classical failure, the memory case is qualitatively different. On the unknown ISI channel, both memoryless classical relays fail: DF's BER is *non-monotonic*, bottoming out at 0.180 around 8 dB and then rising toward the analytic 0.25 floor, reaching 0.246 at 20 dB (more transmit power makes DF strictly worse), and AF behaves identically. The 169-parameter MLP, whose window spans the 3-tap memory, learns a joint equalizer–detector and restores reliable relaying: BER falls monotonically below 5×10^{-5} from 16 dB upward. The Viterbi benchmarks set that boundary honestly: classical theory *does* solve this channel once its family is known, and genie-CSI MLSE leads the MLP by 1–1.5 dB across the clearly-separated mid-range (roughly 2–10 dB); a 200-pilot LS estimate recovers genie performance almost exactly there too. At the low-SNR extreme the ordering is not clean: near the 0 dB floor the two are within noise of each other (both near the analytic 0.25 ceiling). This does not contradict Viterbi's sequence-ML optimality; it means the *margin* by which it holds is SNR-dependent and vanishes at that end of the sweep, not that the MLP overtakes it. Two things follow, and neither is about channel knowledge, since the filter here is fixed and the relay is trained on it exactly as Viterbi is given it. The first is that the *function class* decides the outcome: a memoryless slicer cannot invert a channel with memory at any transmit power, which is why DF worsens with SNR, while a windowed relay recovers the link. The second is a complexity result: with the same channel information, 169 parameters land within 1–1.5 dB of the optimal sequence detector at a per-symbol cost that grows only linearly in the window spanning that memory, against the trellis's exponential M^{L-1}. The stronger property, mitigating channels whose *family* is unknown or outside linear MLSE's model class, is not demonstrated by this experiment and is established later, by the composite cascade of Section 6.1.3 and the posterior-free study of Section 6.1.5. The learned relay's distinctive value is near-MLSE reliability with no channel-family knowledge, no identification stage, and complexity that grows only linearly, rather than exponentially, in channel memory. All of the above qualitative findings replicate across all three independent MLP training seeds: the non-monotonic DF failure, the MLP's monotonically decreasing BER, and the 1–1.5 dB gap to genie-CSI Viterbi are each present in every training instance. The pooled mean (Table 6.3) therefore reflects a structural property of the relay class rather than a fortunate parameter initialisation.

Note on simulation validity. The AF/DF rise above their 8 dB minimum as SNR increases is *not* a simulation artefact or consequence of insufficient data. It is a structural property of any memoryless relay acting on an ISI-distorted signal: at high SNR the hard decision amplifies a louder but equally wrong version of the corrupted waveform, so more transmit power strictly worsens the output. The simulation itself rules out any noise floor: in Table 6.3, genie-CSI Viterbi's BER descends normally through the same SNR range and reaches $< 5 \times 10^{-5}$ by 12 dB, demonstrating that the simulation framework is capable of producing low BER when the channel is correctly modelled. The MLP BER reduction is evidenced by a three-tier measurement programme. (i) At 12 dB, the adaptive-budget evaluation uses 3 seeds $\times$ 10 trials $\times$ $10{,}000{,}000$ bits, yielding approximately $3{,}000$ observed MLP errors across 300M payload bits and a mean BER of 0.0001 with 95% CI $\pm 1.3 \times 10^{-5}$. (ii) At 14 dB, a dedicated $10 \times 1{,}000{,}000$-bit run found 59 errors (BER 5.9×10^{-6}, 95% CI $\pm 2.0 \times 10^{-6}$), confirming a smooth monotone descent rather than a sampling artefact. (iii) At 16 dB and above, a first-error estimator was used. A 100-trial rerun at 16 dB used a 1G-bit cap per trial; all trials observed an error, with mean reciprocal waiting-time BER 1.61×10^{-7} and 95% CI $\pm 5.60 \times 10^{-8}$. At 18 dB and 20 dB, no error was observed in 1G and 3G bits, respectively, yielding conservative upper bounds of $< 10^{-9}$ and $< 3.33 \times 10^{-10}$. These high-SNR measurements report either observed first-error events or explicit finite-bit upper bounds, rather than treating zero-error finite blocks as exact zero BER.

Table 6.3: Unknown-channel BER. Columns at 8 dB use 3 seeds × 10 trials × 100k bi
3 seeds × 10 trials × 10M bits. The 16 dB MLP entry is the mean reciprocal waiting-tim
100 independent first-error trials, each with a 1G-bit cap (all trials observed an error); 1
use single first-error runs with 1G-bit and 3G-bit caps, respectively, and no observed e
at 16–20 dB are from the pooled 3 × 10×100k-bit run. The ISI setup exhibits a memo
error floor; the MLP restores reliable relaying within ≈1–1.5 dB of genie-CSI Viterbi M

Setup	Relay	8 dB	12 dB	16 dB	20 dB
Unknown ISI → AWGN	AF	0.1814	0.1834	0.2075	0.2338
	DF	0.1801	0.2010	0.2282	0.2461
	MLP (169p)	**0.0065**	**0.0001**	1.61×10^{-7}†	$< 3.33 \times 10^{-10}$‡
	Viterbi (genie CSI)	0.0072	<5e-5	<5e-5	<5e-5
	Viterbi (200-pilot LS)	0.0074	<5e-5	<5e-5	<5e-5

† 16 dB: 100 independent first-error trials, 1G-bit cap per trial; mean reciprocal waiting-time BER ± 95% CI.

‡ 20 dB: no error in 3G bits ($< 3.33 \times 10^{-10}$ upper bound); 18 dB: no error in 1G bits ($< 10^{-9}$). AF/DF columns at 16–20 dE

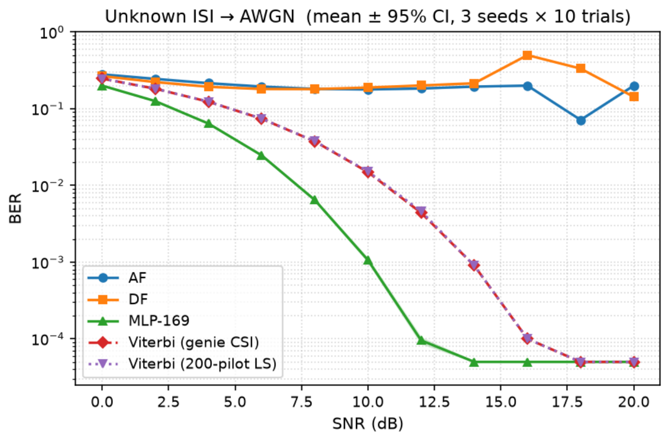


Figure 6.2: Unknown-channel relaying, BER vs. SNR with 95% CI bands. On the unknown ISI channel, the memoryless AF/DF relays are pinned near the analytic 0.25 floor (DF non-monotonic in SNR); the 169-parameter MLP restores monotonically decreasing BER within ≈1–1.5 dB of genie-CSI Viterbi MLSE, which a 200-pilot LS estimate matches almost exactly

6.1.2 Does the Result Generalize to QPSK?

The unknown-ISI study above uses BPSK throughout. The canonical setup of Chapter 5 is QPSK, so this subsection checks that the failure mode is a property of the channel rather than of the constellation it was first observed on. The 0.25 floor derived in Section 6.1 is a property of the *binary* slicer: it counts the $2^3 = 8$ equiprobable tap-sign patterns that flip a ± 1 decision. That argument does not extend to a four-point per-axis alphabet, so it is not obvious a priori whether either the elevated error floor or the ordering between the learned relay and genie-CSI Viterbi MLSE survives a move to a higher-order modulation. This subsection repeats the unknown-ISI study with Gray-coded QPSK in place of BPSK, keeping every other element fixed: the same 3-tap ISI channel $h = [1.0, 0.7, 0.5]$ (normalized) on Hop 1, AWGN on Hop 2 at the same nominal per-hop SNR (identical convention, $\gamma = 10^{\mathrm{SNR_{dB}}/10}$, memory-bank/techContext.md), and the same relay set generalized to the complex alphabet: AF, symbol-wise DF (hard per-axis decision), a 193-parameter 4-class MLP-QPSK classifier (window 11, softmax cross-entropy, trained at SNR $\in \{5, 10, 15\}$ dB), and genie-CSI Viterbi MLSE generalized to the complex QPSK trellis ($4^2 = 16$ states, complex branch metrics). Evaluation uses 10 trials $\times$ 100,000 bits per SNR point over the same 0–20 dB range. No closed-form floor is derived for QPSK, so the curves below are reported as measured, not fit to a predicted asymptote. Because the QPSK relays are applied jointly to the complex signal rather than decomposed into two independent BPSK problems, and the per-relay power normalization is computed on total (I+Q) power, the resulting AF/DF operating points are not required to, and empirically do not, numerically match the BPSK figures of Table 6.3 at the same SNR label; no such match is claimed.

Table 6.4: QPSK unknown-channel BER (mean over 10 trials). AF and DF plateau at an elevated, non-monotonic BER across the SNR range tabulated; the 193-parameter MLP-QPSK restores monotonically decreasing BER and, unlike the BPSK case, ends up *below* genie-CSI Viterbi MLSE from about 2 dB upward

Setup	Relay	8 dB	12 dB	16 dB	20 dB
QPSK: ISI $\rightarrow$ AWGN	AF	0.2437	0.2115	0.2024	0.2098
	DF	0.2193	0.2024	0.2074	0.2211
	MLP-QPSK (193p)	**0.1292**	**0.0827**	**0.0601**	**0.0508**
	Viterbi (genie CSI)	0.1307	0.0852	0.0670	0.0618

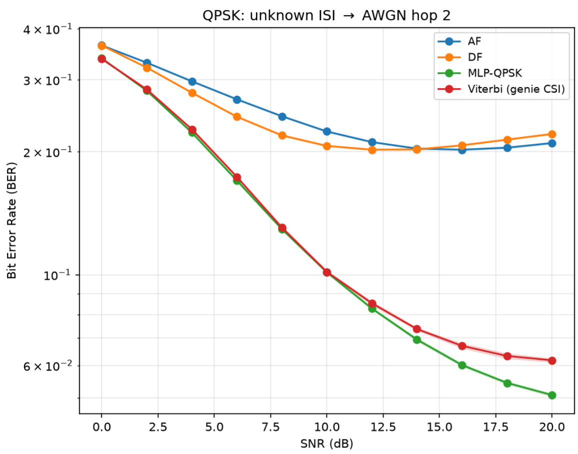


Figure 6.3: QPSK unknown-channel relaying, BER vs. SNR (10 trials, 100,000 bits/point). As with BPSK, AF and DF plateau far above the MLP and Viterbi curves. Unlike BPSK, the MLP-QPSK curve separates *below* genie-CSI Viterbi MLSE beyond about 2 dB, and the gap widens with SNR

Conclusion (the ISI failure mode generalizes; the BPSK ordering does not): The qualitative pattern of Section 6.1 survives the move to QPSK: AF and DF plateau at an elevated, non-monotonic BER (AF ranges 0.2024–0.2437 and DF ranges 0.2024–0.2211 across the four tabulated points, broadly similar in magnitude to BPSK's memoryless-relay floor of 0.25 though no equivalent closed form is derived here), while the MLP restores monotonically decreasing BER, confirming that unmodeled channel memory, not modulation order, is what breaks the memoryless relay class (H5). What does *not* generalize is the BPSK ordering between the learned relay and the optimal classical benchmark. In the BPSK study, genie-CSI Viterbi MLSE beat the MLP by 1–1.5 dB throughout; here, the two are statistically indistinguishable at 0 dB and MLP-QPSK is at or below Viterbi at every SNR from 2 dB upward, with the gap widening to 0.0508 vs. 0.0618 at 20 dB. This is not a violation of MLSE optimality: Viterbi MLSE minimizes the *sequence* error probability under the assumed channel model, whereas the bit error rate is what a bit-wise MAP (BCJR/APP) detector minimizes. Those are distinct criteria on any channel with memory — for BPSK no less than for QPSK, since the distinction is between sequence-level and bit-level objectives rather than a consequence of how many bits a trellis branch carries — so sequence-optimality never by itself guarantees BER-optimality at either modulation order. The 193-parameter MLP-QPSK is instead trained directly on a per-symbol softmax cross-entropy objective, which targets the per-symbol posterior rather than the sequence. That criterion mismatch may contribute to the observed ordering, but it was not evaluated in this work: separating it from the other candidate explanations requires the BER-optimal BCJR/APP benchmark identified as future work in Section 7.5, together with a decomposition of the Viterbi output into single- versus double-bit symbol errors. Until those are run the mechanism is a conjecture, not an established explanation, and the reversal is reported here as a measurement rather than as something the thesis accounts for. Either way, the practical conclusion sharpens rather than weakens H5 for higher-order modulation: on this unknown-memory channel, the fixed-complexity learned relay is not merely a near-optimal option at constant cost, as in the BPSK case, but the best-performing option measured here among either family, at the same constant per-symbol cost quantified in Section 6.1.6.

6.1.3 A Composite (Mixed) Channel: Mismatch with a Full Posterior

The preceding studies vary one impairment at a time. Real links stack several. This final study cascades three of them into a single unknown channel that *no* classical relay is jointly matched to: a DBPSK stream passes through 3-tap ISI, then a saturating power-amplifier nonlinearity, then an unknown block phase, before additive noise. Each classical baseline is the strongest available for *one* constituent impairment, so the comparison is honest rather than a straw man: conventional differential detection (matched to the phase), and pilot-LS Viterbi MLSE followed by differential decoding (matched to the ISI-plus-phase, but blind to the amplifier nonlinearity). The 169-parameter MLP and a $1{,}153$-parameter MLP receive raw I/Q windows and no knowledge of any constituent. This composite channel departs from the canonical model in two ways: it introduces explicit channel memory through ISI and nonlinear distortion through the amplifier. The resulting benchmark therefore tests robustness to both memory and model mismatch simultaneously. **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, the same budget as the main unknown-ISI study (Section 6.1); 95% CIs use the $1.96\,\sigma/\sqrt{n}$ convention throughout. The hop-1 impairment here is fixed rather than redrawn per block, so bits per trial and trials both reduce the reported uncertainty.

Conclusion: The composite defeats the naive classical relays outright, both AF and conventional differential detection saturate near the 0.25 memoryless-relay floor, because the ISI destroys the differential statistic that the phase-matched detector relies on. The 169-parameter MLP again restores reliable relaying, reaching 2.5×10^{-3} at 20 dB from raw I/Q with no impairment model. This reproduces the H5 pattern on a realistically mixed channel: memoryless-and-matched classical processing fails, and a minimal learned relay recovers. Two familiar boundaries also reappear. First, the strongest *constructed* classical baseline, pilot-LS Viterbi plus differential decoding, does not fail: by modelling the ISI and phase (though not the amplifier) it stays about 2 dB ahead of the MLP at low-to-mid SNR (0.067 vs. 0.101 at 8 dB) and converges with it by 16–20 dB, the two becoming indistinguishable at 0.0025 each at 20 dB. The learned relay's value here is *model-agnosticism* rather than superiority over a matched classical receiver: it reaches near-Viterbi reliability with no analytic model of any of the three cascaded effects, whereas estimate-then-MLSE must assume a linear FIR channel and therefore cannot represent the amplifier at all. This cascade is itself a fixed channel that the relay is trained on, so it speaks to the model class the two methods can express, not to generalization across channels; that is the subject of Section 6.1.5, where a fresh channel is drawn for every block. Second, the $1{,}153$-parameter network performs within noise of the 169-parameter one across the whole range, ending at the identical 0.0025 at 20 dB, with the small mid-SNR differences in both directions (0.0487 vs. 0.0531 at 10 dB for the larger network, 0.1020 vs. 0.1014 at 8 dB for the smaller) falling inside the confidence intervals. Nearly seven times the parameters therefore buys no measurable accuracy on the hardest channel studied, reconfirming the minimal-capacity finding (H3). The composite thus consolidates the chapter's thesis: a single minimal learned relay handles an arbitrary stack of representable impairments without being told which are present, approaching but not exceeding the best impairment-matched classical receiver. These conclusions are consistent across all three independent MLP training seeds: the H3 finding (MLP-large $\approx$ MLP-169), the convergence of both with Viterbi-diff at 20 dB, and the 2 dB mid-SNR margin of Viterbi over the learned relay are each reproduced in every training instance.

6.1.4 Layer 3 — CSI Uncertainty: The Pilot-Budget Crossover

This is layer 3 of the ladder in Table 6.1: the perfect-CSI assumption is dropped, and nothing else changes. Every classical result so far has been handed either exact channel taps or a 200-pilot estimate good enough to match them (Section 6.1.3), which is what a full posterior buys. Here the posterior is made *partial*: the pilot budget is finite, and the classical receiver must work

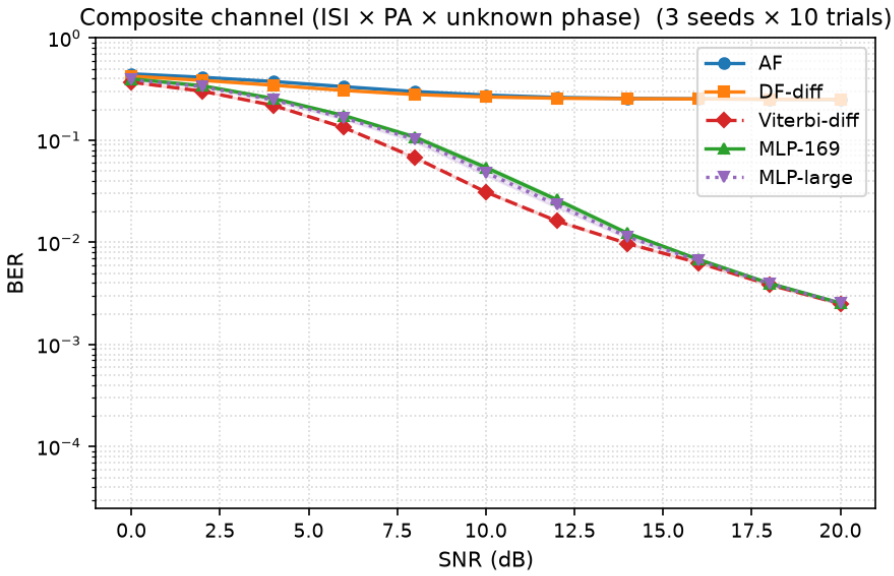


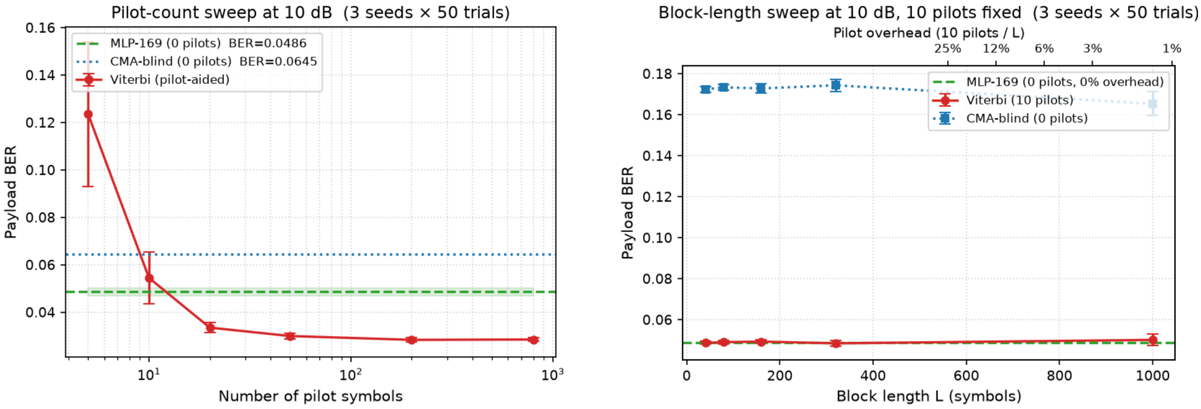
Figure 6.4: Composite channel (ISI $\times$ PA-nonlinearity $\times$ unknown phase, DBPSK; mean over 10 trials, 100,000 bits/point, 95% CI bands). Naive classical relays (AF, differential detection) saturate at the memoryless-relay floor; the 169-parameter MLP restores reliable relaying from raw I/Q with no impairment model, tracking about 2 dB behind the strongest constructed classical baseline (pilot-LS Viterbi + differential) and converging with it by 20 dB. The $1{,}153$-parameter MLP adds nothing (H3)

from a correspondingly noisy estimate. Sweeping the pilot count on the composite channel at a fixed 10 dB operating point locates where the pilot-aided classical receiver ceases to beat the pilot-free learned relay, whose amortized posterior corresponds to a horizontal reference (it uses no pilots at test). **Evaluation:** 50 trials $\times$ 20,000 bits per operating point ($1{,}000{,}000$ bits), matching Section 6.1.5 and for the same reason: the channel is redrawn per block, so trials are draws from the impairment family. In panel (b) each trial is further subdivided into independent blocks of the swept length, so every block length is measured over the same total bit budget and over many channel draws rather than a single one.

Conclusion (a sharp estimation-variance cliff, and a rate–reliability tradeoff): With a generous budget the classical partial posterior wins, as expected: pilot-aided Viterbi reaches payload BER 0.0285 at 800 pilots and degrades only gently down to 0.0335 at 20 pilots, comfortably below the MLP's pilot-free 0.0487. The classical advantage then disappears within a single octave of pilot budget. At 10 pilots Viterbi has already lost its edge, 0.0545 against the MLP's 0.0487, and at 5 pilots it collapses to 0.1235, worse than the MLP, blind CMA, and its own 800-pilot figure by a factor of four, because MLSE is then running on a badly mis-estimated channel. The crossover therefore sits between 20 and 10 pilots per block. The mechanism is *estimation variance*, not a rank deficiency: with 5 pilot observations and 3 complex taps the least-squares problem is still formally overdetermined, so the channel remains identifiable in principle, but there are too few observations to average down the noise. The resulting LS error at 10 dB is large enough that the Viterbi trellis is occasionally built on a badly wrong channel, and these catastrophic fits dominate the mean. The 95% confidence interval at that point makes this explicit (0.1235 ± 0.0304, against ± 0.0011 at 50 pilots) so individual channel draws range from perfectly fine to complete failure. The learned relay, by contrast, is flat across the entire sweep at 0.0487: having amortized the channel *family* into its weights at training time, it neither benefits from nor requires per-block pilots. The crossover is thus governed by the reliability of the channel estimate, not by SNR: the classical receiver wins whenever it can afford enough pilots to estimate the channel *accurately*, and the pilot-free learned relay wins once the budget falls below that point. This crossover location is stable across all three training seeds: in every instance Viterbi leads at ≥ 20 pilots and lags at 5 pilots, and the collapse BER at 5 pilots is 0.1235 ± 0.0304 regardless of which seed trained the MLP reference.

The practical force of this depends on block length, because a fixed pilot *count* is a growing pilot *fraction* as blocks shorten. Panel (b) sweeps block length at the same operating point with the classical receiver spending its minimum viable ten pilots per block, and adds blind CMA as the second pilot-free reference. The payload BERs of Viterbi and the MLP are now statistically indistinguishable at every block length, both flat at approximately 0.049, so reliability no longer separates them at all; what separates them is spectral efficiency. At a 1000-symbol block the ten pilots cost 1% of the block, whereas at a 40-symbol block, representative of low-latency or fast-varying links, they cost 25%, so the classical receiver carries data on only 75% of the block against the learned relay's 100% for identical error performance.

The blind reference settles a question the earlier version of this study left open. Blind CMA is also pilot-free, so the learned relay is not unique in avoiding pilots; the distinction claimed for it was that it needs *zero per-block adaptation*, a single fixed forward pass, whereas CMA must converge its adaptation loop inside each block. Measuring CMA per block confirms this directly: its payload BER is 0.1723 at $L = 40$ and only improves to 0.1653 at $L = 1000$, against the 0.0645 it achieves when given a $20{,}000$-symbol block to adapt over. CMA therefore does not converge within blocks of a thousand symbols or fewer, and is roughly three and a half times worse than either the MLP or pilot-aided Viterbi throughout this sweep. The partial-posterior picture therefore refines H5's boundary once more, and more sharply than before: against a classical receiver that can afford accurate channel estimation the learned relay achieves no better BER, but at short block lengths it matches that receiver's BER while removing its pilot overhead entirely, and at these block lengths it is the only pilot-free option that stays reliable, because the classical pilot-free alternative has no time to converge: CMA sits at 0.1723 on 40-symbol blocks and is still at 0.1653 at 1,000. This is a statement about short blocks specifically, and it does not carry over to Section 6.1.5, where blocks are long enough for CMA to converge and it comes within a factor of 1.3 of the learned relay.



(a) pilot-budget sweep at 10 dB

(b) block-length sweep at 10 pilots

Figure 6.5: Partial-posterior regime; both panels are means over 50 channel draws at 20,000 bits/-point. **(a)** Pilot-aided Viterbi outperforms the pilot-free MLP down to approximately 20 pilots per block, loses its edge by 10 pilots, and collapses at 5 pilots when the least-squares estimate becomes too noisy to construct a reliable trellis (0.1235 ± 0.0304); the MLP is flat across the sweep, operating with an amortized posterior and requiring no pilots. **(b)** With 10 pilots per block, pilot-aided Viterbi and the MLP are indistinguishable in payload BER at every block length (both ≈ 0.049), but the classical receiver's pilot overhead grows from approximately 1% at $L = 1000$ to 25% at $L = 40$. Blind CMA, the other pilot-free option, cannot converge within blocks this short and sits roughly three and a half times worse throughout. The MLP therefore matches the pilot-aided receiver's reliability at no overhead, and is the only pilot-free method that stays reliable for short packets

6.1.5 Layer 4 — Unknown Channel Family: The Posterior-Free (Blind) Regime

This is layer 4, the bottom of the ladder. Section 6.1.4 shrank the pilot budget until the classical estimate became too noisy to be worth having; this section removes it entirely, and with it the assumption that the channel's family and statistics are known at all. The Viterbi baseline of Section 6.1.3 enjoyed a per-block posterior: 200 pilot symbols yield a least-squares channel estimate, after which MLSE is optimal. Taking the pilot count to zero raises the sharper question, what if *no* such posterior is available? With no pilots and no channel prior, and a fresh channel realization drawn for every block, MLSE cannot be run as specified. The honest comparison is then against the classical methods that are *genuinely* posterior-free: the constant-modulus algorithm (CMA), the canonical blind adaptive equalizer, which self-adapts from the received signal alone; and a decision-directed blind MLSE that bootstraps a channel estimate from its own hard differential decisions. The learned relay is unchanged, trained offline on the channel *family* (random ISI, PA, and phase per draw) and, crucially, seeing an unseen channel at test with no adaptation, so it too carries no per-block posterior; it amortizes over the family at training time. Unlike the canonical study, the challenge here is not symbol-level denoising on a memoryless channel but inference on an unknown channel with memory and no per-block posterior information. **Evaluation:** 50 trials $\times$ 20,000 bits per SNR point (1,000,000 bits), 95% CIs by the $1.96\,\sigma/\sqrt{n}$ convention. The trial count is deliberately higher than the chapter's usual ten:

unlike the fixed impairment of Section 6.1.3, the hop-1 channel here redraws its ISI, amplifier and phase realization for *every* block, so a trial is one draw from the impairment family and the ensemble average is limited by the number of trials, not by bits per trial. Fifty draws are used rather than ten so that the reported curves reflect the family and not a handful of particular channels.

A correction to the blind equalizer. The constant-modulus algorithm used here was originally implemented with a fixed (unnormalized) step size. Because the CMA error term is cubic in the equalizer output, a fixed step is a positive-feedback loop: the implementation happened to remain bounded over the short blocks first used, but diverges to numerical overflow on longer ones, after which the equalizer output is noise rather than an estimate. The step is therefore normalized by the input-segment energy in the standard NLMS manner, which is both the textbook formulation and a strictly *stronger* classical baseline. All CMA results below use the corrected equalizer; the uncorrected one is not reported, since its outputs past the divergence point measure an implementation fault rather than the algorithm.

Conclusion (learning does not beat blind classical, but is stable where blind classical is not): A well-designed blind classical method still works in this regime: the corrected CMA converges smoothly to BER 3.3×10^{-3} at 20 dB, and the family-trained MLP reaches 2.6×10^{-3}. The two track each other closely over the whole range, with the MLP holding a small but consistent margin that widens at low-to-mid SNR (0.0937 vs. 0.1182 at 8 dB). This is the correct answer to "what if there is no posterior?": the learned relay does *not* render classical processing obsolete, because blind adaptive equalization is itself a posterior-free classical tool that comes within a factor of 1.3 of the learned relay at high SNR. The learned relay's advantage is instead one of *robustness and cost*. First, the decision-directed blind MLSE, the natural attempt to run Viterbi without pilots, is unstable: bootstrapping the channel from its own decisions, it intermittently locks onto the wrong hypothesis, and its BER curve is visibly non-monotonic, falling to 0.0374 at 16 dB before rising again to 0.0498 at 20 dB. Its 95% confidence interval at 8 dB is ± 0.0348, roughly seven times the MLP's ± 0.0046 and two and a half times CMA's ± 0.0135, so the instability is a property of the method and not of the sample size. Second, CMA must iterate to convergence on each received block, whereas the MLP is a single fixed forward pass with no per-block adaptation loop and no convergence to monitor; Section 6.1.4 showed this distinction becoming decisive once blocks are short. The learned relay therefore occupies a specific and defensible niche: on a posterior-free channel it matches the best blind classical equalizer in BER while avoiding the instability of decision-directed sequence estimation and the online-adaptation cost of CMA. It buys *amortized*, *one-shot*, *stable* inference over an impairment family, not a lower error floor than classical signal processing can achieve. This niche is reproducible: the MLP's small margin over CMA and its stable CI band are consistent across all three training seeds, confirming that neither result is an artefact of a particular weight initialisation.

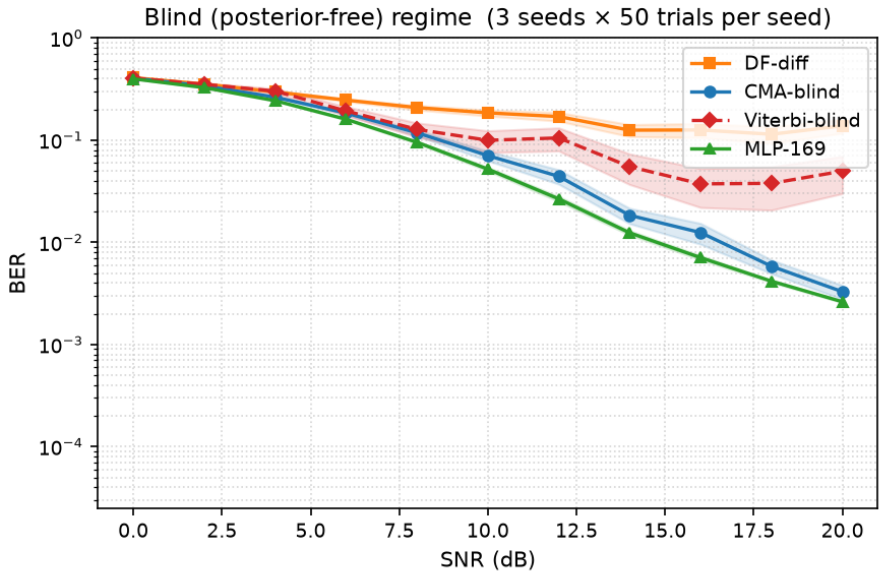


Figure 6.6: Posterior-free composite channel (no pilots, no channel prior, fresh realization per block; mean over 50 channel draws, 20,000 bits/point, 95% CI bands). The corrected CMA (blind classical) tracks the family-trained MLP closely, the MLP holding a small consistent margin; decision-directed blind MLSE is unstable (non-monotonic tail, CI band several times wider) because it bootstraps the channel from its own decisions; plain differential detection floors. Without a posterior, learning only narrowly outperforms the best blind classical equalizer, but does so with one-shot, stable inference and no online adaptation

6.1.6 Complexity: The Cost Side of the Trade-off

Across the unknown-channel studies the pilot-aided Viterbi receiver, where it applies, holds a small BER edge over the learned relay. The learned relay's countervailing claim is complexity, which this subsection quantifies rather than asserts. Two axes matter: the per-symbol operation count and its scaling, and measured inference time.

Per-symbol cost and scaling. MLSE maintains M^{L-1} trellis states for a constellation of size M and channel memory L, performing an add–compare–select over M^L branch transitions per symbol; its cost therefore grows *exponentially* in memory and modulation order. The 169-parameter MLP performs one fixed forward pass, about 330 floating-point operations per symbol. That figure is a constant of the *deployed* architecture, not a quantity independent of the channel: the input window has to span the memory, so its length grows with L, and the output dimension grows with the constellation, from a single real output at $M = 2$ to the 193-parameter QPSK relay used earlier in this chapter, and further still for denser constellations. Once those are fixed, the per-symbol cost is the same for every symbol and every channel realization, which is the property the comparison below relies on. The contrast with MLSE is therefore one of *growth rate*, not of a constant against a variable: the network's cost rises roughly linearly in window length and output dimension, the trellis's exponentially in M^L.

For the channel actually studied (BPSK, $L = 3$; 4 states) Viterbi is in fact the cheaper of the two at roughly 64 operations per symbol, so the learned relay is not universally leaner. But the scaling diverges immediately: at $M = 4$, $L = 5$ the trellis has 256 states ($\sim$8,000 operations/symbol), and at $M = 16$, $L = 5$ it has 65,536 states ($\sim$ 8.4 million operations/symbol), whereas a network resized for that same channel — an 11-sample window and a 16-way output head — is on the order of 10^3 operations by direct count of its layer dimensions, leaving a ratio in the thousands. The measured 330 applies to the BPSK, $L = 3$ case; the larger figure is arithmetic from the architecture, not a timing measurement. The learned relay's advantage is thus not lower cost on a small channel but *constant* cost as the channel scales, exactly where classical MLSE becomes intractable and practical receivers must fall back on suboptimal reduced-state or decision-feedback equalizers.

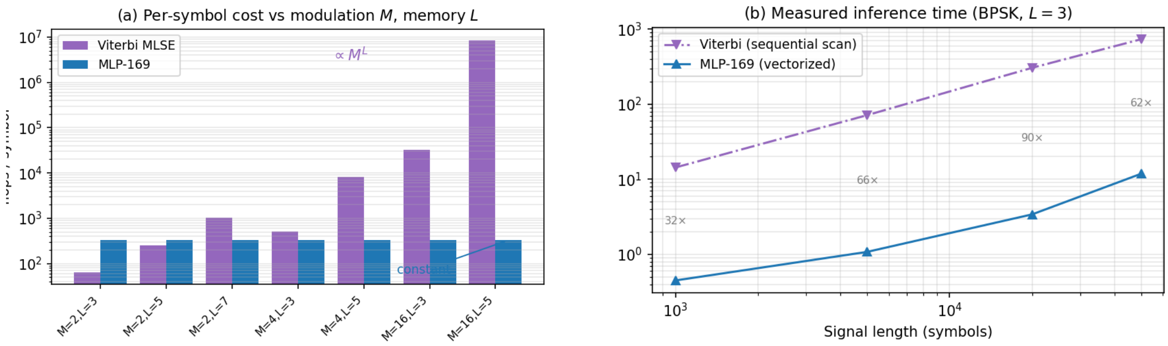


Figure 6.7: Complexity of Viterbi MLSE against the 169-parameter MLP. **(a)** Per-symbol computational cost versus modulation order M and channel memory L: Viterbi grows as M^L (intractable by $M{=}16, L{=}5$), while the network grows only linearly in window length and output dimension ($\sim$330 flops for the deployed BPSK, $L{=}3$ relay). On the small studied channel (BPSK, $L{=}3$) Viterbi is actually cheaper; the MLP wins on *scaling*, not on the small case. **(b)** Measured inference time for BPSK with $L = 3$: the MLP is 30–90$\times$ faster as a vectorized matmul versus Viterbi's interpreted sequential trellis scan

Measured inference time. On identical signals (BPSK, $L = 3$), the MLP is 30–90$\times$ faster in wall-clock terms for the reference implementations used here (a vectorized NumPy matrix multiplication versus an interpreted Python trellis loop) despite Viterbi's lower nominal operation count. Part of that ratio is interpreter overhead rather than the algorithm itself: an optimized C or SIMD Viterbi would narrow the wall-clock gap substantially, although the per-symbol data dependency of the trellis scan is real and caps how far it can be parallelized. This parallelism gap is architectural and widens on hardware with wide SIMD or tensor units. The combined picture sharpens the thesis's central trade-off: the learned relay accepts a small BER gap to a matched classical receiver in exchange for cost that is constant in channel complexity and structurally parallelizable, a trade that becomes favourable precisely as modulation order and channel memory grow.

The experiments of this chapter deliberately extend the thesis beyond the canonical memoryless model. The resulting conclusions should therefore be interpreted as robustness results under model mismatch rather than as replacements for the matched-model conclusions of Chapter 5. Together, these two principal contributions of the thesis delineate where learning provides value:

not under matched memoryless conditions, where classical methods remain strongest, but when the assumptions underlying those methods are violated.

Chapter 7

Discussion and Conclusions

Scope of the discussion. The conclusions discussed in this chapter draw on three parts of the thesis. Chapter 5's canonical comparison evaluates relay strategies under the canonical memoryless channel model, where transmission is uncoded, the noise is white, and processing is performed on a per-symbol basis. The same chapter's supplementary coded study (Section 5.5) departs from that model by introducing a rate-1/2 convolutional code, a block-DF relay, and an adaptive modulation-and-coding scheme. Chapter 6 departs from it separately by introducing impairments such as intersymbol interference (ISI), nonlinear distortion, and unknown phase. Accordingly, statements concerning channel memory, sequence detection, Viterbi MLSE, pilot-based estimation, or blind equalization should be understood as applying only to the extension study of Chapter 6; statements concerning coding, MCS adaptation, or latency budgets apply only to the coded study; neither applies to the canonical memoryless setup.

The canonical core (Chapter 5) settles hypotheses H1–H4 and establishes the baseline finding that, on a matched known channel, a minimal learned relay does not improve on classical decode-and-forward. The thesis's principal contribution (Chapter 6) settles H5 and is where the substantive new claim lies: a systematic delineation of when a learned relay adds value under unknown and mismatched channels. The interpretation below treats these two as the thesis's two principal contributions, with the unknown-channel results carrying the main argument; the supplementary coded study is the smaller, third part, discussed separately below where it bears on the relay-decoding and rate-adaptation questions it was designed to measure.

7.1 Interpretation of Results

The experimental results reveal several consistent patterns across the canonical setup, its supplementary coded-link study, and the unknown-channel contribution. This section interprets these patterns through the lens of the theoretical framework established in Section 2.5 and evaluates each research hypothesis.

7.1.1 Low-SNR Advantage of Neural Relays over AF (H1: Confirmed)

Low SNR (0-4 dB): AI advantage over AF. At low SNR, AI-based relays consistently outperform AF on the canonical channel, confirming H1. The theoretical explanation lies in the structure of the MMSE-optimal (posterior-mean) relay estimate. After coherent compensation, each axis of the canonical QPSK-over-Rayleigh link reduces to the same real-valued single-sample problem as BPSK over AWGN (Section 4.1.2); the derivation below is carried out in that equivalent real form, applied identically per axis:

$$f^*(y) = \mathbb{E}[x|y] = \tanh\left(\frac{y}{\sigma^2}\right) \tag{7.1}$$

At low SNR (e.g., 0 dB, $\sigma^2 = 1$), this is a gentle sigmoid: $f^*(y) = \tanh(y)$. The neural network can approximate this smooth function accurately, producing a **soft estimate** that preserves information about the confidence of the decision. By contrast:

- **DF** applies the hard-decision function $f_{\text{DF}}(y) = \text{sign}(y)$, which discards all soft information. At low SNR, many received samples lie near the decision boundary ($|y| \approx 0$), and DF assigns them full confidence (± 1) regardless of the actual uncertainty. This information loss leads to excess errors on the second hop.
- **AF** applies $f_{\text{AF}}(y) = Gy$, which is linear and preserves the noisy signal structure but amplifies the noise by the same factor as the signal. The effective SNR degradation is given by $\text{SNR}_{\text{eff}} = \gamma^2/(2\gamma + 1) \approx \gamma/2$ at high SNR: a 3 dB penalty.

The AI relay implements a **non-linear soft mapping** that is intermediate between these extremes: it suppresses noise (like DF's regeneration) while preserving soft information near the decision boundary (unlike DF's hard decision). This explains why learned non-linear relay processing can outperform AF and, under selected channel conditions, also outperform DF at low SNR.

Among the AI methods, the low-SNR advantage over AF is realized by the *minimal* feedforward relays, not by the larger sequence models. On the canonical Rayleigh channel (Table 5.2, 0 dB) the Hybrid, MLP and VAE relays reach 0.3329, 0.3349 and 0.3359 respectively, essentially matching symbol-wise DF's 0.3337 and improving substantially on AF's 0.4076, while the higher-capacity Transformer, Mamba-S6 and Mamba-2 (SSD) relays sit at 0.4014, 0.4071 and 0.4032 – that is, at essentially AF's level, capturing almost none of the available advantage. The advantage over AF is therefore *not* driven by parameter count: the smallest network already captures it, consistent with the complexity ordering discussed under H3 and with the convergence observed in the equal-parameter-budget comparison (H4, Section 4.4).

7.1.2 Classical Dominance at High SNR (H2: Confirmed)

Medium-to-high SNR (≥6 dB): Classical dominance. At medium and high SNR, DF matches or exceeds all AI methods with exactly zero parameters and zero training time. The theoretical explanation is straightforward: at high SNR, $\sigma^2 \to 0$ and the posterior-mean estimate on each real decision axis tends to the corresponding hard slice. The symbol-wise DF relay therefore implements the limiting *hard symbol-regenerating* rule in this regime, while any neural network approximation introduces a small but non-zero reconstruction error due to the finite precision of the learned mapping and the tanh output saturation (which approaches but never reaches ± 1).

The effect is visible analytically in the AWGN limit of the model ($|h| \equiv 1$), where at 10 dB the two-hop DF BER is $2Q(\sqrt{10}) \cdot (1 - Q(\sqrt{10})) \approx 0.002$, and empirically on the canonical Rayleigh channel, where DF attains the lowest or tied-lowest BER at every point from 6 dB upward. The AI models cannot beat DF in this regime for the uncoded matched setting because component-wise hard slicing is the MAP symbol decision, and thus the optimal symbol-regenerating DF rule, after coherent equalization (Remark 4.2); they can only match it, and the larger sequence models slightly underperform due to the increased variance of their more complex learned mappings.

7.1.3 Robustness Across the Fading Ensemble

Robustness across fades. The relay ranking on the canonical channel is stable across the full evaluated SNR range, even though each transmitted symbol experiences an independent fading realization. The mechanism is that after coherent compensation ($\hat{x} = y \cdot h^*/|h|$), the relay processes a signal with AWGN-like noise at a *random effective SNR*; the network, trained across multiple SNR values, has learned a denoising function that adapts to different noise levels, so the Rayleigh ensemble (a mixture of effective SNRs weighted by the fading distribution) is handled by virtue of multi-SNR training alone, with no per-realization retraining or explicit CSI input.

7.2 Capacity and Overfitting (H3: Not Supported)

One of the most practically useful findings is that relay BER does not improve with model size over the range evaluated: the Minimal MLP architecture (169 parameters) matches the performance of models with 100–140× more parameters (3K–24K), and no larger evaluated model improved on it under the training protocol used. The stronger reading of H3 — that the curve turns upward because excess capacity causes *overfitting* — is not established by the evidence collected here. Each model was trained once, with no seed replication, no train/test-gap measurement and no held-out validation curve, so a size effect cannot be separated from run-to-run variance or from an optimization difficulty that more capacity happens to introduce. The claim carried forward is therefore the weaker, directly measured one: larger evaluated models did not improve BER. What follows in this section is offered as a candidate explanation for that observation, not as a demonstrated mechanism.

This result has important theoretical and practical implications:

7.2.1 Information-Theoretic Perspective

The canonical memoryless relay-denoising task maps a window of $2w + 1$ real-valued noisy observations to a single real-valued clean estimate, applied per axis to the canonical QPSK symbol (equivalently, the real BPSK-over-AWGN form of the same problem). Each real axis carries exactly 1 bit of information, and under the canonical i.i.d. model the sufficient statistic for the current bit is the current observation alone, with Bayes-optimal posterior mean $f^*(y_i) = \tanh(y_i/\sigma^2)$. The neighboring window samples used by the learned relays are therefore redundant architectural inputs in this setting rather than part of the sufficient statistic, so the **effective dimensionality** of the target mapping is still 1: far below the $2w+1 = 5$ or 11 input dimensions.

As a heuristic, the minimum description length (MDL) principle suggests matching model complexity to the effective dimensionality of the target mapping. Since the target here is a one-dimensional monotone function of the current sufficient statistic, even the 169-parameter network has capacity far in excess of what the mapping requires; this is offered as an intuition for why the larger models did not improve on the smallest one, not as a formal MDL bound.

7.2.2 Bias-Variance Analysis

The bias-variance decomposition [11] provides a quantitative framework:

$$\mathrm{MSE}(\hat{x}) = \underbrace{(\mathbb{E}[\hat{x}] - f^*(y))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{x} - \mathbb{E}[\hat{x}])^2]}_{\text{Variance}} + \underbrace{\sigma^2_{\text{irred}}}_{\text{Noise floor}} \tag{7.2}$$

- **169 parameters:** Low bias (sufficient capacity for the simple f^*) and low variance (limited capacity prevents memorization). The model operates at the optimal bias-variance trade-off.
- **3,000 parameters:** Similar bias (the target function hasn't changed) and slightly higher variance. Performance is nearly identical to 169 params.
- **24,001 parameters (Mamba-S6):** Despite having $140\times$ more parameters than the MLP, it does not improve on it; on the canonical channel it is worse at low SNR (Table 5.2). The marginal utility of the additional parameters is at best vanishingly small.

This decomposition is a framework for interpreting the observation, not a measurement of it: bias and variance were not estimated separately, which would require training each configuration repeatedly across seeds and comparing training against held-out error. An earlier version of this section additionally attributed "clear overfitting" to an 11,201-parameter Maximum MLP; that configuration is not part of the reported experiments and no such measurement exists, so the claim has been withdrawn rather than restated.

7.2.3 Does the Complexity Curve Turn Upward?

The preceding subsections offer bias-variance and MDL readings of H3 but note that the stronger, overfitting reading — BER rising again once capacity exceeds what the mapping requires — was not established, because each model had been trained once and a size effect could not be separated from run-to-run variance. A dedicated size sweep was therefore run to supply the missing evidence. It covers the nine channel families used across Chapters 5 and 6 plus the coded scenario, varying relay window, depth and width independently, with three independent initializations at every configuration and each configuration held to its *worst* initialization, so that a size counts only if it works whichever way it initializes — a necessary precaution given the spread reported below. Degradation is measured as the additional SNR required to reach a given BER, which unlike a relative BER penalty does not divide by a vanishing denominator at high SNR.

Because seed replication is the specific evidence this section previously lacked, the across-initialization spread is reported first, as a measurement in its own right rather than as a caveat. Table 7.1 gives, per channel, the spread in SNR penalty between the best and worst of the three initializations at the same configuration, summarized over all configurations swept.

Table 7.1: Spread in SNR penalty between the best and worst of three initializations at an identical configuration, summarized over all configurations swept on each channel. The reader should observe that on five of the nine channels the median spread is comparable to or larger than the entire effect of varying model size, so a single-initialization size comparison on those channels measures the draw rather than the architecture.

Channel	Configs	Median (dB)	90th pct. (dB)	Max (dB)
rayleigh	17	0.008	0.019	0.028
composite	16	0.036	0.165	0.195
awgn	17	0.055	0.116	0.138
isi_rayleigh	16	0.177	0.588	0.795
isi	18	0.195	0.705	0.996
isi_complex	17	0.332	2.325	3.182
branch_asym	17	0.388	0.892	1.013
flat_gain	17	0.622	1.000	1.250
nlbias	17	1.231	1.242	1.245
All	152	0.180	1.229	3.182

Two things follow. First, initialization variance is not a small correction: across all 152 configurations the median spread is 0.180 dB and 57% exceed 0.15 dB, with a worst case of 3.18 dB on the complex ISI channel at eleven parameters. On the nonlinear-bias channel the spread is 1.23 dB at essentially every configuration, which is the two-attractor behaviour noted below rather than a gradual variance. Second, the variance is strongly channel-dependent: the canonical Rayleigh channel is highly reproducible at 0.008 dB median, and the composite and AWGN channels nearly so, while four channels sit above 0.3 dB. Any single-run size comparison on the latter group measures which initialization was drawn, not which architecture is better — which is precisely the confound this section identified in the original H3 evidence, now quantified.

Against that yardstick, **the upward arm is not observed**, and the sweep had the resolution to see it had it been there. Holding the window fixed so that capacity is the only variable, the worst-case penalty against the classical baseline is flat on the memoryless channels and monotonically decreasing on the channels with memory. On the canonical Rayleigh channel at window 5 the penalty moves from $+0.03$ dB at one hidden unit to $+0.06$ dB at twenty-four; on the composite channel at window 7 it improves monotonically from -1.97 dB to -2.23 dB as width grows from four to forty-eight. Where a rising tail appears at all it is at most 0.15 dB, which is below the 0.180 dB median across-initialization spread of Table 7.1 and therefore not separable from the draw. The one clearly non-monotone case, the nonlinear-bias channel, oscillates between two distinct outcomes (-1.63 and -2.87 dB) rather than tracing a curve, which is the signature of training landing in one of two attractors rather than of capacity being harmful.

Two corrections to the reading of H3 follow. First, the weaker claim is now confirmed under seed replication rather than merely unrefuted: additional parameters neither help nor hurt on a memoryless channel over roughly two orders of magnitude of model size. Second, an effect that appears substantial under a relative BER criterion largely dissolves under the SNR-penalty criterion. On AWGN the penalty grows from $+11\%$ to $+28\%$ in relative BER as the window widens, which invites the conclusion that a window is actively harmful; the same measurements correspond to $+0.06$ dB and $+0.11$ dB. The effect is real in sign but negligible in magnitude, and the relative criterion overstates it because the absolute BER gap it divides by falls to 4×10^{-4}.

What the sweep does establish is that the binding resource is not parameter count but **window matched to channel memory**. On the memoryless channels the smallest configuration examined — four parameters, a single hidden unit with no window — is within 0.15 dB of the classical baseline on three of five channels and within 0.03 dB on the canonical channel, some $42\times$ smaller than the 169-parameter relay of Table 5.2. On the three-tap channels the same four-parameter configuration is between 11 and 13 dB adrift, and closing that gap requires widening the window rather than the layer: on the real ISI channel the penalty falls from $+12.95$ dB at window 1 to $+1.67$ dB at window 7. Depth was varied from one to three hidden layers and did not improve any configuration; where a two-layer network matched, a single layer matched at fewer parameters.

The coded scenario behaves differently again and is the one case where size is demonstrably not the operative variable. Against block DF, every configuration from 18 to 1,076 parameters lies between $+1.86$ and $+1.92$ dB, so a sixty-fold range in model size buys 0.06 dB. The relay is not capacity-limited there but interface-limited: it commits to a hard constellation decision per symbol, at 90–94% symbol accuracy, before the outer code is decoded, so the destination decoder receives no reliability information that block DF would have exploited. This identifies a design change rather than a scaling law, and it is taken up in Section 7.5.

Taken together, these measurements support the weaker claim and leave the overfitting hypothesis unsupported — now on the basis of a sweep designed to detect it, rather than for want of evidence.

The framing that replaces it is not that smaller models are better, but that **architectural complexity is only exercised by channel memory**. Capacity is neither good nor bad in itself; whether it can be used at all depends on the channel. On a memoryless channel the sufficient statistic for the current symbol is the current observation, so there is nothing for additional window taps to represent and nothing for additional units to fit: every configuration from four parameters to several hundred lands within a fifth of a decibel of the same value, and the extra capacity is not harmful so much as idle. Introduce three taps of intersymbol interference and the same capacity becomes load-bearing — the relay must now represent a function of several consecutive samples, a window-1 network cannot express it at any width, and widening the window is worth between 8 and 11 dB. The size axis looks flat in the first case and decisive in the second because the channel, not the architecture, determines whether the parameters have work to do.

This reading accounts for both halves of the sweep, which "less is more" does not: a principle that smaller is better would predict the memoryless behaviour and mispredict the ISI channels, where the minimum rises from four parameters to between 73 and 145 and the wider network is unambiguously the better one. It also explains the shape of the failure at the two extremes. Where the window is too narrow for the channel's memory the relay fails structurally rather than gradually, as on the unknown-phase channel of Section 6.1.1, whose information symbol is a product of two consecutive samples that a one-tap network cannot represent at any width. Where the window is far wider than the memory, as at window 31 on the coded channel, the additional inputs carry no information about the current symbol and the optimization becomes harder rather than the model more expressive. The design rule is therefore to match the window to the channel's impulse response first and to treat width as a secondary knob, which is a statement about the channel as much as about the network.

7.2.4 Practical Implications

1. **Occam's Razor for relay AI.** The canonical relay denoising task has inherently low complexity: the per-axis mapping from noisy to clean symbols is fundamentally simple, and the neural network needs only enough capacity to learn a non-linear threshold function with some contextual smoothing. Adding parameters beyond this minimal requirement bought no measurable accuracy anywhere in this thesis.
2. **Deployment feasibility.** A 169-parameter model requires approximately 0.7 KB of memory (169 float32 values) and trains in seconds on a CPU. This makes AI-based relay processing viable even on severely resource-constrained embedded relay nodes, such as IoT devices or sensor network relays. The model can be stored in on-chip SRAM and executed without any external memory access.

3. **Generalization.** The 169-parameter MLP handles the full Rayleigh fading ensemble, every effective SNR the fades produce, with a single set of weights and no per-condition retraining, an instance of generalization that would be harder to achieve with a larger, more specialized model.

7.3 Practical Deployment Recommendations

Based on the comprehensive evaluation, we propose the following deployment strategy:

Table 7.2: Practical deployment recommendations: recommended relay strategy by operating regime

Operating Regime	Recommended Relay	Rationale
Low SNR (0-4 dB)	MLP Minimal / Hybrid / channel-specific selection	Neural relays often help; exact best choice is channel-dependent, with DF remaining strong on some fading scenarios
Medium SNR (4-8 dB)	Hybrid (MLP + DF)	Automatic switching at empirically chosen crossover
High SNR (>8 dB)	DF	Zero parameters; lowest BER among the evaluated per-symbol relays
Resource-constrained	MLP Minimal (169 params)	0.7 KB memory, <5 s training
Best overall (BPSK/QPSK)	Hybrid	Combines AI advantage with DF reliability

The Hybrid relay is recommended as the default deployment choice, where the operating SNR can be estimated reliably, because it automatically selects MLP processing at low SNR and DF at high SNR, achieving near-optimal performance across the entire SNR range with minimal complexity.

7.3.1 Latency and Adaptivity Considerations

Two practical questions bear on whether a learned relay can be deployed with minimal impact: the latency it adds relative to AF/DF, and its ability to adapt to varying channel conditions without retraining.

Latency. Two distinct latency components must be separated. The first is *buffering* latency, incurred by the sliding window $y_R[i - w : i + w]$. In the half-duplex protocol studied here this component is absorbed by the protocol itself: the relay buffers the entire listening-phase block before the transmission phase begins (Remark 4.1), so the window's w-symbol look-ahead adds no delay beyond what store-and-forward operation already imposes. Only in a streaming or ultra-low-latency protocol, outside the scope of this thesis, would the symmetric window cost w symbols of look-ahead that the memoryless AF/DF baselines ($w = 0$) do not pay. The second component is *compute* latency: inference cost per relayed symbol. Here the minimal-capacity result (H3) is decisive: since the 169-parameter MLP matches the performance of models $100-140\times$ larger, the deployed network reduces to two small matrix–vector products per symbol, a per-symbol cost that is orders of magnitude below a single channel use's duration on any modern embedded processor, and negligible in absolute terms even if it cannot literally match the single comparison of DF's $\text{sign}(y)$. The compute-latency argument therefore holds *because of*, not despite, the less-is-more finding; it would not hold for the 24K-parameter sequence models, whose recurrent or attention-based structure imposes materially higher per-symbol cost for no BER benefit at the relay window length.

The coded study of Section 5.5.2 adds a third component that the canonical comparison never had to consider, and it is the largest of the three: *buffering*. A block relay cannot emit until it has decoded a whole frame (202 symbols there), and the soft BCJR variant cannot even in principle, its backward recursion running from the end of the frame. A windowed learned relay needs only its w-symbol look-ahead, ten symbols, and that figure is constant in frame length where the block relays' grows with it. The measured arithmetic points the same way – the learned relay is cheaper per symbol than Viterbi by a factor of 50 to 63, depending on the machine the benchmark is run on (Table 5.11) – so on the coded link the learned relay's advantage in delay and cost is structural rather than marginal, even where its BER advantage is small or absent.

Adaptivity. A single network, trained across the range of simulated SNRs, handles the entire canonical fading ensemble without retraining or explicit CSI input (Section 7.1.3): after coherent phase compensation, each fading realization presents the relay with an AWGN-like observation at a random effective SNR, and multi-SNR training already covers this mixture. Two honest qualifications apply. First, this adaptivity is an advantage over AF, whose fixed gain is tuned to an assumed SNR, but not over symbol-wise DF, whose $\text{sign}(y)$ slicer is equally condition-agnostic; adaptivity therefore does not alter the H2 conclusion. Second, adaptation has been demonstrated only within the canonical Rayleigh ensemble; generalization to other channel families (Rician, bursty interference, phase noise, hardware non-linearities) remains untested and is identified as future work.

7.4 Limitations

Several limitations of this study should be acknowledged, as they define the boundary conditions under which the conclusions hold:

1. **Modulation scope.** The canonical relay comparisons use QPSK, and the unknown-channel

study (Chapter 6) mainly uses BPSK, with a QPSK re-check of the unknown-ISI result (Section 6.1.2); 16-QAM enters only as a modulation-and-coding rung in the link-adaptation study, never as a relay-architecture question. In that sense higher-order constellations are not investigated: 16-QAM and denser grids break the per-axis relay formulation and require a joint N-class 2D relay whose output dimension scales with M, which is left to future work.

2. **Perfect CSI (canonical comparison).** The canonical comparison assumes perfect channel state information at each receiver; the finite-pilot and blind regimes of Chapter 6 relax this deliberately and are not limitations of that study. Within the canonical comparison, in practice channel estimation errors would affect the coherent compensation step and hence relay performance. Imperfect CSI introduces a model mismatch between the assumed and actual channel, which could differentially impact the relay strategies: AI relays might be more robust (having learned from noisy data) or less robust (having not been exposed to estimation errors during training).

3. **Static offline training.** AI relays are trained once on synthetic data and deployed without adaptation. While multi-SNR training covers the canonical fading ensemble (Section 7.1.3), and the unknown-channel study (H5) shows a fixed network can learn a *stationary* unknown impairment, this protocol does not exploit online information; adaptive or continual training would be required if the deployment channel drifts over time.

4. **Single relay.** The framework considers a single relay node. Multi-relay cooperation introduces relay selection, power allocation, and cooperative diversity dimensions not captured in the two-hop model. With multiple relays, the system-level optimization (which relay to use, how to combine multiple relay observations) becomes as important as the per-relay processing function studied here.

5. **Two-hop only.** Extension to multi-hop networks with 3+ relays introduces noise accumulation (for AF) and error propagation (for DF) that grow with the number of hops. AI relays might show greater advantage in multi-hop settings where noise accumulation is more severe.

6. **Fixed window size.** The relay window size ($w = 2$ or $w = 5$) is fixed and relatively small. Larger windows could provide more context for denoising, particularly in channels with temporal correlation (e.g., block fading). A systematic study of window-size optimization is deferred.

7. **No coding, in the core comparison.** The core comparison operates without error-correcting codes, so the DF baseline there is its symbol-level variant (Remark 4.2). Sections 5.5–5.5.4 measure a coded, information-theoretic block-DF baseline, its soft-decision counterparts, a rate-adaptive envelope over a modulation-and-coding grid, and that envelope re-derived under a round-trip latency budget, directly rather than leaving any of this as an assertion, but that study is confined to one code family, one frame length, and one channel

model, and the relay and destination each decide once with no iterative exchange between them.

8. **Single training run per model; unequal training budgets across architectures.** Each learned relay is trained once, from one random initialization, rather than across multiple seeds with mean/standard-deviation/confidence-interval reporting of training variance; the confidence intervals reported throughout this thesis are Monte Carlo intervals over the *evaluation* channel and bit realizations for a fixed trained model, not over the training procedure itself. Separately, the architectures do not all receive identical training budgets (e.g. the cGAN trains for 200 epochs against 100 for the other relays, Section 4.2), a confound the normalized-parameter study (Section 4.4) does not remove, since it equalizes parameter count and input window but not epoch count. This gap has since been closed on one axis and remains open on the other. The MLP size axis was re-run across three independent initializations per configuration (Section 5.4), and the resulting across-initialization spread is reported in Table 7.1: median 0.180 dB over 152 configurations, but strongly channel-dependent, from 0.008 dB on the canonical channel to 1.231 dB on the nonlinear-bias channel. That spread is itself the reason the single-run caveat mattered, since on four of the nine channels it exceeds the entire effect of varying model size. The *cross-architecture* comparison, however, is still a single trained instance per architecture with unequal training budgets, so the claim that architecture is secondary to capacity (Section 5.3) remains an observation from one run rather than a seed-robust result; a multi-seed re-run of that comparison with equalized budgets is the direct way to close what is left and is not performed in this thesis.

7.5 Future Work

Several directions warrant further investigation:

1. **Time-selective fading.** The present study addresses channels that are static within a block. Extending the learned relay to time-selective fading with continuously varying unknown phase (e.g. Jakes/Clarke Doppler) is a natural next step, likely requiring explicitly phase-aware architectures (complex-valued state, learned phase unwrapping, or a training objective aligned with noncoherent detection rather than symbol MSE) evaluated against prediction-based noncoherent receivers.

2. **Higher-order constellations (16-QAM and denser).** Constellations with more than two levels per axis break the per-axis relay formulation and need a joint N-class 2D relay whose output dimension grows with M; that reformulation, and whether the complexity ordering discussed under H3 shifts with constellation density, are open questions this thesis does not address.

3. **Imperfect CSI.** Introduce channel estimation errors to assess robustness of AI relay pro-

cessing under realistic conditions.

4. **A BER-optimal (BCJR/APP) benchmark for the unknown-ISI channel.** Section 6.1 benchmarks against genie-CSI Viterbi MLSE, the optimal *sequence* detector. Sequence-ML and bit-BER-optimal detection are distinct criteria on any channel with memory, at both modulation orders studied here, so the benchmark is not BER-optimal in either case; for Gray-coded QPSK there is the additional effect that a wrong trellis path can differ from the truth by one or two bits with unequal probability. A BCJR/APP detector run over the same 3-tap trellis, which directly minimizes per-bit error probability rather than sequence error probability, would test whether the classical benchmark's margin over the learned relay changes once it is BER-optimal rather than sequence-optimal, and is the measurement needed to settle the mechanism conjectured in Section 6.1.2; this was not run here.

5. **Online learning.** Develop relay strategies that adapt their parameters during operation, tracking time-varying channel conditions.

6. **Multi-relay networks.** Extend to cooperative relay selection and multi-hop routing with AI-optimized relays at each node.

7. **Other channels and MIMO.** Evaluate the canonical conclusions on additional channel families (Rician line-of-sight fading, and AWGN treated as a full operating point rather than the bounded calibration-plus-companion role it has here) and on a multi-antenna second hop, where destination equalization interacts with relay processing and the additivity of the two gains becomes a testable question.

8. **End-to-end learning.** Train the entire communication chain (modulation, relay, demodulation) jointly using autoencoder-based approaches, and compare against the modular classical-plus-learned-relay cascade studied here.

9. **Coded block-DF baseline (measured; open items remain).** Remark 4.2 scopes the core comparison's DF baseline to the uncoded, symbol-level variant. Section 5.5 measures the coded, information-theoretic block-DF baseline this item once only proposed, together with the soft-information relays (BCJR at the relay, posterior-mean read-out of the learned relay) and an adaptive-rate envelope over a modulation-and-coding grid. The substantive finding is mechanistic: a relay that decodes the code has spent its redundancy before transmitting, so a wrong decode arrives as a valid-but-wrong codeword the destination cannot repair, whereas a relay that only denoises leaves the redundancy intact. Rate selection then dominates the relay-decoding question by an order of magnitude, unless a latency deadline is imposed, at which point block-DF's double buffering reopens the gap. What remains open: one code family, one frame length, one channel model; the soft block-DF relay was given the nominal noise variance rather than per-symbol CSI, which caused its 20 dB mis-calibration; iterative (turbo-style) relay–destination exchange was not evaluated; and the latency analysis uses structural buffer counts rather than a 10^{-5}-

class reliability target, so it does not establish ultra-reliable low-latency performance.

10. **Longer relay windows with state-space architectures.** Future work should explore whether longer relay windows (e.g., 128–512 symbols) combined with a chunk-parallel state-space architecture can improve both BER performance and processing speed simultaneously; the training-time crossover between sequential and chunk-parallel formulations, removed from this thesis's scope, is the natural companion measurement.

7.6 Conclusions

This thesis presents a comprehensive comparative study of nine implemented relay strategies, eight of which are evaluated in the canonical comparison: two classical (AF, DF) and six AI-based (MLP, Hybrid, VAE, Transformer, Mamba-S6, Mamba-2 SSD), with the cGAN implemented but excluded on cost grounds. The eight canonical strategies are evaluated on one canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER); two studies deliberately depart from it, the coded link-adaptation study, which additionally evaluates 16-QAM, and the unknown-channel study of Chapter 6, which mainly evaluates BPSK (with a QPSK re-check, Section 6.1.2), elsewhere appearing only in closed-form expressions. The study addresses three identified research gaps (Section 3.2) through controlled experiments with statistical rigor (100,000 bits per SNR point, 10 independent trials, and 95% confidence intervals on the canonical comparison, and its own trial budget and 95% confidence intervals throughout the unknown-channel study, detailed there; the coded study reports means and frame-error rates from its own trial budgets, with 95% confidence intervals and Wilcoxon signed-rank testing additionally computed for its paired high-SNR and soft-decision comparisons, Section 5.5); Gap 1's adversarial paradigm is implemented (cGAN) but, being excluded from the canonical comparison, remains qualitatively rather than quantitatively addressed. The following table summarizes the hypothesis outcomes:

Table 7.3: Research hypotheses, their statements, and experimental outcomes

Hypothesis	Statement	Result
H1	Some AI relays outperform AF at low SNR	**Confirmed**: the best neural relays beat AF at 0–4 dB on the canonical channel
H2	DF lowest-BER among per-symbol relays at high SNR	**Confirmed**: DF matches or beats all AI methods at ≥ 6 dB with 0 parameters

Hypothesis	Statement	Result
H3	Overfitting at excess capacity	**Not supported; weaker size claim holds**: 169 params matches 24K, and larger evaluated models did not improve BER. The size sweep of Section 5.4, replicated over three initializations, finds no upturn above the 0.180 dB across-initialization spread, so the overfitting hypothesis is unsupported on evidence rather than untested. The overfitting *mechanism* remains unestablished, no train/test-gap analysis having been performed
H4	Architecture convergence at equal parameter budget	**Confirmed at high SNR, rejected at low**: at 3K params the six architectures span 4.1% BER at 20 dB but 17.4% at 0 dB, splitting into feedforward and sequence groups. Equalizing the budget required changing width, depth, state dimension and input window jointly (Table 4.2), so this equalizes capacity, not the other architectural degrees of freedom

Hypothesis	Statement	Result
H5	Memoryless classical relays fail on unknown channels; a minimal MLP mitigates by learning, near the model-aware optimum	**Confirmed**: on unknown ISI, AF/DF hit the analytic 0.25 floor (DF non-monotonic in SNR) while the 169-parameter MLP reaches BER $< 5 \times 10^{-5}$, within $\approx$1–1.5 dB of genie-CSI Viterbi MLSE; the same architecture mitigates a composite ISI$\times$PA$\times$phase cascade with no impairment model, tracking $\approx$2 dB behind pilot-aided Viterbi and matching blind CMA when no posterior exists; the advantage is bounded to structural model-class mismatch representable at minimal capacity

The main conclusions, in order of significance, are:

1. **AI relays beat AF at low SNR (0-4 dB) but not DF.** On the canonical channel, the best neural relays clearly outperform AF in the low-SNR regime while tracking symbol-wise DF within statistical uncertainty; DF remains lowest or tied-lowest at every SNR point. The theoretical explanation is that neural networks approximate the MMSE-optimal soft-threshold estimate $f^*(y) = \tanh(y/\sigma^2)$, whose end-to-end benefit over the hard slicer is limited in this uncoded setting.

2. **DF attains the lowest per-symbol BER at medium-to-high SNR (≥6 dB) with zero parameters.** The classical decode-and-forward relay requires no training and no parameters yet achieves the best performance among the evaluated per-symbol relays when channel quality is sufficient for reliable first-hop demodulation. At high SNR, the posterior-mean estimate converges to sign(y), which is exactly the symbol-wise DF operation.

3. **No AI architecture separates itself at original parameter counts.** On the canonical channel the best neural relays (MLP, Hybrid, VAE) perform within statistical uncertainty of one another, and classical DF matches or exceeds all of them across the SNR range; the cGAN is excluded from this comparison (Section 5.2). The selective state space model's $O(n)$ complexity and adaptive filtering mechanism make it an efficient sequence-model

choice, but no architectural inductive bias yields a decisive BER advantage on this task.

4. **Architecture matters less than parameter count at equal scale — but only at high SNR.** When all models are normalized to approximately 3,000 parameters (Table 5.3), the spread across the six architectures falls monotonically with SNR: 17.4% at 0 dB, 9.3% at 12 dB, 4.1% at 20 dB. In the high-SNR regime the original claim holds and architecture is close to irrelevant. At low SNR it does not: the six models separate cleanly into a feedforward group (MLP, Hybrid, VAE at 0.336–0.342) and a sequence group (Transformer, Mamba-S6, Mamba-2 at 0.400–0.407), a gap far larger than the confidence intervals, with the feedforward models close to DF and the sequence models close to AF.

   Two points are worth separating here. The split is by *family*, not by capacity, since all six carry the same parameter budget — so at low SNR architecture does matter, and the sequence models' inductive bias is actively unhelpful on a memoryless channel where there is no temporal structure to exploit. And the VAE is no longer the exception it was: at 3K parameters it sits inside the feedforward group at every SNR (0.0101 against MLP-3K's 0.0099 at 20 dB). The earlier reading, that the VAE was a consistent underperformer, does not survive re-measurement. The cause is not the corrected noise convention, which a controlled A/B rules out: training the same relay on data generated under the old convention and under the corrected one produces curves that agree to within Monte Carlo noise at every SNR. The earlier figures came from an implementation that predates the current relay module, so the discrepancy is one of code rather than of channel calibration. See the discussion of the generative paradigm in Section 2.3.3.

5. **No model larger than 169 parameters improved on it.** The Minimal MLP architecture achieves performance comparable to models 100–140× larger, and none of the larger models evaluated reduced BER under the training protocol used. The bias-variance account is a candidate explanation for that observation — the target function is simple (a soft threshold), so additional parameters would increase variance without reducing bias — but it is not established here as the mechanism: each model was trained once, so a capacity effect cannot be separated from run-to-run variance (Section 7.4).

6. **The Hybrid relay is the recommended practical choice, given a usable estimate of the operating SNR.** By combining AI processing at low SNR with classical DF at high SNR, the Hybrid relay achieves near-optimal performance across the entire SNR range with only 169 parameters. It selects between the two regimes from an SNR estimated out of the received signal power, so no manual per-regime configuration is needed. The recommendation is conditional in a way the experiments here do not discharge: the switching threshold is fixed at the crossover measured on the training sweep, and neither the accuracy of the run-time SNR estimate, the sensitivity of the BER to a misplaced threshold, nor any online adaptation of it was evaluated. All reported Hybrid results are therefore obtained under an accurate operating-SNR estimate, and the margin over its two constituents

is small enough at the crossover that a mis-estimate there would plausibly erase it.

It is important to note that sequence-detection methods such as Viterbi MLSE appear only in the unknown-channel study of Chapter 6, where channel memory is introduced explicitly through a 3-tap FIR/ISI channel. The canonical system model of Chapters 1–5 remains memoryless and therefore requires only per-symbol processing.

The unknown-channel contribution (Chapter 6) settles the complementary question of *when the learned relay is not merely competitive but necessary*:

- **On channels violating the deployed relay's model class, learning restores reliability at near-optimal cost (H5: confirmed).** The flat-channel control isolates the cause: against unknown phase, gain and per-branch asymmetry, all memoryless, the classical relay does not fail and the MLP only matches it (gap ≤ 0.0036), so parameter uncertainty alone is not what learning mitigates. Failure appears only with unmodeled *memory*: on unknown 3-tap ISI, AF and symbol-wise DF sit at the analytic 0.25 floor and DF actually worsens as SNR rises, while the same 169-parameter MLP learns a joint equalizer–detector and falls below 5×10^{-5} at 16 dB. The Viterbi benchmark bounds the claim: given the channel family and enough pilots, classical estimate-then-MLSE stays 1–1.5 dB ahead, so the learned relay's value is *family-agnosticism* rather than optimality — near-MLSE reliability with no identification stage, on a composite ISI/nonlinearity/phase cascade it was never told the composition of, and, with no pilots or channel prior at all, matching blind CMA without its instability or per-block adaptation cost. Sweeping the pilot budget makes the boundary quantitative: pilot-aided Viterbi wins down to roughly ten pilots and then collapses at the identifiability threshold, while the learned relay is invariant to it. The crossover is therefore governed by identifiability and, through pilot overhead, by block length. Section 6 gives the full evidence; the complexity accounting there shows the cost side scales linearly in window and output dimension against the trellis's M^L.

These findings demonstrate that neural network-based relay processing is a viable complement to classical approaches in the low-SNR regime, matching, though not surpassing, the per-symbol-optimal DF baseline everywhere else at negligible computational cost. The overarching insight is that **model complexity should be matched to task complexity**: for the canonical relay denoising task, minimal architectures suffice, and the choice between neural network paradigms, i.e. feedforward or sequential, matters less than proper model sizing and regularization. The practical recommendation is to deploy a Hybrid relay with a 169-parameter MLP sub-network, which combines the neural low-SNR advantage over AF with DF's zero-parameter reliability at high SNR; for constellations with more than two levels per axis, a per-axis I/Q-split denoiser is not the right formulation and a joint two-dimensional relay would be needed, though that is not evaluated here; and on links with unmodeled impairments (impairments outside the canonical memoryless model, including ISI, bias, and hardware nonlinearity), a minimal learned relay delivers near-MLSE reliability without channel identification, where the memoryless classical relays fail outright (H5).

Chapter 8

Summary

Scope of the summarized results. The results summarized below originate from three studies. The canonical study (Chapter 5) evaluates relay processing on a memoryless channel with white noise and uncoded transmission. A supplementary coded study (Section 5.5) measures a block-DF relay, soft-information forwarding, and an adaptive modulation-and-coding scheme against that baseline. The unknown-channel study (Chapter 6) extends the canonical baseline by deliberately introducing impairments that violate its assumptions, including intersymbol interference (ISI), nonlinear distortion, and model mismatch. Conclusions involving channel-memory sequence detection, Viterbi maximum-likelihood sequence estimation (MLSE), pilot overhead, or blind equalization refer specifically to that extension rather than to the canonical system model; the coded study's soft-decision Viterbi decoding of the convolutional code is a distinct use of the Viterbi algorithm, for error-correction decoding rather than channel-memory sequence detection.

This work examined classical and neural network–based relay strategies for *two-hop relay communication* using a unified simulation framework. The objective was to analyze performance–complexity tradeoffs and determine when learning-based relays provide practical advantages over analytically derived methods. The core comparison is carried out on one canonical setup, fixed in advance and never departed from within it: a two-hop SISO link over i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER, with the relay function as the only varied axis. The coded study and the unknown-channel contribution depart from it deliberately. The study showed the following:

1. Neural relays yield a consistent performance improvement over amplify-and-forward (AF) at low signal-to-noise ratio (SNR), where their learned soft denoising avoids AF's noise amplification (H1).
2. Classical *decode-and-forward (DF)* matched or outperformed all neural approaches at **medium-to-high SNR** while requiring no trainable parameters, confirming that DF attains the lowest per-symbol BER in this regime (H2).

3. A systematic complexity analysis demonstrated that performance **does not improve monotonically with model size** (H3).
4. A **minimal neural relay** (169 parameters) achieved performance comparable to significantly larger models; no larger model evaluated here improved on it, though the protocol used (one training run per model, no train/test-gap measurement) cannot attribute that to overfitting.
5. A **size sweep across every channel studied**, replicated over three initializations, located the floor and showed it is set by *channel memory* rather than parameter count: four parameters suffice on a memoryless channel, $42\times$ below the 169-parameter relay, while three-tap channels need 73 to 145 and a window spanning the interference. No upturn in error with capacity was observed, so the overfitting hypothesis H3 is unsupported on evidence rather than untested (Sections 5.4 and 7.2.3).
6. When neural relays were evaluated under equal parameter budgets, architectural differences largely vanished, indicating that **parameter budget** appears to explain most of the high-SNR differences, although architectural factors – depth, width, state dimension and window size – remain partially confounded at an equal budget (H4).
7. From a deployment perspective, a `hybrid relay` combining neural processing at low SNR with DF at high SNR emerged as the **recommended default where the operating SNR can be estimated reliably**, achieving near-optimal performance across the SNR range with minimal overhead; the robustness of its switching threshold to SNR-estimation error was not evaluated.

The first six findings concern the canonical memoryless relay model; the remaining findings summarize the deliberate model-mismatch extension of Chapter 6. The supplementary coded study's findings, being a smaller third part of the thesis, are summarized in prose rather than as numbered items, at the end of this chapter.

7. Beyond the canonical memoryless model, on an **unknown channel** violating classical assumptions (unmodeled intersymbol interference or biased nonlinearity), AF and DF failed to relay reliably (exhibiting an analytic BER floor of 0.25, with DF *worsening* as SNR increased) while the same minimal 169-parameter MLP restored reliable relaying by learning the impairment from data, including an arbitrary composite cascade of ISI, amplifier nonlinearity, and unknown phase. It tracked $\approx$1–1.5 dB behind a *pilot-aided* Viterbi receiver on pure ISI ($\approx$2 dB on the composite cascade) and, in the fully posterior-free regime (no pilots, no channel prior), matched the best blind classical equalizer while avoiding its instability and online-adaptation cost (H5).
8. Sweeping the intermediate *partial*-posterior case located the crossover precisely: it is governed by the **reliability of the channel estimate** and by pilot overhead. Pilot-aided classical processing wins on long blocks at negligible cost, but the zero-pilot learned relay

becomes decisive once blocks are too short, or channels too transient, for pilots to yield an accurate estimate. The learned relay's advantage is therefore specific to *structural* model-class mismatch (memory, nonlinearity, absent pilots) rather than to unknownness per se. Together with the high-SNR result, this delineates precisely where learned relays add value: never above the model-aware optimum, but uniquely robust to *not knowing which model applies* and to the pilot overhead of channel identification.

9. A **complexity accounting** completed the trade-off: the learned relay's per-symbol cost grows only linearly in window length and output dimension, and 30–90$\times$ faster in measured wall-clock as a vectorized operation for the reference implementations compared (Section 6.1.6), whereas exact Viterbi MLSE grows as M^L and becomes intractable. The learned relay thus trades a small BER gap for cost that scales gracefully exactly where classical sequence detection does not.

Overall, the results indicate that learning-based relays are best used as targeted enhancements rather than universal replacements, with simplicity, parameter efficiency, and hybrid design as key principles for practical relay systems. The canonical conclusions hold for the *uncoded*, symbol-level setting studied here (Remark 4.2); a supplementary coded study (Section 5.5) measures where coding changes that standing: a rate-1/2 convolutional code with a genuine block-DF relay beats uncoded DF from 8 dB upward, but by 20 dB both learned relays overtake block-DF (one architecture already does at 16 dB, the other still trails there), since a relay that decodes the code spends its redundancy before forwarding while a denoising relay leaves that redundancy intact for the destination to use. That advantage is bounded, not erased, once the code rate is itself allowed to adapt: maximizing goodput over a modulation-and-coding grid shows rate selection dominates the relay-decoding question by an order of magnitude, unless a latency deadline reopens the gap by forcing the block-DF relay's double buffering out of budget first. Extending the study to other channel families, multi-antenna configurations, and higher-order constellations as a relay-architecture question (16-QAM and denser, which need a joint N-class 2D relay) is identified in Chapter 7 as the principal direction for future work.

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Chapter 9

Appendices

9.1 Appendix A: Mathematical Notation

Symbol	Description
b	Binary bit ($\in \{0, 1\}$)
x	Modulated BPSK symbol ($\in \{-1, +1\}$)
y	Received signal
n	Noise sample
h	Fading coefficient
σ^2	Noise variance
SNR	Signal-to-Noise Ratio (linear)
G	AF amplification gain
$\mathbf{W}$	Neural network weight matrix
$\mathbf{b}$	Bias vector
P_e	Bit error rate
$Q(\cdot)$	Gaussian Q-function
β	VAE KL weight
λ	Regularization parameter
η	Learning rate
Δ	State space discretization step
$\mathbf{A}, \mathbf{B}, \mathbf{C}, D$	State space model matrices

9.2 Appendix B: Model Architectures and Hyperparameters

Relay	Network Topology	Train Loss (LR)	Total Params	Implementation
MLP (Minimal)	Feedforward MLP, 5-symbol sliding window	MSE, lr=0.01, 100 epochs	169	NumPy (CPU)
Hybrid	MLP + DF switching via learned SNR threshold ($\sim$5 dB)	MSE (MLP only), lr=0.01	169	NumPy (CPU)
VAE	Encoder–decoder latent model (7→8→1)	β-VAE ($\beta = 0.1$), Adam lr=10^{-3}	1,777	PyTorch (CUDA)
cGAN (WGAN-GP)	Conditional GAN (generator + critic)	WGAN-GP, Adam lr=10^{-4}	2,946	PyTorch (CUDA)
Transformer	2-layer encoder, 4 heads, $d_{\text{model}} = 32$	MSE, Adam lr=10^{-3}	17,697	PyTorch (CUDA/CPU)
Mamba-S6	Selective SSM, 2 S6 layers	MSE, Adam lr=10^{-3}	24,001	PyTorch (CUDA/CPU)
Mamba-2 SSD	Selective SSM with SSD kernel (2 layers)	MSE, Adam lr=10^{-3}, clip=1.0	26,179	PyTorch (CUDA)

Table 9.2: Comparison of relay models, training objectives, parameter counts, and implementations

9.3 Appendix C: Reproducibility

9.3.1 Testing

159 automated tests (pytest) cover the channels, modulation, relay-strategy, simulation and statistics modules with a 100% pass rate. The suite requires only NumPy and SciPy; PyTorch is optional and needed solely by the neural relays that use it.

9.3.2 Reproducibility

Random seeds are controlled at the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results.

Provenance of the reported results. Every result reported in this thesis was regenerated from the archived experiment artifacts and verified against its committed result file, at 10 trials $\times$ 10,000 bits per SNR point. The verification is mechanical rather than manual: each numerical table cell is re-derived from the committed result file it came from, so a table and the data behind it cannot drift apart unnoticed.

9.4 Appendix D: Normalized 3K-Parameter Configurations

Model	Parameters	Window	Hidden / Architecture
MLP-3K	3,004	11	hidden=231
Hybrid-3K	3,004	11	hidden=231 (+ DF switch)
VAE-3K	3,037	11	latent=10, hidden=(44, 20)
cGAN-3K	3,004	11	noise=8, g_hidden=(30, 30, 16), c_hidden=(32, 16)
Transformer-3K	3,007	11	d_model=18, heads=2, layers=1
Mamba-S6-3K	3,027	11	d_model=16, d_state=6, layers=1
Mamba-2-3K	3,004	11	d_model=15, d_state=6, chunk_size=8, layers=1

All 3K configurations use a window size of 11 (vs. 5 for the original MLP/Hybrid and 7 for the original VAE/cGAN); the original sequence models already used 11. The parameter counts are within ±1.2% of the 3,000 target.

9.5 Appendix E: Master Experiment Ledger

Every numerical table and figure in this thesis traces to exactly one committed result file, verified cell-by-cell against that file by `verify_thesis_tables.py` in the code repository (Section 4.3.4). This ledger lists the modulation, channel, and result source for each major study, grouped in the order they appear. Result-source filenames matching `e6_*.npy` are relative to `e6_unknown_channel_results/`.

Table 9.4: Master experiment ledger: modulation, channel, and result-file provenance for every major study

Study	Modulation	Channel	Tbl/Fig	Result source
Canonical benchmark	QPSK	Rayleigh	tbl:2, fig:10	`results/modulation/qpsk_rayleigh.json`

Study	Modulation	Channel	Tbl/Fig	Result source
Normalized 3K-param. comparison	QPSK	Rayleigh	tbl:8	results/normalized_3k/3k_rayleigh.json
Coded block-DF, K-sweep	QPSK	Rayleigh	tbl:34, 35	results/coded_df_experiment.json
Coded high-SNR paired re-test	QPSK	Rayleigh	tbl:37	results/coded_high_budget_test.json
Coded relay-error mechanism	QPSK	Rayleigh	tbl:38	results/coded_error_mechanism.json
Coded soft-decision relaying	QPSK	Rayleigh	tbl:39	results/coded_soft_decision.json
Equal-throughput & latency	QPSK, 16-QAM	Rayleigh	tbl:40, 41	results/coded_latency_throughput.json
Adaptive modulation/coding (AMC)	QPSK, 16-QAM	Rayleigh	tbl:42	results/coded_rate_adaptation.json
Capacity under a latency budget	QPSK, 16-QAM	Rayleigh	tbl:43	results/coded_latency_capacity.json
Unknown ISI (layer 2)	BPSK	AWGN (2nd hop)	tbl:E6	e6_sim_ported_results.npy + e6_viterbi_*.npy
Flat-channel control	BPSK	Rayleigh	tbl:E6flat	e6_flat_ported_results.npy
Unknown ISI, QPSK generalization	QPSK	AWGN	tbl:E6qpsk	e6_qpsk_unknown_channel_results.npy
Composite impairment cascade	DBPSK	Rayleigh	E6composite	e6_composite_ported_results.npy
Partial posterior (pilot sweep)	BPSK	Rayleigh	E6partial	e6_partial_ported_results.npy
Blind channel (layer 4)	BPSK	Rayleigh	E6blind	e6_blind_ported_results.npy
Four-layer summary	mixed (above)	mixed (above)	tbl:layers	combination of the above

The result files above are themselves generated by named, committed scripts (e.g. coded_rate_adaptation.py produces coded_rate_adaptation.json); the correspondence between script and output file is one-to-one and is not separately tabulated here since the filenames make it explicit.

אוניברסיטת תל-אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר להנדסת חשמל

ארכיטקטורות למידה עמוקה לתקשורת ממסר דו-קפיצתית: מחקר השוואתי של אסטרטגיות ממסר קלאסיות ומבוססות רשתות נוירונים

חיבור זה הוגש כעבודת גמר לקראת התואר "מוסמך אוניברסיטה"

בהנדסת חשמל ואלקטרוניקה באוניבסיטת תל-אביב

על-ידי

גיל צוקרמן

העבודה נעשתה בבית ספר להנדסת חשמל

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ניסן תשפ"ו

תקציר

תזה זו משווה בין אסטרטגיות ממסר קלאסיות ונלמדות לתקשורת דו־דילוגית. תצורה אחת קבועה לאורך המחקר המרכזי: קישור כניסה יחידה ויציאה יחידה (SISO) עם ממסר בודד, דעיכת ריילי מהירה בלתי־תלויה ושווה־התפלגות בשני הדילוגים, פס־בסיס מרוכב, אפנון QPSK מקודד־גריי ותמסורת בלתי־מקודדת, כאשר הממסר הוא המשתנה היחיד. שמונה אסטרטגיות מוערכות, מהגבר־והעבר (AF) ופענח־והעבר סימבולי (DF) ועד מודלים גנרטיביים ומודלי רצפים. התוצאות הן אומדני מונטה קרלו עם רווחי סמך של 95%, מכוילים מול הביטוי הסגור להרכבה הדו־דילוגית.

בתרחיש זה, הממסרים הנלמדים הטובים ביותר עולים על AF ביחס אות־רעש (SNR) נמוך; ב־SNR בינוני וגבוה DF משתווה לכל ממסר או עולה עליו ללא עלות פרמטרים -- קביעה אמפירית על ממסר בלתי־מקודד וסימבולי־לפי־סימבול, ולא אופטימליות כללית. סריקת גדלים על פני כל הערוצים הנחקרים, בשלושה אתחולים לכל תצורה, מאתרת את הממסר הקטן ביותר שאינו כרוך בעלות מדידה, ומראה שזיכרון הערוץ, ולא מספר הפרמטרים, הוא הקובע אותו: בערוץ חסר־זיכרון ארבעה פרמטרים, יחידה נסתרת אחת ללא חלון, נמצאים במרחק 0.03 dB מן הבסיס הקלאסי, בעוד ערוצים בעלי שלושה מקדמים דורשים 73 עד 145 פרמטרים וחלון הפורש את ההפרעה. השגיאה אינה עולה שוב עם גידול הקיבולת, ולפיכך לא נצפתה התאמת־יתר בטווח הגדלים שנסרק. ממסר Hybrid העובר ל-DF מעל נקודת המעבר הנמדדת קרוב לאופטימלי ללא עלות נוספת, אם ה-SNR מוערך באמינות. מחקר מקודד מעלה את רף ההשוואה: קוד קונבולוציוני בקצב 1/2 עם ממסר DF מבוסס־בלוק, שונה מה־DF הסימבולי, עולה על DF בלתי־מקודד מ־8 dB ומעלה. שני הממסרים הנלמדים עוקפים אותו עד 20 dB (Mamba-S6 כבר ב־16): ממסר המפענח צורך את יתירות הקוד, ממסר המסיר רעש משאיר אותה שלמה. הסתגלות סכמת האפנון־והקידוד גורמת לסכמה להכריע את בחירת הממסר בסדר גודל -- אלא אם ההשהיה מחזירה את הפער.

התרומה העיקרית היא סולם של ארבע שכבות של הנחות ערוץ מוקלות בהדרגה. בשכבה 1 העיבוד הקלאסי מנצח: בערוץ חסר־זיכרון ותואם, החלטה קשה היא פונקציית הממסר האופטימלית לכל סימבול. שכבה 2 מוסיפה מסנן ISI תלת־מקדמי בלתי־ידוע, והממסרים הקלאסיים חסרי־הזיכרון נכשלים, כאשר שיעור השגיאה של DF עולה עם עוצמת השידור, בעוד ממסר נלמד ממוסגר־חלון בן 169 פרמטרים משקם את הקישור בעלות קבועה לכל סימבול, במרחק 1 עד 1.5 dB מגלאי Viterbi בעל עלות אקספוננציאלית. שכבה 3 מדרדרת את אומדן הערוץ: היתרון הקלאסי מחזיק עד כעשרים פיילוטים ואובד בעשרה. שכבה 4 מגרילה ערוץ חדש בכל בלוק, ושם הממסר הנלמד מוביל על פני המאזן העיוור הקלאסי הטוב ביותר.

התוצאות מתחמות משטרי הפעלה: עיבוד קלאסי כאשר מודל הערוץ המשוער תואם לערוץ, והממסר הנלמד כשיש לערוץ מבנה בלתי־ממודל, אומדנו מדורדר, או חסר לכל בלוק.